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SECTION 042 GENERAL ADMINISTRATIVE REQUIREMENTS

042.1 TERMS AND DEFINITIONS

042.1.1 The term "these specifications" refers to the document entitled, "Homeland Security Cutter – Light (HSC-L) Specification".

042.1.2 The term "section" refers to a part of these specifications designated by a single 3-digit number. Reference to sections of these specifications is made by calling out the appropriate 3-digit Expanded Ship Work Breakdown Structure (ESWBS) number.

042.1.3 The term "herein" shall mean within this entire specification and the Statement of Work.

042.1.4 The term “as shown on the drawings” shall mean that there is additional information on the contract drawings, contract guidance drawings and the project peculiar documents.

042.1.5 **Contractor** - The Contractor is the firm holding a prime contract with the Government for design or construction of the HSC-L.

042.1.6 The phrase “be approved” or “for approval” used herein shall be interpreted to mean approval from the Government Contracting Officer or Contracting Officer’s Representative (COR) in accordance with the terms and conditions of this contract.

042.1.7 The term "days" shall mean calendar days unless otherwise specified herein.

042.1.8 "Exterior location", "Weather location" and other similar terms shall mean exposed to the elements and left unprotected by the boat’s interior environment.

042.1.9 FCC is the Federal Communications Communication.

042.1.10 GFP means Government Furnished Property- It includes both Government Furnished Equipment and Government Furnished Information - Items and information in the possession of, or acquired by the Government and delivered to, or otherwise made available to, the Contractor. These items are included in Schedule A and Schedule C of the contract.

042.1.11 "Interior location" and other similar terms shall mean protected by the boat's interior environment.

042.1.12 Standards Bodies are entities that establish standards for the industry such as IEEE, ASTM, ABYC, NFPA, ISA, etc.

042.1.13 USCG is the United States Coast Guard.

042.1.14 The term "approved waiver request" shall mean: Any non-conformance, found to depart from specified requirements during or after manufacture, which is considered usable as is or after rework by an approved method.

042.2 GENERAL

042.2.1 These specifications describe the requirements for a Homeland Security Cutter- Light including the general features, functions, performance, and arrangements, but not necessarily the details of the design. Together with the contract drawings and contract guidance drawings, hereafter referred to as “the drawings,” and other referenced documents, they define the boat as to principal dimensions, arrangement, and power.

042.2.2 Material and equipment under these specifications shall be new and unused. Equipment shall be current production models with existing equipment and repair parts support. Materials or equipment referred to by trade or brand name or manufacturer's model or by Federal/National Stock Number represent, to the best knowledge of the Government, current production models on 5 January 2026.

042.2.3 Where a commercial manufacturer's type, model, or other designation is referenced herein, together with the term "or equal", such identification is intended to be descriptive, but not restrictive, and is used to indicate the applicable characteristics as they existed on 5 January 2026.

042.2.3.1 If the term "or equal" does not appear after the requirement of a commercial manufacturer's type, model, or other designation, it shall be read as only the specific commercial manufacturer's type, model, or other designated material or equipment will suffice.

042.2.4 Approval shall be obtained from the Contracting Officer prior to the use of other than referenced document(s) in lieu of the requirements of the primary reference cited. This includes superseding documents if applicable. Later versions of product specifications may be used upon notifying the Contracting Officer and are approved unless specifically rejected within 45 days. Proposals for substitutions for primary and the above sub-tier references shall be made by letter from the Contractor to the Contracting Officer. To ensure that sufficient information is available to permit evaluation of alternatives, the Contractor shall furnish descriptive information necessary to substantiate that the alternative offered is cost effective and meets the performance, life, safety, interchangeability, maintainability, and reliability of that prescribed by the referenced document(s).

042.2.4.1 Equivalent equipment - Where equipment is specified herein by a reference document, the Contractor may propose equivalents to these references.

042.2.4.1.1 In order for the proposed item to be acceptable, an equipment equivalency certificate shall be prepared and be approved.

042.2.4.1.2 The equivalency certificate shall include, and approval will be based on, the following equipment attributes:

042.2.4.1.2.1 Meet the performance requirements specified herein, and possess similar:

042.2.4.1.2.1.1 Dimensions.

042.2.4.1.2.1.2 Weight.

042.2.4.1.2.1.3 Power, HVAC, cooling water, and other support services.

042.2.4.1.2.1.4 Suitability for marine service.

042.2.4.1.2.1.5 Material.

042.2.4.1.2.1.6 Reliability and Maintenance features.

042.2.4.1.2.1.7 Vendor furnished training.

042.2.4.1.2.1.8 Vendor furnished service.

042.2.4.1.2.1.9 Data requirements.

042.2.4.1.2.1.10 Maintenance and support costs, expected life, and crew workload.

042.2.4.1.2.1.11 Environmental and Safety characteristics including airborne noise and vibration.

042.2.4.2 To the extent that the above similarities are not met, the Contractor shall explain why they are not essential to the proper functioning and support of the equipment.

042.2.4.3 The equivalency certificate shall identify support, auxiliary, or subsystem equipment or material that would be changed or modified as a result of acceptance of the equivalent equipment.

042.2.4.4 The equivalency certificate shall identify discrepancies with the requirements of these specifications that would result from the acceptance of the equivalent equipment.

042.2.4.5 Unless otherwise specified, measurements, calculations, and mathematical data shall be presented in U.S. Customary units of measure. In addition, written data produced in support of this effort or required as a deliverable, shall be in the English language.

042.2.4.6 If the Contractor proposes to modify a product to make it conform to the specified requirements, he shall include in his proposal a clear description of such proposed modifications and clearly mark any descriptive material accordingly. The Contracting Officer will be the final judge of the suitability of the item submitted.

042.3 REFERENCED DOCUMENTS

042.3.1 Definitions

042.3.1.1 "Contract drawings" are drawings which delineate design features of the boat. No departure from details of a contract drawing shall be made without specific approval of the Government. Minor rearrangements to a space defined by a Contract Drawing (± 6 inches for bulkheads except main transverse and main longitudinal bulkheads, ± 10 inches for accesses and ± 12 inches for equipment and furniture and other items within a space) may be made, to clear interferences or accommodate changes in sizes of those components, provided that operability, maintainability, man-machine interfaces, and service envelopes are not adversely affected.

042.3.1.2 "Guidance drawings" are drawings identified as "Guidance Drawing" which illustrate design features of the ship. A guidance drawing does not necessarily depict, nor is it intended that it depict, features and details of the systems and structures to which it relates. It serves the purpose of providing information which, when utilized in conjunction with applicable specification requirements, contract drawings, project-peculiar drawings, and other information, will result in an acceptable design. Departure from specific details may be made without approval, provided an acceptable design results. Guidance Drawings will not necessarily be updated or revised to reflect future modifications or changes to these specifications.

042.3.1.3 "Expanded Ship Work Breakdown Structure" (ESWBS) is a method of ship weight and system classification defined by NAVSEA 0900-LP-039-9010. Reference to ESWBS or SWBS in these specifications shall mean the Expanded Ship Work Breakdown Structure.

042.3.1.4 "Industry standards" are documents promulgated by recognized non-Government organizations (such as ANSI or ASTM) that establish engineering and technical limitations and applications for items, materials, processes, methods, design, and engineering practices.

042.3.1.5 "Primary references" and "sub tier references": Any referenced document which is directly cited for use or application by an identifying number or name in these specifications or in any contract drawing, is a primary reference for each particular use or application for which it is so cited. Any document which is cited in a primary reference, or in a document referenced therein, is a subtier reference. A document which is a subtier reference in one application may be a primary reference in another application when it is cited directly

042.3.1.6 Where any primary or sub-tier reference uses language such as: “recommends, should, or similar,” this shall be interpreted as a “shall” requirement in the reference document itself.

042.3.2 In the event that the Contractor discovers any error or inapplicability in the identification of reference documents, the Contractor shall notify the Contracting Officer within 5 days of that discovery.

042.4 EFFECTIVE ISSUE

042.4.1 The effective issue of a reference document refers to that issue of a reference document which forms a part of these specifications. Whenever first tier references are identified within these specifications by revision or amendment, that issue shall be the effective issue for the particular use or application cited. These specifications do not prescribe effective issue requirements for sub-tier references. Sub-tier references shall be those in effect on the date of the first tier referenced document version used. The effective issue of references not specifically identified by issue or amendment shall be that in effect on 5 January 2026 unless otherwise specified. The effective issue of contract drawings, guidance drawings and project peculiar documents is the revision listed in Table I. Where no revision is indicated, the original issue is intended.

042.5 ORDER OF PRECEDENCE

042.5.1 Order of Precedence of requirements within the Expanded Ship Work Breakdown Structure Sections of the Homeland Security Cutter- Light (HSC-L) Specification is as follows:

042.5.1.1 Requirements within a SWBS Section applicable to the entire HSC-L (for example, SWBS Section 070 - General Requirements for Design and Construction) state the broadest requirements and apply to systems and equipment, unless otherwise specified in the more detailed SWBS Sections, in which case the detailed SWBS Section requirements apply.

042.5.1.2 Specific requirements within a lead, 1-digit SWBS Section (for example, a SWBS Section which includes the general or common requirements for a number of similar systems, such as SWBS Section 500 - Auxiliary Systems, General) are the next broadest category of requirements, and shall apply to systems and equipment in its grouping as well as other ship systems where applicable, unless otherwise specified in detail system SWBS Sections.

042.5.1.3 Specific requirements within a 2-digit SWBS Section (for example, a SWBS Section which includes the details of a particular system such as SWBS 530, Fresh Water Systems) are the next broadest category of requirements and shall apply to systems and equipment in its grouping, unless otherwise specified in detail system SWBS Sections.

042.5.1.4 Specific requirements within a detailed, 3-digit SWBS Section (for example, a SWBS Section which includes the details of a particular system such as SWBS 533, Potable Water Systems) are the most detailed requirements. Requirements in these detailed system SWBS Sections apply for the system covered in the detailed SWBS Section.

042.5.1.5 The omission in one SWBS Section of details covered in another SWBS Section shall not be considered a conflict.

042.5.1.6 In the event that two or more requirements of equal precedence are found to apparently conflict, the Contractor shall formally identify the conflict to the Government for clarification, or correction of the specifications, as appropriate.

042.5.2 Order of Precedence of Documents

042.5.2.1 In case of conflict between the Specification and referenced documents, the following order of precedence applies:

042.5.2.1.1 Contract Drawings.

042.5.2.1.2 HSC-L System Specification

042.5.2.1.3 Primary references

042.5.2.1.4 Contract Guidance Drawings.

042.5.2.1.5 Sub-tier references.

042.5.2.2 The order of precedence only applies in the event there is a conflict between requirements.

042.5.2.3 In the event that requirements are found to apparently conflict, the Contractor shall formally identify the conflict to the Government for clarification, or correction of the specifications, as appropriate.

042.5.3 The following drawings and form a part of these specifications to the extent specified herein.

042.5.3.1 Contract Drawings

Drawing Number	Title	Revision
	TBD	

042.5.3.2 Guidance Drawings

Drawing Number	Title	Revision
	TBD	

SECTION 070 GENERAL REQUIREMENTS FOR DESIGN AND CONSTRUCTION

070.1 GENERAL

070.1.1 This section contains general technical requirements for the design and construction of a Homeland Security Cutter - Light . The information contained in this section sets forth general requirements for design, workmanship, and installation unless otherwise specified herein. The cutter shall incorporate proven technology.

070.1.2 Equipment, machinery, and materials shall be suitable for extended exposed stowage and reliable operation in the marine environment.

070.1.3 The Contractor shall comply with Original Equipment Manufacturers (OEM) installation instructions and recommendation of components unless there is a conflict with the HSC-L contract requirements, in which case the USCG will adjudicate the conflict.

070.2 PRINCIPAL DESIGN FEATURES

070.2.1 **Physical Constraints**

070.2.1.1 Draft – Not to exceed 7 feet at Full Load ATON Condition at end of service life. Draft is measured from the bottom- most extremity including propeller(s), appendages, and hull in the full load condition.

070.2.1.2 Air Draft – Fixed height not to exceed 20 feet above the Minimum Operating Condition waterline.

070.2.2 **Propulsion Plant Architecture**

070.2.2.1 The boat shall have 2 main propulsors.

070.2.2.2 The boat and its systems shall operate on the following fuels:

070.2.2.2.1 NATO F-76 fuel in accordance with MIL-DTL-16884;

070.2.2.2.2 Marine Gas Oil conforming to ISO 8217 Grade DMA with a maximum sulfur content of 0.5 mass % and a maximum Fatty Acid Methyl Esters content of 0.5 vol. %; and

070.2.2.2.3 ASTM D975 Grade Number 2 Ultra Low Sulfur Diesel with a maximum sulfur content of 15 ppm.

070.2.3 **Speed, Range and Endurance**

070.2.3.1 The HSC-L shall be capable of maintaining speed of at least 10 knots in seas with significant wave height of less than one foot, while outfitted for Full Load ATON.

070.2.3.2 Range: The boat shall be able to travel 720 nautical miles (nm) in seas with significant wave height of less than one foot, at 10 knots with a ten percent (10%) fuel reserve.

070.2.3.3 Endurance: The HSC-L shall be capable of conducting unreplenished icebreaking for at least three days to include: fuel, potable water, waste water, food, and solid waste.

070.2.4 **Seakeeping, Dynamic Stability and Maneuverability**

070.2.4.1 At speeds and loading conditions in significant wave height of one foot, the boat shall be able to maintain a straight course ahead.

070.2.4.2 The boat, while using no more than 1/3 of the engine's available power, shall be capable of maneuvering astern in a straight course and perform backing turns.

070.2.4.3 The HSC-L shall be capable of surviving in seas with a significant wave height of six feet, while outfitted for Full Load ATON.

070.2.4.4 The HSC-L shall be capable of transiting in seas with a significant wave height of three feet, while outfitted for Full Load ATON.

070.2.4.5 The HSC-L shall be capable of anchoring in 40 knot winds, in a depth of water that is at least five times its draft.

070.2.4.6 The HSC-L shall be capable of maintaining speed of at least 10 knots in seas with significant wave height of less than one foot, while outfitted for Full Load ATON.

070.2.4.7 The HSC-L shall enable the operator to maintain station, utilizing propulsion and steering, for the purpose of servicing floating ATON, in seas with a significant wave height of three feet.

070.2.4.8 The HSC-L shall be capable of completing an 180 degree turn in waterways as narrow as 140 feet without using ground tackle.

070.2.5 **Service Life and Annual Hours of Operation**

070.2.5.1 The HSC-L shall be designed to operate at a planned 700 hours per year.

070.2.5.2 The HSC-L shall be designed for a minimum service life of 30 years.

070.2.6 **Accommodations**

070.2.6.1 HSC-L shall provide accommodation as shown in the drawings.

070.2.7 **Mission Profiles**

070.2.7.1 Individual sorties may have considerably different percentages depending on the mission conducted. The HSC-L mission profile is an average breakdown and shall be used to establish the propulsion system duty rating.

070.2.7.2 Table 070-1: Mission Profile

070.2.7.2.1	Percentage of Mission	Speed Condition
070.2.7.2.2	33%	Transit
070.2.7.2.3	17%	Icebreaking Full Power
070.2.7.2.4	17%	Icebreaking Half Power
070.2.7.2.5	33%	Anchored

070.2.8 **Towing**

070.2.8.1 The HSC-L shall be capable of towing a vessel of at least equivalent displacement astern, while outfitted for Full Load ATON.

070.2.8.2 The HSC-L shall be capable of being towed by another vessel while outfitted for Full Load ATON.

070.2.9 **Flooding and damage control**

070.2.9.1 The HSC-L shall have survival systems including an alarm system to sense flooding and alerting the crew, a bilge system to remove water, and sufficient access to make temporary repairs to the hull or system.

070.2.10 **Ice Breaking**

070.2.10.1 Level ice shall have a flexural strength of 110 lb/in^2.

070.2.10.2 The HSC-L shall be capable of breaking level ice of 12 inches thickness at a continuous speed of 3 knots ahead.

070.2.10.3 The HSC-L shall be capable of turning 180 degrees in level ice of eight inches thickness by conducting a star maneuver.

<u>070.3</u> <u>LOADING CONDITONS</u>

070.3.1 DEFINTIONS: Loading condition definitions shall be in accordance with the Society of Allied Weight Engineers Recommended Practice No. 12C, modified as follows:

070.3.1.1 "Full Load ATON Condition" (or "Maximum Operating Condition"): The boat is complete and ready for service in every respect equivalent to a departure condition for ATON work. Including: light ship condition, plus full fuel, fresh water, lube oil, crew and effects, stores and maximum ATON cargo load, including buoys, sinkers and associated equipment. (SAWE Condition A)

070.3.1.2 "Full Load Icebreaking Condition": The boat is complete and ready for service in every respect, equivalent to a departure condition for icebreaking work. It is light ship condition, plus full fuel, fresh water, lube oil, crew and effects, stores, with Zero cargo load or ballast.

070.3.1.3 "Minimum Operating Condition" (Min Op): The boat is complete and ready for service in every respect with the assumed loading that may occur at the end of normal operations and arrival back to port. It is light ship condition, plus 10% fuel, 10% fresh water with the 90% weight of water moved to the black/grey water tank location, lube oil, crew and effects, and no ATON load or associated equipment. (SAWE Condition F)

070.3.1.4 "Full Load End of Service Life (EOSL) Condition": shall include the full load condition (SAWE Condition A) plus service life weight and KG allowances.

070.3.1.5 "Minimum Operating EOSL Condition": shall include the Min Op condition (SAWE Condition F) plus service life weight and KG allowances.

070.3.2 Loading Condition Limits:

070.3.2.1 Absolute trim limits shall be no more than 0.5 ft down by the bow and 1.0 ft down by the stern in any operating condition.

070.3.2.2 Absolute list limits shall be no more than 1/2-degree port and 1/2-degree starboard for any loading condition. ATON loads shall be assumed to have net zero transverse center of gravity and shall not be shifted to correct lightship list angle.

070.4 OPERATING ENVIRONMENT

070.4.1 The boat shall be capable of operating in fog, heavy snow, heavy rain, and topside icing conditions.

070.4.2 The boat's equipment and systems shall be designed to operate without degradation for the following temperatures:

070.4.2.1 Minimum Ambient Air Temperature of –20°F dry bulb

070.4.2.2 Maximum Ambient Air Temperature of 95°F, dry bulb

070.4.2.3 Ambient seawater temperature between 28°F and 90°F, wet bulb

070.4.2.4 Ambient freshwater temperature between 32°F and 90°F**,** wet bulb

070.4.2.5 Equipment and machinery shall operate with dynamic roll of as much as 45° and dynamic pitch of as much as 10° (both single amplitude)

070.4.2.6 Equipment and machinery in exposed locations shall not be installed in areas where it would operate outside its specified limits

070.5 WORKMANSHIP

070.5.1 Workmanship and quality of construction shall meet the limits as outlined in the ABS Guide and Repair Quality Standard for Hull Structures During Construction Part 5.2.

070.5.2 Penetrations through weathertight and weather boundaries shall be made with metallic stuffing tubes. Multi-cable transits may be used in lieu of stuffing tubes in other than weather boundaries

070.6 LIFTING POINTS AND EQUIPMENT REMOVAL

070.6.1 Padeyes of rated capacity shall be installed to permit removal of machinery components from the boat or to onboard maintenance facilities. Lift point placement and removable routes shall consider drawing package.

070.6.2 Lifting points shall be fitted so that parts may be lifted and moved with the minimum disconnection of piping, cables, ventilation ducts, and other installed equipment.

SECTION 071 ACCESS

071.1 GENERAL

071.1.1 The design and arrangement of the HSC-L shall, under operational conditions, provide for safe and convenient access to boat spaces, and for safe and convenient service, maintenance, and operation of controls, systems, machinery, and components. Access shall be provided to pockets, through swash bulkheads and non-tight structures within tanks and elsewhere to permit inspection, care and preservation of the structure.

071.1.2 Primary and secondary egress routes from spaces with fixed fire suppression systems shall be equipped with access that open outward.

071.1.3 Unless otherwise specified clear headroom shall not be less than 6 feet 6 inches. Machinery, piping, lighting fixtures, wireways, ducts, operating rods, brackets, tracks and other items shall be kept clear of normal routes of access. For exceptions, which shall be subject to approval, suitable guards or padding shall be installed.

071.1.4 Ingress and egress of internal spaces shall be free of trip or snag hazards that may impede the safe evacuation of the space.

071.1.5 The HSC-L shall have access to support routine maintenance and inspections.

071.1.5.1 Access to compartments containing equipment, machinery, spare parts shall be provided to facilitate their maintenance and removal with hand tools and without disassembling the component.

071.1.5.2 Access shall be arranged to be clear of piping, wire ways, ducts, and other obstructions.

071.1.5.3 Equipment accessed for routine maintenance completed daily or underway, such as dipsticks and strainers, shall be located so they are readily accessible.

071.1.5.4 Equipment shall be arranged so that each component shall be accessible and removable independent of another component.

071.1.5.5 Access shall allow compartment inspection for damage control while underway without the use of tools.

071.1.6 Access shall be provided to allow any diesel engine to be moved to a location for removal by vertical lift without:

071.1.6.1 Disassembly of the engine, except engine-driven accessories;

071.1.6.2 Removal or disassembly of other major equipment; and

071.1.6.3 Cutting of shell structure.

071.1.7 Access shall be provided to bilge areas. Bilge access points shall provide access to the lowest point of the bilge to ensure it can be reached for maintenance, checked for water content, completely dewatered, and wiped clean of any residue.

071.1.8 Access to fuel tanks, fuel hoses, and fittings shall be provided for repair and maintenance without removing any permanent structural component.

071.1.9 Access shall be provided to tanks, cofferdam, and voids through watertight closures and fasteners that can be opened and re-secured with non-powered hand tools. Such openings shall be sized and placed to allow visual inspection of the interior of the space.

071.1.10 The diesel engines shall be arranged such that the maintenance access needed to perform required annual (or more frequent) routine service is present without removing the engines or removing or disassembling any other equipment, machinery, or structure.

<u>071.2 CONSOLE DESIGN</u>

071.2.1 The console design shall provide for modular replacement of components to mitigate impacts of expected obsolescence, particularly of electronics. This may include incorporation of features such as hinged or bolted console faceplates, equipment racks, or bolt-on sub-assemblies

071.2.2 Equipment within the console shall be arranged so any equipment can be removed without removing any other equipment.

SECTION 073 NOISE, VIBRATION AND RESILIENT MOUNTS

<u>073.1 DEFINITIONS</u>

073.1.1 **A-Weighted Sound Level (dB(A))** - The frequency weighted sound pressure level obtained with the sound level meter set to A-weighting.

073.1.2 **Hazardous Noise** – Hazardous noise is any sound that exceeds 84 dB(A).

073.1.3 **Speech Interference Level (SIL) -** Speech Interference Level is the arithmetic average of the sound levels measured in the 500, 1000, and 2000 Hz octave band center frequencies.

073.1.4 **Transit** – The propulsion plant maintaining sustained speed, normal ship service equipment and lighting, HVAC fans at maximum setting, buoy handling machinery secured, steering equipment in operation.

073.1.5 **ATON** – The propulsion plant idling, normally ship service equipment and lighting, HVAC fans at maximum setting, buoy handling machinery in operation, steering equipment in operation.

<u>073.2 AIRBORNE NOISE</u>

073.2.1 Steady state airborne noise acceptance levels for each category of cutter spaces are as shown in Table 073-1. Airborne noise category assignments for cutter spaces are specified in Table 073-2. Spaces not listed in Table 073-2 shall have the same airborne noise category as a listed space that supports similar functions.

073.2.1.1.1 Acceptance levels shall not be exceeded at any measurement location with machinery and equipment operating simultaneously.

073.2.1.1.2 Transit and ATON operating conditions shall be assessed separately.

073.2.2 Compartments and topside areas with hazardous noise levels (i.e., any area 84 dB(A) or above) shall have a warning sign at each entrance indicating the presence of a noise hazard with indication of the appropriate hearing protection required in accordance with ASTM F1166.

073.2.3 Areas that exceed 96 dB(A) shall be labeled “DANGER, NOISE HAZARD AREA – DOUBLE HEARING PROTECTION MUST BE WORN” at each entrance. Labeling shall comply with ASTM F1166.

073.2.4 Airborne sound insulation properties for acoustic insulation within the accommodation area shall be in accordance with ABS Guide for Crew Habitability on Workboats, Section 5.2.

Table 073-1: Maximum Steady State Airborne Noise Acceptance Levels for Ship Spaces (dB)

Airborne Noise Category	Octave Band Center Frequencies (Hz) (Note 1)								
	32	**63**	**125**	**250**	**500**	**1000**	**2000**	**4000**	**8000**
A-3	87	81	75	72	(SIL LEVEL 64)			63	62
A-12	77	71	65	62	(SIL LEVEL 54)			53	52
B	87	81	75	72	69	66	65	63	62

D	99	94	89	84	79	79	79	79	79
F	110	105	100	95	(SIL LEVEL 65)			80	80
G (Notes 2-4)	115dB(A)								

NOTES:

(1) A 2 dB excess is allowed in any one octave band except where SIL levels apply. A-weighted noise level measurements are measured for information only.
(2) Noise limit applies underway.
(3) Double hearing protection required in spaces that exceed 110 dB(A).
(4) Maximum allowable dockside noise level in Category G spaces, with diesel engines, propulsion system, and thrusters secured is 84 dB(A).

Table 073-2 Typical Airborne Noise Categories for Ship Spaces and Topside Location

Category	Compartment
A-12	Pilothouse
B	Crew Berthing
	Water closet
	Water closet and shower
D	Galley Space (Note 1)
	Passageways
	Workshop (Note 1)
F	Buoy Deck, Working Deck (Note 2)
G	Main Machinery Room

NOTES:
(1) Measured when installed equipment is in operation.
(2) Measured under conditions that are representative of the mission performed at that location.

<u>073.3</u> <u>WHOLE-BODY VIBRATION</u>

073.3.1 Steady state whole-body vibration shall be in accordance with ANSI S2.25. A lower limit of 2 Hz is acceptable.

073.3.2 Transient whole-body vibration while maneuvering shall not exceed steady state vibration by more than 30

<u>073.4</u> <u>MECHANICAL VIBRATION</u>

073.4.1 The propulsion system vibration acceptability levels for torsional, lateral, and longitudinal vibration of conventional shaft line systems shall be in accordance with the following.

073.4.1.1.1 Propulsion system vibration shall comply with ANSI S2.27 for torsional, lateral, and longitudinal vibration, as well as diesel engines and thrusters

073.4.1.1.2 The exclusion for icebreakers in ANSI S2.27 Section 1 does not apply

073.4.1.1.3 The requirements of ANSI S2.27 Annex A for hard turns to port and starboard at sustained speed shall apply

073.4.1.1.4 IACS M53, DNV-RU-SHIP 4-2-2 or ABS MVR 4-2-1/5.9 are acceptable alternative methods to ANSI S2.27 for determining diesel engine crankshaft torsional stress limits

073.5 HULL STRUCTURAL AND MAST VIBRATION

073.5.1 Vibrations outside the limits allowed in this section resulting from the excitation of a resonant frequency local to any equipment and machinery by the propeller or other exciting force shall be corrected by local stiffening of structure, installation of suitable mountings, modification of components, or other effective means.

073.5.2 Mast Vibration

073.5.2.1.1 Masts shall be free of global or local resonances that would be excited during normal operations.

073.5.2.1.2 As a minimum, the following resonance avoidance bands shall be used as mast vibration design criteria.

073.5.2.1.2.1 A frequency band defined by the propeller blade rate up to 115% maximum shaft rpm

073.5.2.1.2.2 For rigidly mounted variable speed engines, a frequency band defined by the largest diesel engine first and second order excitation frequency from 85% engine idle speed to 115% rated engine speed.

073.5.2.1.2.3 A frequency band defined by +/- 10% of the rotating frequencies of the equipment mounted on the mast

073.5.2.1.3 Mast fundamental frequency shall not fall within +/- 25% of the hull's 2-noded or 3-noded vertical natural frequencies, or the hull's fundamental torsional natural frequency.

073.5.2.1.4 Mast vibration shall not exceed a broadband RMS velocity of 0.6 in/sec

073.5.3 Structural Vibration

073.5.3.1 Vibration limits for local structures shall be in accordance with ABS Guide for Vibration of Machinery Equipment and Related Structures, Section 2, "Vibration Limits for Local Structures"

073.5.3.1.1 From 1 Hz to 5 Hz, the vibratory displacement for each peak response component (vertical, transverse, and longitudinal directions) shall not exceed a peak displacement of 0.04 in

073.5.3.1.2 From 5 Hz to 100 Hz, the vibratory velocity for each peak response component (vertical, transverse, and longitudinal directions) shall not exceed a RMS velocity of 1.2 in/s

073.5.4 Mountings, Foundations, Connections

073.5.4.1.1 Machinery and equipment shall comply with the ABS Guide for Vibration of Machinery Equipment and Related Structures for resilient mounting, foundation natural frequency separation, and flexible connections.

073.5.4.1.1.1 Design shall ensure separation of natural frequencies from excitation sources (propeller blade rate, diesel firing frequencies, hull modes) as required by the ABS Guidance Notes on Ship Vibration.

073.5.4.1.1.2 For propulsion machinery foundations, avoid resonance within ±20% of blade rate or firing frequencies

SECTION 074 WELDING AND FABRICATION

074.1 GENERAL

074.1.1 Weld Procedure Specifications (WPS), Procedure Qualification Records (PQR), and welder qualifications shall be completed and maintained in accordance with American Welding Society (AWS) D1.1 Structural Welding Code – Steel, and other material specific AWS codes and guidance, as required. This documentation shall be provided if requested by the USCG. This data shall include:

074.1.1.1 WPS applicable to the contract, along with a summary list and current revision dates;

074.1.1.2 Welder certification documentation including, at a minimum, welder name and WPS(s) certified to perform.

074.1.1.3 Welder continuity documentation that provides proof of proper certification maintenance; and

074.1.1.4 Supporting PQRs.

074.1.2 The Contractor shall perform fabrication/erection inspection and testing as necessary prior to assembly, during assembly, during welding and after welding to ensure that materials and workmanship meet the requirements of the contract.

074.1.3 Prior to welding, the welder or a qualified inspector shall ensure proper joint preparation, fit up, and cleanliness.

074.1.4 STANDARDS

074.1.5 Welding, brazing, and related procedures (joint design, joint strength calculations, edge preparation, fabrication, and records) shall be in accordance with AWS standards using approved procedures and certified welders, brazers, and solderers, except as specifically directed herein.

074.2 NON-DESTRUCTIVE INSPECTION

074.2.1 Individuals performing non-destructive inspections shall be certified to one of the following requirements:

074.2.1.1 AWS Senior Certified Welding Inspector; or

074.2.1.2 AWS Certified Welding Inspector.

074.2.2 Penetrant testing (PT) shall be performed on 100% of shell and deck plating butt groove welds and on 50% of all shell and deck plating seam groove welds.

074.3 RESTRICTIONS

074.3.1 Gas Metal-Arc Welding that applies short circuiting arc transfer technique shall not be used unless the process and application are specifically approved by the USCG.

SECTION 075 MECHANICAL FASTENERS – SECTION UNDER DEVELOPMENT

SECTION 076 RELIABILITY, MAINTAINABILITY AND AVAILABILITY (RM&A) - SECTION UNDER DEVELOPMENT

SECTION 077 SYSTEM SAFETY- SECTION UNDER DEVELOPMENT

SECTION 078 MATERIALS

078.1 GENERAL

078.1.1 All structural steel shall be in accordance with ABS Marine Vessel Rules (MVR), with guidance from CH 6 -1-4 Strengthening for Navigation in Ice, Class IC First Year Ice. Special attention is to be paid to welds in structures attached to side shell, such as transverse bulkheads, decks, frames, web frames and side shell stringers, within the ice belt, which are to be of double continuous weld.

078.1.2 Unless otherwise specified in the design artifacts, materials shall conform to the standards below.

078.2 MISCELLANEOUS REQUIREMENTS

078.2.1 Wood in any form shall not be used.

078.2.2 Materials that contain asbestos and refractory fiber materials (also termed ceramic fiber and aluminum-silica material) shall not be used.

078.2.3 For chemicals listed in NAVSEA Hazardous Material Avoidance Process, T9070-AL-DPC-020/077-2 (provided as GFI):

078.2.3.1 The prohibited chemicals listed in Appendix A, Table A-1, shall not be used.

078.2.3.2 Restricted chemicals listed in Appendix A, Table A-2, shall be documented to identify quantity, application, and location used.

078.2.3.3 Restricted chemicals listed in Appendix A, A.2 "NAVSEA LIST OF TARGETED CHEMICALS (N-LTC) APPLICATION EXCEPTION NOTES" do not need to be documented.

078.2.3.4 Valves, fittings, and other components containing copper or copper- bearing alloys shall not be used, except for items enclosed within OEM equipment or electrical wiring.

078.2.3.5 Bedding compounds such as polysulfide and polyurethane sealants may be used. Where waterproof adhesive bonds are required, a polyurethane compound shall be used.

078.2.3.6 Chlorinated polymers (e.g., PVC, CPVC) shall not be used.

078.3 ELECTROLYTICALLY DISSIMILAR METALS AND CORROSION PROTECTION

078.3.1 Direct contact of electrolytically dissimilar metals is prohibited. Electrolytic corrosion shall be prevented by insulating dissimilar materials from each other.

078.3.2 Mercury Exclusion

078.3.2.1 Equipment using mercury shall not be installed on the boat.

078.3.2.2 Equipment containing mercury, including fluorescent lights, shall not be aboard the boat during construction or testing.

SECTION 079 STABILITY AND BUOYANCY

079.1 DEFINITIONS

079.1.1 Design Waterline (DWL). The projected waterline (draft) of the ship when loaded to its designed capacity at end of service life, typically the Full Load condition.

079.1.2 Forward Perpendicular (FP). The longitudinal reference point located at the intersection of the design waterline with the stem. The FP location shall be agreed upon in preliminary design and not varied regularly.

079.1.3 Aft Perpendicular (AP). The longitudinal reference point located at the intersection of the design waterline with the stern or transom. The AP location shall be agreed upon in preliminary design and not varied regularly.

079.1.4 Virtual KG (KGv). The KGv is the load condition's vertical center of gravity (KG) plus the virtual rise in KG due to free surface effects of liquids plus the inclining experiment tolerance (see section 096).

079.2 GENERAL

079.2.1 The boat shall be designed in accordance with the Procedures Manual for Stability Analysis of U.S. Navy Small Craft, NSCSES 6660-99, except as modified by this section.

079.2.2 The boat shall be designed and constructed to withstand the following hazards:

079.2.2.1 60 knot beam winds, with a 25 degree roll back angle in the intact condition.

079.2.2.2 48 knot beam winds combined with rolling, and the required amount of topside icing.

079.2.3 Topside Icing weight and center of gravity shall be calculated as follows:

079.2.3.1 The ice density shall be 56.20 lb/ft3

079.2.3.2 3" of ice on surfaces forward of the pilothouse (including the forward face of the house).

079.2.3.3 1.5" of ice on surfaces aft of the forward face of the house.

079.2.4 Lifting heavy weights over the side with the maximum lifting moment of any crane.

079.2.4.1 Righting arm A1 Area for lifting of heavy weights shall be limited by angle of unrestricted downflooding or angle at which ½ of the available freeboard is submerged.

079.2.5 Towing another vessel astern.

079.2.5.1 Towing heeling arms shall be calculated with the heeling arm equation of 46 CFR 173.095.

079.2.5.2 Towing Heeling arms curves shall be evaluated against the righting arm curve of the vessel using the same criteria for high-speed turning.

079.2.6 Flooding of any single subdivision below main deck.

079.2.6.1 Lesser extents of asymmetric tank damage shall also be considered.

079.2.6.2 Only passive open cross-flooding arrangements without manually operated or automated valves shall be considered for implementation

079.3 DETAIL REQUIREMENTS

079.2.1 The boat, at delivery and at the end of service life, including service life weight and KG allowances, shall satisfy the intact and damage stability criteria for the operating trim range for every operating condition between and including the Full Load ATON and Minimum Operating Condition without ATON. See Section 070 for the loading condition definitions.

SECTION 088 HUMAN SYSTEMS INTEGRATION

088.1 GENERAL

088.1.1 The system design shall reflect Human Systems Integration processes that ensures the effective utilization of manpower, minimizes life cycle costs, supports effective training, and can be operated, maintained, and supported within the crew's physical capabilities and skill sets.

088.2 HUMAN FACTORS ENGINEERING

088.2.1 Human Engineering design criteria for the boat, hull mechanical and electrical (HM&E) systems, subsystems and equipment used in operations and for maintenance shall be in accordance with ASTM F1166.

088.2.2 Controls, displays and human computer interfaces shall be in accordance with ASTM F1166.

088.2.3 The boat shall be designed to accommodate 5th percentile female to 95th percentile male in accordance with the North America anthropometric data sets in ASTM F1166 for operational, maintenance, and damage control tasks while wearing personal protective equipment (PPE) and arctic/ cold weather clothing and using the required tools.

088.2.4 Controls and displays shall be placed such that the crew can perform the mission essential tasks during operating conditions requiring seated and/or standing positions, where applicable (ex. coxswain, crane operator, etc.).

088.2.5 Material handling shall be in accordance with ASTM F1166, section 16.

088.2.6 Ladders, Stair Ladders, and Stairs shall be in accordance with MIL-STD-1472 Section 5.11.8.

088.2.7 Visibility from helm locations shall be in accordance with ABYC H-1, Field of Vision from the Helm Position with the following modification:

088.2.8 A properly seated crewmember in the helmsman position shall not have a blind spot greater than one boat length from the Pilot house over the bow while the boat is at rest in calm water at the performance weight condition.

088.3 WORKFORCE AND PERSONNEL

088.3.1 The boat shall be capable of being operated by no more than 6 crew.

088.3.2 Bridge watch-station tasks shall be capable of being performed by one person for open water transit (i.e., coxswain/helm tasks) and 2 persons for ATON operations (i.e., coxswain/helm tasks and crane operator tasks).

SECTION 096 MASS PROPERTIES

096.1 GENERAL

096.1.1.1 Definitions shall be in accordance with the Society of Allied Weight Engineers Recommended Practice No. 12C modified as follows: .

096.1.1.2 “Transverse lever”: Transverse levers to starboard are positive in sign convention.

096.2 Weight and Stability Limits

096.2.1 The displacement of Full Load ATON Condition at delivery shall not exceed the Full Load Condition displacement at delivery of the Allocated Baseline Weight Estimate adjusted by the values agreed upon for Contract Modifications and changes to Government Furnished Material.

096.2.2 The KGv of Full Load Condition at delivery, including free surface corrections and a 0.20-foot margin for the stability test margin, shall not exceed the Full Load Condition KGv at delivery of the Allocated Baseline Weight Estimate adjusted by the values agreed upon for Contract Modifications and changes to Government Furnished Material.

096.3 Margins

096.3.1 Weight and KG margins shall be applied to lightship condition, per Table 096-1:

Table 096-1: Weight and KG Margins

Margin Name	Weight	KG	Notes
Design and Building (D&B)	-	-	Inclusive of Detail Design weight margins
Contract Modification (CM)	.7%	.5%	Government Controlled Margin
Government Furnished Material (GFM)	0.3%	0.2%	Government Controlled Margin
Service Life Allowance (SLA)	7.5%	5.0%	Government Controlled Weight Allowance

096.3.2 Weight margins shall only be applied at the vessel LCG, so as not to affect trim, and on centerline (TCG=0). Weight shall start at vessel vertical center of gravity and raised vertically to provide the moment required for the required KG Margin.

096.3.3 Margins shall only be depleted to compensate for an increase in the Light Ship weight or increase to the variable loads. Conversely, a decrease in the Light Ship weight or variable load shall result in a corresponding increase in the appropriate margin account.

096.3.4 Government controlled margins, CM and GFM, and the service life allowance, shall be carried through from concept design to delivery and shall only be consumed at the direction of the Contracting Officer.

SECTION 100 GENERAL REQUIREMENTS FOR HULL STRUCTURE

100.1 GENERAL

100.1.1 This section addresses additional requirements for hull structure including loading conditions, permissible stresses, definition of the hull girder structure, hull structural material, openings, fairness and smoothness.

100.1.2 The structure is defined as the hull bottom, sides, deck, and superstructure including longitudinal stiffeners, keelsons, girders, foundations, transverse frames, stringers, bulkheads, and hull appendages.

100.1.3 The hull and superstructure shall be of welded steel construction.

100.1.4 There shall be no "hard spots" or sharp-cornered notches. Cuts, doublers, and inserts in primary structure shall have radiused corners. Concentrated loads shall be dispersed by means of doublers, inserts, brackets, and gussets.

100.2 MAIN REQUIREMENTS

100.1.5 The structural shall be in accordance with the drawings.

100.1.6 The structure has been designed to ABS MVR Part 6-1-4, Structural Requirements for First Year Ice Class. The boat shall be built to Ice Class IC.

100.1.7 Where ABS MVR Part 6-1-4 is silent, the structure shall be designed and built in accordance with ABS MVR Part 3

100.1.8 Structure not specifically addressed by the ABS rules or in this specification shall be designed to withstand the loads which can be expected in service, including the effects of ship motions and environmental loads, using recognized industry standards and allowable stresses. Loads shall be combined as appropriate to represent worst case loading conditions. The structure shall withstand loads due to docking, towing, side impact loads (fendering), dry docking, heavy seas and other operational loads.

100.1.9 Interference of plating butts and seams with longitudinal stiffeners, bulkheads, decks and structural members and fastenings that attach to shell plating shall be avoided

100.1.10 Girders and deck longitudinals shall be continuous through transverse structures.

100.1.11 Butts shall be at least 3 inches, but no more than 12 inches from transverse structure.

100.1.12 Seams shall be at least 3 inches from longitudinal elements of the structure.

100.1.13 Full penetration welds shall be provided for butts and seams of the keel, bottom shell, side shell, working deck, and transom. Welded joints in the keel, keelsons, girders, engine foundations, and bottom longitudinals shall be full penetration welds at the webs as well as the flanges.

100.1.14 Flanged Plate sections are prohibited.

100.1.15 Lapped brackets and connections, except lapped collars, are prohibited in primary longitudinal and transverse structure.

100.1.16 Structural details shall be designed to maintain continuity and minimize stress concentrations. Abrupt changes in section shall be avoided.

100.1.17 Beam and column ends shall land on other structural framing members.

100.1.18 Thickness differences between connecting plates of more than 1/8 inch shall be tapered at a slope of 4 to 1.

100.1.19 Plate inserts shall have corner radii no less than one tenth the minimum dimension of the insert but need not exceed 12 inches.

100.1.20 Design Loads

100.1.20.1 Wind Loads on Exposed structure shall be 30 lbs. per square foot.

100.1.20.2 Exposed Ice loads shall be 7.5 lbs. per square foot.

100.1.20.3 An Ice Loads Monitoring System that meets ABS Guides Ice Loads Monitoring Systems (2021) shall be installed.

100.1.21 Bulkhead stiffeners shall be kept out of passageways, showers, washrooms and water closet spaces.

100.1.22 Swage bulkheads are prohibited.

100.1.23 Openings and Penetrations

100.1.23.1 Penetrations are defined as holes in a ship with an area of less than 2 ft$^{2.}$ Openings are defined as holes in a ship with an area of greater than 2 ft^{2}.

100.1.23.2 Structural penetrations shall be in accordance with NSRP 0490.

100.1.23.3 There shall be no penetrations in stiffener or girder flanges

SECTION 114 HULL APPENDAGES

__114.1 GENERAL__

114.1.1 Hull appendages are included in the USCG provided design package as shown on the drawings.

114.1.1.1 Skeg shall meet ABS MVR 3-2-2/12.

114.1.1.2 Skegs shall be watertight.

114.1.1.3 Skegs shall have ends faired into the hull.

114.1.1.4 Skeg design loading shall be consistent with other bottom loads including loads associated with ice and vessel docking.

114.1.2 Rudders shall have ice horns installed to prevent damage to rudders, stocks, seals, and bearings while navigating and backing and ramming in ice.

114.1.3 Fendering System

114.1.3.1 Fendering shall be D shaped EPDM and installed all around the main deck sheer line, except in way of the buoy deck.

114.1.3.2 Paint shall be applied to the hull prior to final fender installation.

SECTION 115 STANCHIONS

__115.1 GENERAL__

115.1.1 Stanchion design and locations shall be as shown on the drawings.

115.1.2 Stanchions shall not interfere with the removal of equipment or maintenance.

SECTION 160 SEACHEST

__160.1 GENERAL__

160.1.1 Sea chests shall be located per the drawings.

160.1.2 Plating of any seachest shall not be less than the adjacent shell plating.

160.1.3 Strainer plates shall be fitted on sea chests. The total open area of the sea chest strainer plate for each sea chest shall not be less than four (4) times the cross-sectional area of suction piping drawing from the sea chest where the screen is fitted.

160.1.4 A means to clear debris using water shall be provided for each sea chest.

160.1.5 Each sea chest shall contain a vent with a valve to prevent the formation of air or gas pockets and provide clear passage for air to escape via vent pipes to the main deck.

SECTION 167 HULL STRUCTURAL CLOSURES

167.1 GENERAL

167.1.1 The locations of doors and hatches are shown in the drawings. Doors and hatches shall be marine grade and have an equivalent water, weather, and fire rating as the structure in which they are installed.

167.2 HATCHES

167.2.1 Scuttles shall have a minimum clear opening diameter of 25 in. and shall be operable from above and below.

167.2.2 At least one manhole in each tank or void shall be in the highest practicable point in the uppermost level of the tank or void.

167.3 DOORS

167.3.1 Doors shall meet the requirements of ABS MVR Part 3.

167.3.2 Doors, structural and non-structural shall have hold open fittings and bumpers. Positive latching devices shall be provided for doors that open outward to hold them open.

SECTION 170 MASTS

170.1 GENERAL

170.1.1 Masts and vertical appendages that may be used shall be retractable by 2 crew members in 5 minutes using existing onboard equipment and shall support the following items:

170.1.1.1 Required navigation lights including Restricted Ability to Maneuver (RAM) lights.

170.1.1.2 Day shapes, including RAM shapes,

170.1.1.3 National Ensign (15 in. by 24 in.),

170.1.1.4 USCG ensign (12 in. by 18 in.),

170.1.1.5 Required navigation and C4IT systems.

170.1.2 The ensigns shall not interfere with any electronics or lights. Fittings and foundations shall be constructed to prevent tearing of ensigns or their rigging.

170.1.3 Masts and vertical appendages shall be weathertight and designed to withstand the loads from ice accretion.

170.1.4 Masts shall not be susceptible to excitation from synchronous boat motions, propeller motion, or wind induced excitation.

SECTION 180 FOUNDATIONS

180.1 GENERAL

180.1.1 The structural design package provided will include foundations for major equipment such as the diesel engines and the crane. See design package calculations and drawings for details. This section addresses other foundations required.

180.1.2 Strength and rigidity of foundations shall be suitable to withstand design loads and distribute such loads into the structure, while maintaining accurate alignment of the equipment. The design shall include the following:

180.1.2.1 Weights and dimensions of machinery and equipment, including liquids at operating levels.

180.1.2.2 Load resulting from boat operating motions, attitudes and ice shock loads.

180.1.2.3 Vibration of machinery including main engine, drive train, and propeller .

180.1.2.4 Loads imposed by the operation of machinery and equipment.

180.1.2.5 In exposed locations, loading caused by ice accretion and wind shall be included.

180.1.2.6 Equipment manufacturer's requirements and recommendations.

180.1.2.7 Foundations shall contain no pockets which can retain liquids.

180.1.2.8 Foundations shall be arranged with sufficient clearance for servicing equipment without dismantling other components.

SECTION 200 GENERAL REQUIREMENTS FOR PROPULSION PLANT

200.1 GENERAL

200.1.1 Rated Power Definition. The Rated Power is the maximum power output at which the engine is designed to run continuously at its rated speed (rpm) between center section overhaul intervals specified herein. The Rated Power shall be adjusted for Ambient Reference Conditions using the methods in ISO Standard 3046-1.

200.1.2 The Contractor shall obtain and provide certification letters from the OEM, verifying that the equipment is installed on the boat in compliance with OEM requirements.

200.1.3 The propulsion system's diesel engines shall be identical to each other, including the direction of rotation, except for placement of controls and external components

200.1.4 The starboard propeller shall have right-hand rotation and the port propeller left-hand rotation.

200.1.5 Propulsion system components shall be current, in-production models that have been operated by the owner of the vessel after acceptance in commercial marine, naval service, or U.S. Government marine service for at least two years with at least 500 operating hours at a rated power greater or equal to that being offered.

200.1.6 The propulsion system shall be in accordance with ABS MVR Part 4 and Part 6-1-4 Ice Class 1C.

200.2 PROPULSION SYSTEM

200.2.1 The arrangement of the propulsion system and components are as shown on the drawings and the Critical Equipment List. The propulsion system consists of two diesel engines linked to reverse/reduction gears (or reduction gears). The reverse/reduction gears (or reduction gears) shall drive propellers.

200.2.2 Propulsion systems' and genset(s)' equipment and components shall be designed, tested, and installed in accordance with OEM's requirements and recommendations. Equipment and components shall comply with uses designated by the manufacturers' warranty so as to not void the manufacturers' warranties.

200.3 OPERATION

200.3.1 The propulsion plant shall be configured so that a failure of any single main diesel engine will not cause the complete loss of propulsion power.

200.3.2 The propulsion plant shall be arranged to ensure continued safe operation of either main diesel engine in the event that a single service pump is unavailable. The systems shall provide adequate cross-connection, redundancy, or bypass capability such that loss of one pump or one pump train does not prevent operation of either engine at rated continuous power.

200.4 ARRANGEMENT AND INSTALLATION

200.4.1 A lube oil sampling port shall be installed in each propulsion plant component lube oil system with an operating volume greater than five gallons.

200.4.2 Upon final installation of diesel engine equipment foundation bolts, a suitable torque seal indicator shall be applied to each foundation bolt and nut.

200.5 ENDURANCE FUEL LOAD

200.5.1 Endurance fuel load shall be determined in accordance with NAVSEA DPC 200-1 for the "operational presence time" condition and "economic transit" for the range and mission profile specified in 070. Tankage shall be sized with the boat at Full Load EOSL Displacement.

200.5.2 A 10% burnable fuel reserve margin shall be added to the calculated burnable fuel load.

200.5.3 The tailpipe allowance shall be 5%.

200.6 VIBRATION

200.6.1 The propulsion system and genset(s) shall operate without restriction and shall have no barred speed range(s).

SECTION 202 MACHINERY CENTRALIZED CONTROL

202.1 GENERAL

202.1.1 The MCMS will provide main propulsion (monitoring only), auxiliary, ships service and emergency electrical distribution system control and monitoring as installed.

202.1.2 The primary location for MCMS display of detailed equipment status and alarm information shall be in the pilothouse

202.1.3 The MCMS operates on the basis that the propulsion and auxiliary machinery spaces are unmanned.

SECTION 233 DIESEL ENGINES

233.1 GENERAL

233.1.1 Two main diesel engines are specified to be the CAT C18B, Medium Duty Rating, or equal. The two main engines selected in the provided design package and arrangements for the HSC-Light shall be installed per OEM requirements.

233.1.2 Two diesel generators are specified to be CAT 4.4 or equal. The provided design package and arrangements for the HSC-L shall be installed per OEM requirements. See Sect. 311 for additional generator requirements.

233.1.3 The diesel engines shall be furnished with accessories necessary to meet the requirements of this specification.

233.1.4 Power, torque and RPM of the engine shall not exceed those specified by the OEM for the intended service.

233.2 FAST LUBE OIL CHANGE SYSTEM (FLOCS)

233.2.1 The boat shall come equipped with Fast Lube Oil Change System (FLOCS) connections for removal of lube oil from the propulsion system and gensets.

233.2.2 The FLOCS system shall consist of required piping, hoses, and fittings configured for compatibility with FLOCS 15 Oil Evacuation Unit, Aeroquip Part No. FF9315-01, and NSN 4930-01-191-6166.

233.2.3 The FLOCS connections shall consist of a flexible suction hose connected to the lube oil sump drain connection(s) and a 3/4-inch male quick disconnect, positive shut-off coupling on the other end of the hose.

233.2.4 The FLOCS suction hoses shall be of sufficient length for the quick disconnect, positive shut-off coupling to extend above the engine compartment cover, and to an area accessible for its use.

233.3 DETAIL REQUIREMENTS

233.3.1 Propulsion diesel engine and generator set diesel engine mounting arrangements shall provide the ability to shim the engine at each support foot or provide height adjustable resilient mounts.

SECTION 241 PROPULSION REDUCTION GEARS

241.1 GENERAL

241.1.1 Each reduction gear shall be capable of transmitting maximum rated engine power and torque in the ahead and astern direction when operating in any condition, including rapid changes in power, RPM, and thrust, in every operating environment. The reduction gears shall be as shown on the drawings.

241.1.2 Each propeller shaft shall be electrically isolated from the rest of the propulsion system at the gear output flange. A resistance of 10,000 ohms or greater shall be measured across the flange joint and coupling to the shaft.

241.1.3 The gearboxes shall be configured with a trolling clutch, or equivalent feature, to permit low-speed operations, below 5 knots. Repeated cycling of the clutches in and out of gear is not permitted to obtain low speed operation. An indicator light should be installed at the control station to indicate use.

241.2 ASSEMBLY AND INSTALLATION

241.2.1 The gears shall be assembled, installed, cleaned, tested, preserved, adjustment of gears and their accessories, and control and indicating equipment installed per OEM requirements.

SECTION 242 MAIN PROPULSION COUPLINGS AND CLUTCHES

242.1 GENERAL

242.1.1 The clutch shall be capable of frequent, consecutive ahead and astern engagements during icebreaking maneuvers, at idle or low maneuvering speed, for the duration of normal operations without exceeding its allowable operating temperature or degrading clutch performance. The clutch manufacturer shall document the duty cycle and any limitations associated with such operation.

242.2 DESIGN REQUIREMENTS

242.2.1 Flexible couplings shall be used to connect the main engine flywheel to a reduction gear input shaft and designed to accommodate alignment tolerances and relative motions of the propulsion units. The coupling may be required to be torsionally resilient as determined by vibration analysis. The coupling selected shall be approved by the manufacturers of the engine and the driven machinery.

242.2.2 Between the MDE output shaft and the first reduction gear mesh, the coupling arrangement shall be designed to provide damping of transient torsional loads and to protect the drivetrain from rapid torque reversals associated with icebreaking operations.

242.2.3 A fluid coupling or other slip-capable coupling may be used to permit controlled slip under transient overload conditions without damage to the reduction gear or engine should the propeller become suddenly impeded or arrested by ice.

SECTION 243 MAIN PROPULSION SHAFT

243.1 GENERAL

243.1.1 Shafting arrangements shall be as shown on the drawings.

243.1.2 Straightness and surface finish(es) shall be in accordance with the requirements of ABYC P-6, *Propeller Shafting Systems*, or the requirements of interfacing component manufacturers (shaft grounding systems, bearings, watertight seals), whichever has the more restrictive requirements.

243.1.3 Shafts fabricated from stainless steel or corrosion-resistant alloys with a Pitting Resistance Equivalent Number (PREN) less than 28 shall not be left bare in seawater service. Such shafts shall be protected by a continuous reinforced fiberglass wrapping in accordance with the requirements of ABS MVR 4-3-2.

243.1.4 Where a propulsion shaft contains more than one geometric discontinuity within a region less than four shaft diameters, including but not limited to:

243.1.4.1 Shoulders or diameter steps;

243.1.4.2 Keyways, spline roots, or axial slots;

243.1.4.3 Oil holes or radial drilled passageways;

243.1.4.4 Undercuts, shrink-fit seats, or circlip grooves.

243.1.4.5 Stress relief grooves associated with couplings.

The combined stress fields shall be evaluated with an FEA under maximum loading conditions to determine local peak stresses and fatigue life.

243.2 INSTALLATION REQUIREMENTS

243.2.1 The shafting system shall be aligned to within the limits set forth by the component manufacturers. Alignment is defined as the positioning of the propulsion engine, intermediate shafting, couplings, joints, and bearing in each system to satisfy requirements of each component in the system.

243.2.1.1 During initial alignment, shafting system bolt stops in each of the components shall be set such that the available clockwise and counterclockwise adjustments are equal.

243.2.2 Intermediate shaft support bearings shall be installed when required by mission parameters. Bearings shall be oil lubricated, journal babbitted bearings or split roller bearings.

243.2.2.1 Journal bearing loading shall not exceed 75 psi for disc lubricated bearings or 150 psi for pressure lubricated bearings, computed by dividing the static bearing load in pounds by the product of the journal length and the outer diameter. The bearing journal length to diameter ratio shall not be less than one.

243.2.2.2 Split roller bearings shall be designed for a minimum L-10 life of 60,000 hours, in accordance with ISO 281:2007, at rated shaft rpm and the worst case loading condition.

SECTION 244 MAIN PROPULSION SHAFT BEARINGS AND SEALS

244.1 GENERAL

244.1.1 The number and location or bearings for the main propulsion stern tube, line shaft, thrust bearings and propellers shall be as shown on the drawings.

244.1.2 Bearings and seals shall be;

244.1.2.1 Water lubricated type;

244.1.2.2 Elastomer lined;

244.1.2.3 Non-metallic outer shell type; and

244.1.2.4 Type approved by a USCG-recognized marine classification society.

244.1.3 Propeller shaft main seals shall be Chesterton 442 mechanical split seals with Inflatable Safety Seals (ISS) or equal.

SECTION 245 PROPELLERS – SECTION UNDER DEVELOPMENT

245.1 GENERAL

245.1.1 The Propellers shall be provided under one of the following procurement approaches, to be determined by USCG prior to contract award:

245.1.1.1 The propeller will be specified as a Brand “X”, Model “Y”, including right-hand and left-hand variants, as defined in this specification.

245.1.1.2 The propellers design will be furnished as Government Furnished Information (GFI) in accordance with a USCG-developed Procurement Package Data (PPD) for propeller acquisition.

245.1.1.3 The Contractor shall procure right-hand and left-hand propellers in accordance with the Government-provided Procurement Package Data (PPD).

245.1.2 No alternatives are allowed unless otherwise approved by USCG

245.2 DESIGN REQUIREMENTS

245.2.1 The propeller shall be a fixed pitch monoblock design.

245.2.2 Propeller shaft shall meet the following:

245.2.2.1 Voids in the hub, hub counterbore, and fairwater shall be filled with corrosion preventive compound Termalene EP Multipurpose Extreme Grease 2 (Bel-Ray Product Code 72440), or equal.

245.2.2.2 The propeller and propeller shaft shall be sealed from water at the forward end of the propeller hub by a rubber packing ring and gland. The propeller hub shall be counter-bored to a depth sufficient to provide a seal between the shaft sleeve and propeller hub.

245.2.2.3 Positive locking devices shall be provided for the propeller connection.

SECTION 251 COMBUSTION AIR SYSTEM

251.1 GENERAL

251.1.1 The combustion air ventilation shall:

251.1.1.1 Meet the diesel engines' OEM requirements and recommendations;

251.1.1.2 Meet ABYC H-32 Ventilation of Boats Using Diesel Fuel requirements;

251.1.1.3 Supply combustion air to the diesel engines from the space in which the diesel engine is located;

251.1.1.4 Provide a water and air separation mechanism.

251.1.2 The compartments with engine(s) shall:

251.1.2.1 Maintain air temperature measured at the engine air intake to be not more than 30°F above ambient when engines are at full rated load;

251.1.2.2 The compartment ventilation and distribution system may be combined with the combustion air ventilation system to achieve the above.

251.1.2.2.1 Air supply to engine compartments shall not be from accommodation spaces.

251.1.3 The combustion air ventilation system shall be fitted with fire dampers that are controlled outside the engine space.

SECTION 252 PROPULSION CONTROL SYSTEMS

252.1 GENERAL

252.1.1 A propulsion/steering control and monitoring system that has provisions for starting, stopping, monitoring, and controlling the propulsion and steering system shall be provided.

252.1.2 The system shall be in accordance with ABYC P-28, Electric/Electronic Control Systems for Propulsion and Steering and meet the requirements in 46 CFR Parts 61 and 62. The built-in redundancy shall be either electric/electronic or hydraulic.

252.1.3 The propulsion control shall:

252.1.3.1 Use independent lever controls per shaft; and

252.1.3.2 Provide the ability to increase no-load (declutched) engine speed to facilitate engine warm-up;

252.1.3.3 Have two operator-selectable idle speeds:

252.1.3.3.1 Low-idle function to temporarily reduce engine speed to allow for a reduced boat speed of 5 knots or less at low idle for close quarters maneuvering and docking; and

252.1.3.3.2 High-idle function to enable mooring in fast currents and station keeping during buoy operations.

252.1.4 The primary steering control shall:

252.1.4.1 Be a steering wheel on the primary helm and

252.1.4.1.1 Have a diameter between 14 - 16 inches diameter

252.1.4.1.2 Have a minimum spoke diameter (or thickness) of 0.75 inches. The minimum spoke diameter (or thickness) may be achieved through building up the spoke diameter (or thickness);

252.1.4.1.3 Have a maximum rim diameter of 1.125 inches;

252.1.4.1.4 Have a minimum of four spokes;

252.1.4.1.5 Be non-magnetic;

252.1.4.1.6 Take no more than 1 pound of manual force to turn the steering wheel; and

252.1.4.1.7 Take no more than 2.5 revolutions of the wheel to swing the rudders from the starboard rudder stop to the port rudder stop. The number of turns shall be the same to swing the rudders from stop to stop in either direction.

252.1.5 The secondary helm shall be a tiller joystick, and

252.1.5.1.1 Have a motion/travel range that does not exceed 45 degrees from center in either direction.

252.1.5.1.2 Have haptic means through a detent to identify the center position.

252.1.5.1.3 Be spring loaded to return to center after release.

252.1.5.1.4 Have a handgrip length between 4 and 7 inches; and

252.1.5.1.5 Have a handgrip diameter between 1 and 2 inches.

252.1.6 The propulsion control lever and steering tiller joystick shall have a palm rest that can be used as a handhold or bracing point.

252.1.7 Local engine control shall be provided in the engine room for use during maintenance.

252.2 PROPULSION AND STEERING CONTROL RESPONSES

252.2.1 While HSC-L is operating, the throttle response times (delays) shall meet the following:

252.2.1.1 Be less than or equal to 5 seconds from maximum ahead shaft RPM to clutched in shaft RPM astern without stalling the engines.

252.2.1.2 Be less than or equal to 1 second shifting from neutral to ahead or astern shaft RPM.

252.2.2 For systems that are being controlled or monitored, there shall be no discernible time lag between a change in the respective systems' condition and their indication on display.

252.3 EMERGENCY SHUTDOWNS

252.3.1 Each propulsion engine shall be provided with an emergency shutdown device that can be activated to immediately stop the flow of fuel to the engine injectors and secure the ignition power to the engine control module (ECM) at each operator station.

252.3.2 This emergency shutdown shall remain functional in the event of propulsion control system failure.

252.3.3 The devices shall be of a type that requires manual resetting before the engine can be restarted.

252.3.4 The devices shall be guarded to minimize accidental activation.

252.4 MONITORING AND ALARM SYSTEM

252.4.1 The propulsion/steering control and monitoring system panels shall provide a means to monitor, alarm, and display engine and reduction gear operation. An OEM engine panel meeting these requirements may be provided in lieu of separate gauges. The system shall monitor, alarm, and display:

252.4.1.1 Engine RPM;

252.4.1.2 Fuel Flow (total and per engine).

252.4.1.3 Fuel Consumption (total and per engine).

252.4.1.4 Jacket Water Temperature.

252.4.1.5 High Jacket Water Temperature Visual Indicator with Alarm Tone.

252.4.1.6 Engine Lube Oil Pressure.

252.4.1.7 Engine Low Lube Oil Pressure Visual Indicator with Alarm Tone.

252.4.1.8 Voltmeter.

252.4.1.9 Low Voltage/Alternator Output Visual Indicator with Alarm Tone.

252.4.1.10 Engine Hour Meter.

252.4.1.11 Raw Water Flow (Note: An indicator shall be provided to indicate loss of exhaust system cooling water supply.

252.4.1.12 Reduction Gear Lube Oil Pressure.

252.4.1.13 Reduction Gear Low Lube Oil Pressure Visual Indicator with Alarm Tone.

252.4.1.14 Reduction Gear Lube Oil Temperature.

252.4.1.15 Reduction Gear High Lube Oil Temperature Visual Indicator with Alarm Tone; and

252.4.1.16 Propulsion System/Shaft Overspeed Alarm.

252.4.2 In the adjacent space with access to a compartment with diesel engines, a propulsion/steering control and monitoring system panel with emergency stop feature for each diesel engine shall be provided for local monitoring.

252.4.3 The propulsion/steering control and monitoring system displays shall be separate from the navigation multi-function displays.

SECTION 256 MACHINERY COOLING SYSTEMS

256.1 DESCRIPTION

256.1.1 The main engines and gensets will have closed loop cooling using keel coolers. The auxiliary equipment and reduction gears will be cooled by a freshwater loop. The freshwater loop(s) will consist of a titanium plate heat exchanger cooled by seawater, head tank, pump, main and branch piping. Temperature in the loop will be controlled by a 3-way modulating valve on the freshwater side of the heat exchanger. Each service will contain a temperature control valve (either on the process or freshwater side), isolation valves, flowmeter, and a constant flow fitting. Freshwater cooling services will include, but are not limited to, reduction gears, chillers, deck equipment HPU, and steering gear HPU. The seawater cooling system will serve the freshwater loop heat exchanger(s), shaft seals, and stern tubes. It will take water from sea chests located at the forward end of the keel cooler pockets and deliver it to each service using orifice plates to accomplish balancing. Seawater discharged from the plate heat exchangers will pass through a three-way valve that regulates the sea chest suction temperature by modulating the amount recirculated to the sea chest or sent overboard.

256.2 HEAT EXCHANGERS

256.2.1 General

256.2.1.1 Independent seawater cooled plate type heat exchangers or keel coolers shall be provided to cool each freshwater loop.

256.2.1.2 Seawater cooled plate heat exchangers nozzles that are in contact with seawater shall be titanium. The seawater inlet and outlet piping serving the heat exchanger shall be galvanically isolated from the titanium plates with non-metallic hoses or non-conductive rubber flex couplings at each fluid connection. The plate heat exchanger frame shall be electrically isolated from the foundation.

256.2.1.3 Plate type heat exchangers for closed-loop freshwater service shall be constructed using stainless steel plates.

256.2.2 Keel Cooler Design, Construction, and Testing

256.2.2.1 Keel cooler material shall be of 90/10 copper nickel and shall be galvanically isolated from the hull. Keel coolers shall be provided with drain plugs to permit draining the coolers when hauled and changing coolant.

256.2.2.2 Keel coolers shall be installed on the shell bottom per USCG design guidance recessed into hull pockets for protection. Hull pockets shall have marine growth prevention systems and antifouling protection.

256.2.2.3 Keel cooler pockets shall be faired at least 45 degrees in the direction of water flow when moving ahead and shall be faired at both ends.

256.2.2.4 Guards and fairing plates shall be fitted flush to the hull to fend off ice, debris, and buoy chain from the coolers. Guards shall be removable to allow maintenance and cleaning of the coolers by a diver.

256.2.2.5 Keel cooler pockets shall be fit with vent pipes, allowing collected air to escape. Vent piping connection shall be at the highest point of the pocket and shall continue to the weather deck to terminate in a gooseneck.

256.3 FRESHWATER COOLING REQUIREMENTS

256.3.1 General Requirements of Freshwater Cooling Systems

256.3.1.1 Each heat exchanger shall accomplish temperature regulation via a 3-way temperature regulating valve modulation. Regulating valves shall have manual overrides or a bypass. The temperature setpoint shall be adjustable.

256.3.1.2 Temperature control for the freshwater loop shall be accomplished on the freshwater side of the seawater heat exchanger.

256.3.1.3 Where multiple services are served by a common pump, each service shall meter the design flowrate using an automatic balancing valve and shall be located in the combined return piping downstream of the 3-way control valve. Design flow rates shall account for any percent error inherent to the automatic balancing valve.

256.3.1.4 The system shall be provided with permanently installed flow meters for each supply to individual equipment. Each meter shall have a measurement capacity of 30 to 120 percent of the design flow rate and shall read in gal/min. Meter assemblies shall be provided with flanged or union ends. Flow meters shall be either an electromagnetic or ultrasonic type.

256.3.1.5 High points in each cooling loop shall be fitted with automatic vent valves or a deaerator system to purge air from the system. System low point drains shall be installed throughout the cutter as necessary to facilitate draining the system down for layup.

256.3.2 Head Tanks

256.3.2.1 For each closed loop, a head tank shall be provided with level controls and indication.

256.3.2.2 Head tanks shall be sized in accordance with ASHRAE Handbook HVAC Systems and Equipment to accommodate the system fluid expansion from minimum to maximum fluid temperatures.

256.3.2.3 The inlet and outlet connections of each head tank shall be connected to the system via locked open cutout valves.

256.3.2.4 Head tank(s) shall have a sight glass, vent, overflow, drain, and a funnel for fill and addition of chemical treatment. The tank location shall be accessible for routine inspection.

256.3.2.5 Head tanks shall be piped to the suction side of the circulation pumps.

256.3.2.6 Head tanks shall be fitted with level controls,, sight glass, vent, overflow, and drain in accordance with Section 505.

256.3.2.7 The filling system shall be a pump drawing suction from the fill tank. The cut-in and cut-out commands shall be via level controls allowing 2-minute pump run time, and the required volume shall be added to the required acceptance volume.

256.3.2.8 The thermal fluid shall be pre-mixed in a tank that contains not less than 200% of the system volume. The tank need not be dedicated to the thermal fluid system and may be combined with the diesel engine jacket water replenishment tank.

256.3.2.9 The injection point of the filling system to each system shall be to the expansion tank or suction side of the pump.

256.3.2.10 The fill pump shall be sized to have a max fill time of 1 hour

256.3.3 Freshwater Cooling Pumps

256.3.3.1 Each cooling pump shall be provided with a pressure gage, transducer, and cutout valve in the associated suction and discharge.

256.3.3.2 A pump recirculating line sized for at least five percent of associated pump rated flow capacity or the minimum required by the pump manufacturer for seal cooling, whichever is greater, shall be installed for each pump.

256.3.3.3 The total capacity of the freshwater machinery cooling pumps shall be at least the highest overall system demand for that loop based on all shipboard operating conditions.

256.3.3.4 Engine driven pumps are allowable if integral to the OEM design. Each engine freshwater cooling loop shall be provided with an independent circulation pump, heat exchanger, and head tank.

256.3.3.5 Where standalone (non-gear driven) pumps are used for engine cooling, the pump shall automatically start with the engine.

256.3.3.6 Each cooling loop pump shall be provided with a pressure or compound pressure gage, transducer, and cutout valve in the associated suction and discharge.

256.3.3.7 Additives for corrosion and freeze protection shall be added for protection against freezing at temperatures down to ambient rated temperatures in accordance with Section 070, and in alignment with manufacturers recommendations. The additives shall be specifically accounted for in system design, as reduced fluid heat transfer coefficients relative to water will reduce system cooling capacity.

256.3.4 Seawater Supply to Freshwater Loop

256.3.4.1 Seawater cooled plate type heat exchangers shall be piped to allow the heat exchanger to be backflushed. This will allow the reversal of seawater flow through the heat exchanger and routine backflushing without removal from service.

256.3.4.2 Butterfly valves shall not be used in seawater systems.

256.3.4.3 System Flow Balancing & Distribution

256.3.4.4 Automatic balancing valves shall not be used in seawater systems.

256.4 SEAWATER COOLING SYSTEM

256.4.1 Arrangement

256.4.1.1 Strainers shall be installed in supply piping to equipment, instruments, and flow control components to protect the smallest flow passage from clogging. Components shall be sized and selected so that strainers are unnecessary whenever possible.

256.4.1.2 A duplex basket strainer or two simplex strainers piped in parallel and each having their own inlet and outlet isolation valves shall be installed on the suction side of each seawater pump.

256.4.1.3 Strainer basket perforations shall be smaller than the largest sphere which can pass through the pumps without damaging the pump. The total clear area of basket perforations shall be not less than three times the area of the strainer discharge connection. Strainer perforations shall be no smaller than one half the size of the smallest flow path served.

256.4.1.4 Emergency seawater cooling services shall be supplied from the seawater washdown system. The connection from the pump shall be provided with a root valve in its associated takeoff, a duplex or self-cleaning strainer, and a pressure reducing station to include pressure relief valve. Alternate configurations meeting the intent and performance of the specification may be submitted for USCG review.

256.4.1.5 System low point drains and high point vents shall be installed throughout the cutter as necessary to facilitate draining the system down for layup.

256.4.1.6 Seawater cooled plate type heat exchangers shall be piped to allow the heat exchanger to be backflushed. This will allow the reversal of seawater flow through the heat exchanger and routine backflushing without removal from service.

256.4.1.7 Butterfly valves shall not be used in seawater systems.

256.4.1.8 Differential pressure gauges shall be installed to measure pressure drop across each strainer.

256.4.2 System Flow Balancing & Distribution

256.4.2.1 Automatic balancing valves shall not be used in seawater systems unless otherwise approved by USCG.

256.4.2.2 System balancing in a distributive seawater system shall be accomplished by using fixed orifice plates for each branch or flow path in the system. See Section 505 for orifice plate sizing criteria, calculation of cavitation indices, and requirements for multi-stage orifice plate arrangements. Orifice plates shall be fitted with a bypass globe valve, and a differential pressure gauge. Orifice plates shall be installed in a straight run of pipe at least 10 pipe diameters downstream, and 5 pipe diameters upstream, of any fitting or component.

256.4.2.3 Where equipment requires temperature regulation, the temperature regulation shall be accomplished on the process side. Where this is not possible due to the media being cooled, the system shall use modulating three-way valve control at each heat exchanger service in the mixing or diverting configuration. The bypass at each control valve shall contain a fixed orifice plate equivalent to the pressure drop of the respective heat exchanger. Modulating valves shall actuate using a 4-20 mA control signal, and shall be capable of fully bypassing flow around the equipment when offline unless otherwise approved by USCG.

256.4.2.4 Except for sea chest temperature regulating valves, where direct acting 3-way temperature regulating valves are installed for seawater systems, modulation control via refrigerant filled capillary tube is prohibited when installed in a compartment with a maximum design temperature above 95°F.

256.4.2.5 Regulating valves shall have manual overrides or a bypass. The temperature setpoint shall be adjustable.

256.4.2.6 Each installed pump shall be fit with a minimum flow bypass set to flow 5% of the pump rating, or that volume which is required by the manufacturer to keep it in recommended operating range even when deadheaded, whichever is greater.

256.4.3 Piping Requirements, Design Fluid Velocity

256.4.3.1 The minimum pipe velocity in seawater piping shall be 3 ft/s, however 0 ft/s is permitted when equipment is shut down. Fluid velocities designed in the 0 to 3 ft/s range are not permissible when equipment and branches are in use.

256.4.3.2 Common inlet and outlet piping to and from pumps and various pieces of equipment is permitted if all possible combinations of online and shutdown equipment being served in the common inlet/outlet piping will not result in fluid velocities in the 0 to 3 ft/s range. Where this is not possible, independent pumps and piping are required.

256.4.3.3 Sea chest recirculation and return piping will have fluid velocities dependent on temperature regulating valve position. Only in this piping are fluid velocities permitted to fall within the 0 to 3 ft/s range.

256.4.4 System Arrangement

256.4.4.1 The diagrammatic arrangement (Figure 256-1 and 256-2) shall be used as guidance for auxiliary services. Alternate configurations meeting the intent and performance of the specification may be submitted for USCG review.

Figure 256-1: Machinery Cooling System (Part 1)

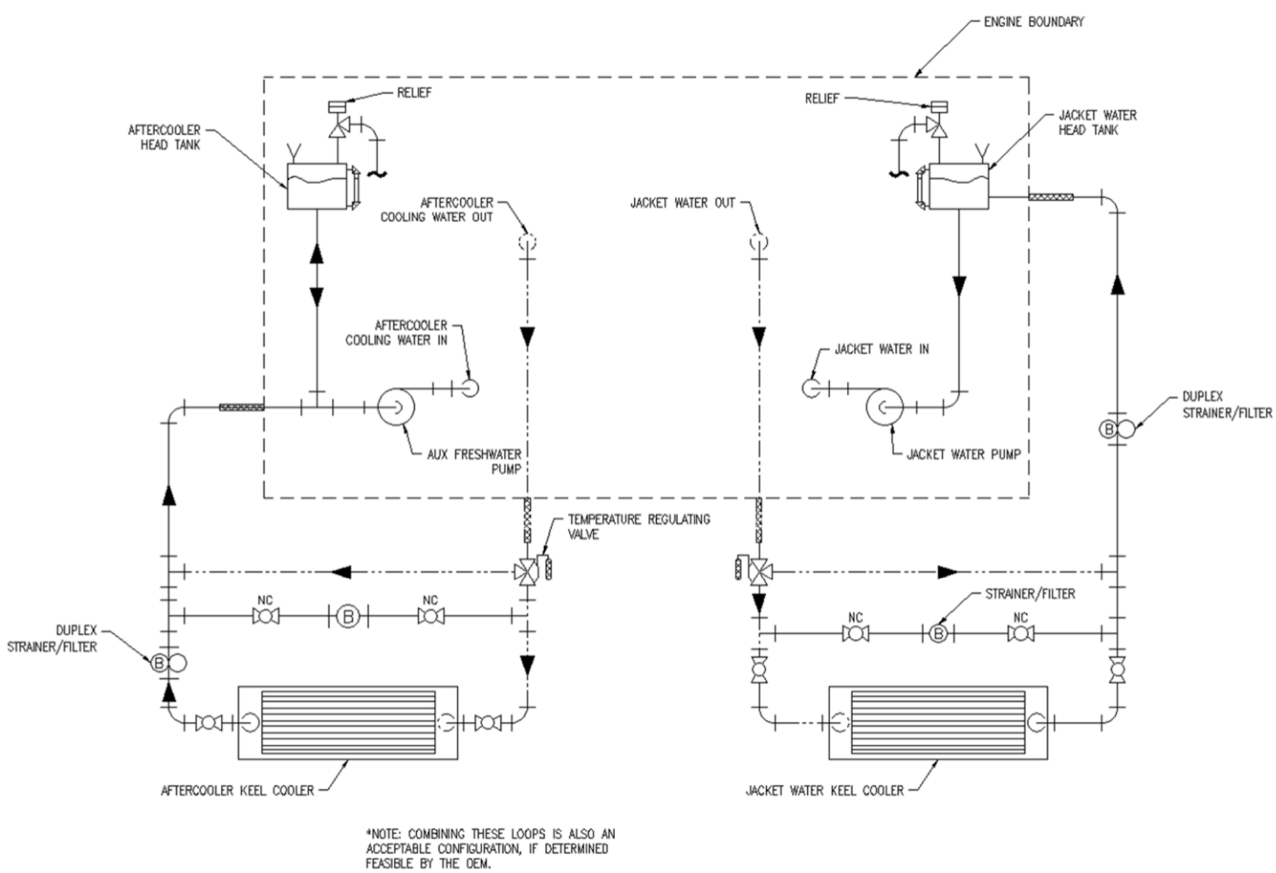
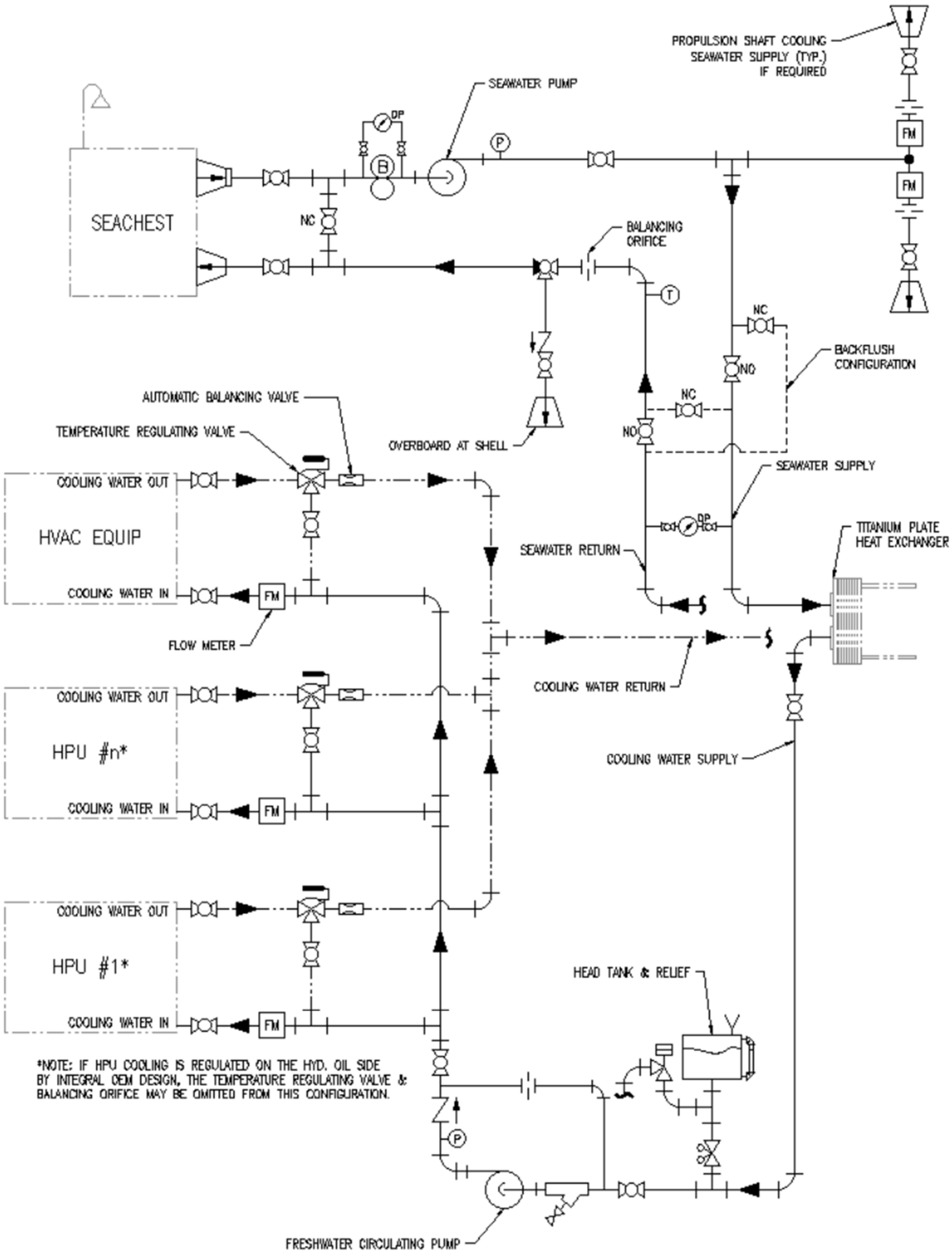


Figure 256-2: Machinery Cooling System (Part 2)

DISTRIBUTION STATEMENT A - Distribution Unlimited



DISTRIBUTION STATEMENT A - Distribution Unlimited

256.5 SEA CHESTS AND OVERBOARDS

256.5.1 General

256.5.1.1 Arrangement shall be in accordance with the drawings.

256.5.1.2 Overboard discharges shall be located at least one foot below the light ship design waterline.

256.5.1.3 The design of the sea chest intakes shall prevent blockage due to freezing or ice.

256.5.1.3.1 Sea chests shall be located clear of seams and butts in shell and inner bottom plating, and clear of welded joints wherever practicable.

256.5.1.3.2 Sea chests shall be fitted at the forward ends of keel cooler recessions, where applicable, to mitigate ice ingestion.

256.5.1.4 Vent Pipes shall be installed

256.5.1.4.1 Sea chest vent pipes shall terminate on the weather deck.

256.5.1.4.2 Means shall be provided to prevent ice from entering or plugging the vent pipes under all operating conditions.

256.5.2 Sea Chest Strainer Plate & Openings

256.5.2.1 Each sea chest shall be fitted with a removable perforated strainer plate. No parts shall project beyond the shell. Openings shall be fitted with appurtenances such as hinges and bolts to allow underwater hull surveys. An anti-seizing compound shall be used to ensure ease of removal. Tooth type lock washers shall be installed to provide positive electrical contact between the hull and the strainer.

256.5.2.2 The diameter of holes or width of slots shall be no more than 1 inch. Slots shall be oriented with the length in the fore and aft direction. The total open area of the sea chest plates for each sea chest shall not be less than four (4) times the cross-sectional area of all suction piping drawing from the sea chest where the plate is fitted

256.5.2.3 Strainer plates and bars shall be secured by CRES or bronze fasteners and self-locking nuts with wire.

256.5.2.4 Prior to installation, sea chests and overboard discharge fittings shall be coated with the underwater paint system. Removable strainer plates and bars shall be protected with the full underwater hull coating system.

256.5.2.5 The suction piping shall be located near the bottom of the sea chest. The suction pipe shall be configured to maximize flow and minimize clogging by ice. The total open area of the perforations along the length of suction pipe shall be at least equal to the cross-sectional flow area of the suction piping itself.

256.5.2.6 A baffle shall be provided in each sea chest not fitted to a keel cooler recess. The baffle shall be arranged to mitigate the recirculated seawater from the coolers entering the sea chest from escaping through the shell openings.

256.5.3 Blowdown/Ice Purge

256.5.3.1 Each sea chest shall be served by a secondary seawater source to clear blockages. The piping shall be arranged to dislodge and clear ice away from the vent, grating holes, or slots.

256.5.3.2 If the secondary water source is capable of producing more than 40 psi, a relief valve shall be installed at the blowdown branch to ensure the sea chest is not subjected to overpressure.

256.5.3.2.1 The relief valve discharge piping shall discharge overboard via a swing check valve and normally open cutout valve.

256.5.3.2.2 A pressure gauge shall be installed on each sea chest.

256.5.3.2.3 A label plate shall be installed near the hose valve, inscribed as follows, "CAUTION – DO NOT PERMIT PRESSURE TO EXCEED 35 PSI WHEN FLUSHING SEA CHEST."

256.5.4 Recirculation

256.5.4.1 Seawater systems shall be configured to recirculate water discharged from the coolers to the sea chest to limit icing of the sea chest, strainers, and suction piping. This shall be accomplished by a 3-way diverting valve capable of sending 0-100% of the water to overboard or back to the sea chest. Temperature regulation shall be controlled by a thermocouple or sensing bulb in the sea chest suction piping upstream of the duplex strainer. The recirculation waster piece in the sea chest shall be configured to fully heat the sea chest and efficiently return all of the recirculated water back to the sea chest suction piping with minimal loss through the sea chest grating.

256.6 SHAFT SEALS AND STERN TUBES

256.6.1 General

256.6.1.1 Details of shaft seals and stern tube arrangements shall be as shown on the drawings.

256.6.2 Seawater Service

256.6.2.1 Shaft seals and stern tube cooling water shall be supplied from a continuously running seawater system, where required.

256.6.2.2 Seawater to shaft seals and stern tubes shall include self-cleaning strainers with differential pressure gauge, high/low flow alarm switch, pressure switch, balancing orifice plate(s), cutout valve, and valved drain, in that order. A valved bypass shall be provided around the constant flow control device.

256.6.2.3 Each shaft seal and stern tube assembly shall have its own set of components dedicated to that shaft.

256.6.2.4 The seawater supplied to the seal housing shall meet the filtration level recommended by the equipment manufacturer.

256.6.2.5 The seawater supply flow rate to the shaft seal and stern tube assembly shall be measured with an electromagnetic flow meter, ultrasonic flow meter, or differential pressure gauge across an orifice meter to monitor flow and provide alarm to the pilothouse.

256.7 STRAINERS

256.7.1 General

256.7.1.1 Strainers shall be installed in supply piping to equipment, instruments, and flow control components to protect the smallest flow passage from clogging.

256.7.1.2 The total clear area of basket perforations shall be not less than three times the area of the strainer discharge connection.

256.7.1.3 Strainer perforations shall be no smaller than one half the size of the smallest flow path served.

256.7.1.4 Y-type strainers, where specified, shall be provided with valved drain lines.

256.7.1.5 All strainers shall be located such that personnel can access and clean the baskets while standing on deck plates without the use of ladders or stools.

256.7.2 Seawater Service

256.7.2.1 Heat exchangers fitted with an internal strainer shall be mounted such that the strainer can be removed without interference or introducing the strained materials back into the cooler during removal.

256.7.2.2 Differential pressure gauges shall be installed to measure pressure drop across each strainer.

256.7.2.3 A duplex basket strainer or two simplex strainers piped in parallel and each having their own inlet and outlet isolation valves shall be installed on the suction side of each seawater pump. Differential pressure gauges shall be installed to measure pressure drop across each strainer.

256.8 SEA/HULL VALVES

256.8.1 General

256.8.1.1 Sea/hull valves shall be full port ball or gate valves.

256.8.1.2 A sea/hull valve shall be installed in each pipe connecting to a sea chest or overboard discharge connection and shall be located as close as practical to the structural penetration.

256.8.1.3 Sea/hull valves using stud bolts to attach the valve to the hull fitting shall not require removal of the stud bolts to remove the valve. Where more than one pump is connected to a sea suction or overboard discharge, an additional valve shall be installed in each branch.

256.8.1.4 Where a sea/hull valve is attached to a steel connection, a steel waster piece shall be installed. Shell and sea chest penetrations and waster pieces shall be fabricated and installed in accordance with USCG Drawing YD-505-003.

SECTION 257 JACKET WATER HEATING SYSTEM

257.1 GENERAL

257.1.1 The engines shall have an electric jacket water heating system.

257.1.1.1 The jacket water heating system shall be thermostatically controlled to heat the jacket water to a minimum temperature required by the OEM when the outside air temperature is at -20°F. The temperature shall be measured at the jacket water temperature gauge sending unit.

SECTION 259 EXHAUST SYSTEMS

259.1 GENERAL

259.1.1 For each engine, a dedicated exhaust piping system routes the hot and loud exhaust gases through a silencer to quieten and remove sparks, and then on to an area above the upper deck to discharge safely away from personnel, equipment, and air intakes.

259.1.2 The exhaust system shall be installed as shown on the drawings, the diesel engine OEM requirements and recommendations. The exhaust system shall be designed and installed so the back pressure does not exceed that specified by the engine manufacturer.

259.2 EXHAUST SYSTEMS

259.2.1 Piping penetrations, braces and hangers shall thermally isolate exhaust system components from the boat's structure and allow for differences in expansion to prevent overstress.

259.2.2 The diesel exhaust piping shall have flexible sections that are supported by dynamic hangers to account for thermal expansion and movement. The exhaust piping shall be connected to the diesel by a flexible bellows system, meeting the flange loading requirements of the diesel engine manufacturer. Expansion bellows shall be provided as necessary in the system and the entire system shall be supported and insulated. The piping weight, after the first bellows, shall be supported by the pipe hangers.

259.2.3 Exhaust silencers - Silencers for diesel engines shall be of the dry, spark-arresting type.

259.2.4 Exhaust systems shall be provided with drains at the low points.

259.2.5 The following minimum parameters shall be used in the selection of expansion joints:

259.2.5.1 Design based on 5000 cycles of full rated movement.

259.2.5.2 Design based on 1000 cycles of movement due to start-up and shut-down.

259.2.5.3 Design based on 100,000 cycles of movement due to system temperature fluctuations.

SECTION 300 ELECTRIC PLANT, GENERAL

300.1 General

300.1.1 The HSC-L electrical systems shall follow the guidance provided by the CG design package; electrical one-line diagrams and Electrical Power Load Analysis (EPLA).

300.1.2 The HSC-L shall be designed and constructed to the following requirements: IEEE-STD 45 (2002), 46 CFR subchapter J, 46 CFR §62: Vital System Automation, 46 CFR 58.25-65 and 58.25-70 for steering and ABS Rules for Building and Classing Steel Vessels for Service on Rivers and Intracoastal Waterways.

300.2 Design Standards and Criteria

300.2.1 All electrical components shall resist corrosion, maximize longevity, and withstand the rigors of use in the operating environment.

300.2.2 All connections not inside a watertight enclosure shall use IP 67 or better connectors.

300.2.3 All electrical enclosures and electrical enclosures' penetrations shall be rated IP 67 or better.

300.2.4 All enclosure penetrations shall be on the bottom of the enclosure.

300.2.5 All cable terminations shall be provided with strain relief.

300.2.6 All electrical cables that penetrate any watertight envelope or bulkhead on the boat shall be contained in a watertight system.

300.2.7 All electrical connections and wire runs shall be accessible for maintenance and troubleshooting without removing components or equipment.

300.2.8 Electrical equipment bonding and personnel safety grounding shall be in accordance with MIL-STD-1310(series).

300.2.9 The electrical equipment shall be capable of operating simultaneously with electronics equipment without causing interference to any electronic equipment or the magnetic compass.

300.2.10 Electrical installation shall ensure that personnel are not exposed to electrical hazards.

300.2.11 Electrical Panels, motor control centers and associated feeder cables shall be sized to accommodate a 10% growth margin.

300.2.12 Continuity of power to emergency services shall be maintained under all conditions.

300.2.13 Enclosures for electrical equipment shall be watertight (NEMA 4X).

300.2.14 Circuit breakers shall not be used as switches for lighting or other equipment.

301.3 Shore Power – SECTION UNDER DEVELOPMENT

301.3.1 Shore Power interface receptacles shall be provided and protected using circuit breakers.

301.3.2 The 240 VAC and 120 VAC electrical power distribution system shall not use the HSC-L hull, structure, or engine block to complete any circuit.

SECTION 302 ELECTRIC MOTORS AND ASSOCIATED EQUIPMENT

302.1 Performance Requirements

302.1.1 Electric motors with associated controllers, brakes, master switches, pilot devices and other associated equipment shall be provided as required for the driven equipment.

302.1.2 If brake horsepower needed by the driven equipment does not coincide with the NEMA motor standard nominal horsepower size the next larger size motor shall be used.

302.1.3 Electrical motors and associated equipment shall be installed in accordance with manufacturer instructions.

302.1.4 If brake horsepower needed by the driven equipment does not coincide with the NEMA motor standard nominal horsepower size, the next larger size motor shall be used.

302.1.5 COTS Motor starters and controllers shall be from the same manufacturer.

SECTION 303 PROTECTIVE DEVICES FOR ELECTRCIRCUITS

303.1 General Requirements

303.1.1 Protective devices shall include but not be limited to: circuit breakers, fuses, thermal tripping devices, current-limiting devices, reverse power protection devices, and power switches.

303.2 Performance Requirements

303.2.1 Each unit of equipment and circuits shall be protected from short circuit currents and thermal overloads.

303.2.2 The selection, arrangement, and performance of various protective devices shall provide a complete coordinated protective system having the following characteristics:

303.2.2.1 High speed clearing of low impedance faults.

303.2.2.2 Maximum continuity of service under fault conditions to be achieved by the selective operation of the various protective devices. Proper localizing of a fault condition to restrict outages to the equipment affected.

303.2.3 The minimum rating for circuit breakers shall be 15 amps.

303.2.4 Single pole circuit breakers shall not be used. All circuit breakers shall be a minimum of double pole devices.

303.2.5 Circuit breakers shall be from the same manufacturer.

303.2.6 Generator and shore-tie circuit breakers shall include integral under voltage trips (UVT).

303.2.6.1 Miniature Circuit Breaker (MCB) shall only be used as supplementary breakers (as defined by NEC 2008) and NOT as a branch circuit breaker.

303.2.6.2 Circuit breakers within a controller and automation equipment enclosures may have circuit breaker frame sizes of less than 15 amps for supplementary circuit protection (as defined by NEC)."

SECTION 304 ELECTRIC CABLES

304.1 Design Standards and Criteria

304.1.1 Cables shall be sized to continuously carry the full-load current of equipment connected.

304.1.2 Where parallel cables are utilized to increase the current carrying capacity of the circuit or to reduce the voltage drop, parallel conductors shall have the same cross-sectional area and length. For three phase circuits using cables in parallel, each cable shall contain all three phases.

304.1.3 Cable shall be installed in a neat and orderly manner presenting a straight run. Cable shall be tightly fitted to the overhead or bulkhead. Pipes shall not be routed through cable runs.

304.1.4 Whenever cable connections are concealed behind panels, sheathing or other surface material, access panels shall be provided. Panels shall be labeled to identify the concealed cable connections.

304.1.5 Cable conductors within equipment or enclosures shall have sufficient length to facilitate tracing, troubleshooting, and opening and closing of hinged doors to prevent conductor damage. Surplus wire length shall be provided to relieve all tension and vibration, to allow for disconnection and servicing, and to permit multiple wires to be fanned at terminal studs.

304.1.6 Cable connections shall be accessible.

304.1.7 Cables in machinery compartments shall not be installed in or behind insulation or behind protective sheathing.

304.1.8 Cables shall be high-flex stranded tinned annealed copper.

304.1.9 Cable servicing essential systems shall comply with the fire resistant requirements of IEC Publication 60331 or another recognized standard and shall be watertight regardless of whether the cable transits through a watertight bulkhead.

304.1.10 Battery cable connections shall be of the hex nut, lock washer type. Wing nuts shall not be used.

SECTION 305 ELECTRICAL AND ELECTRONICS DESIGNATING AND MARKING SYSTEMS

305.1 General

305.1.1 Information plates, label plates, and tags shall be installed for equipment, circuits, conductors and cables for power, lighting, electronics, interior communications, and other electrical systems in accordance with SFLC TS-305.

305.1.2 Nameplates for motors, generators, transformers, control apparatus, switchboards, heaters, brakes, and clutches shall be in accordance with applicable nameplate sections of IEEE-STD-45, "Recommended Practice for Electrical Installations on Shipboard. Installations on Shipboard.

SECTION 310 SHIP SERVICE GENERATORS

310.1 General

310.1.1 Ship service diesel generators shall be Qty two (2) CAT 4.4 marine diesel gensets with electronic control systems or equal. Generators shall be configured to provide 65KW of 240VAC wired in an ungrounded 3 ϕ delta configuration at 60Hz. The generator shall be provided with the following options from the manufacturer.

310.1.1.1 MGGP 200 gauge panel and interface harness

310.1.1.2 Engine shutdown sensors & shutdown controller

310.1.1.3 EMCP 4 engine control panel

310.1.1.4 Double wall high pressure fuel lines

310.1.1.5 AC voltage monitoring

310.1.1.6 Glow plugs

310.1.1.7 Factory installed generator space heater kit

310.2 Performance

310.2.1 The maximum basic operating load shall be the largest of the summer, winter, anchor, shore, cruising, functional, or emergency operating conditions as calculated in provided EPLA plus starting the largest non-maneuvering motor with one generator inoperable.

310.2.2 Manual and automatic paralleling of the generators shall be provided.

310.3 Monitoring Alarm and Controls

310.3.1 Emergency stop controls shall be provided for ship service generators and shall trip and lock open the generator output circuit breakers. Each stop control shall alarm in the engine room and pilothouse.

310.3.2 Local generator control panels shall be designed and installed to allow full front access. Panel face shall be hinged and shall be capable of being locked in the fully-open position.

310.3.3 Ship Service Generator Control shall be capable of operating as a standalone system separate from other machinery plant control systems.

SECTION 313 STORAGE BATTERIES

313.1 General

313.1.1 Batteries shall be Absorbed Glass Mat (AGM).

313.2 Performance

313.2.1 Each installed battery charger, alternator, and voltage regulator shall be compatible with the batteries and battery bank configuration selected, shall have battery temperature compensation, and shall be capable of completely recharging the attached battery bank in a maximum of eight (8) hours without exceeding a safe charging rate. Battery chargers/alternator voltage regulators for AGM batteries shall use three-stage charging.

313.2.2 Each battery charger that is attached to a diesel engine starting battery bank shall have a disconnect device which automatically isolates the charger from the battery bank during engine starting. A cross connect shall be provided to allow engine starting from each battery bank similar engine type without affecting the operation of any running engines.

313.2.3 Each battery charger shall have an audible alarm function that provides charger output failure indication.

313.2.4 Each battery comprising a battery bank shall have equal characteristics and be from the same manufacturer.

313.3 Engine Starting Battery Systems

313.3.1 This section defines the requirements for the dedicated 24VDC starting battery systems for all Main Propulsion Diesel Engines (MPDEs) and Diesel Generator Sets, hereafter referred to as "engines."

313.3.1.1 Each engine shall be equipped with its own dedicated starting battery system. Each starting battery system shall have crossover starting capability with engines of similar types for redundancy. Engine starting systems are independent from all other DC systems on the vessel, including the ship's service battery banks.

313.3.1.2 Each engine starting battery system shall consist, at a minimum, of:
a) A dedicated battery bank.
b) An automatic, AC-powered battery charger.
c) All necessary cabling, overcurrent protection, and disconnects.

313.3.1.3 Each starting battery bank shall be sized with sufficient capacity to provide, at a minimum, six consecutive starts for its respective engine without being recharged.

313.3.1.4 The battery bank and charger for each starting system shall be installed in close proximity to the respective engine to minimize the length of the starting cables and reduce voltage drop during engine cranking.

313.3.1.5 Volt meters shall be provided for starting batteries. The meters shall have a minimum range of 0VDC to 36VDC. The voltmeters shall be located within eyesight of the battery bank selector switches.

313.4 Representative Example System

313.4.1 A notional example of a diagram depicting engine and battery charging is included in Figure 313-1 below.

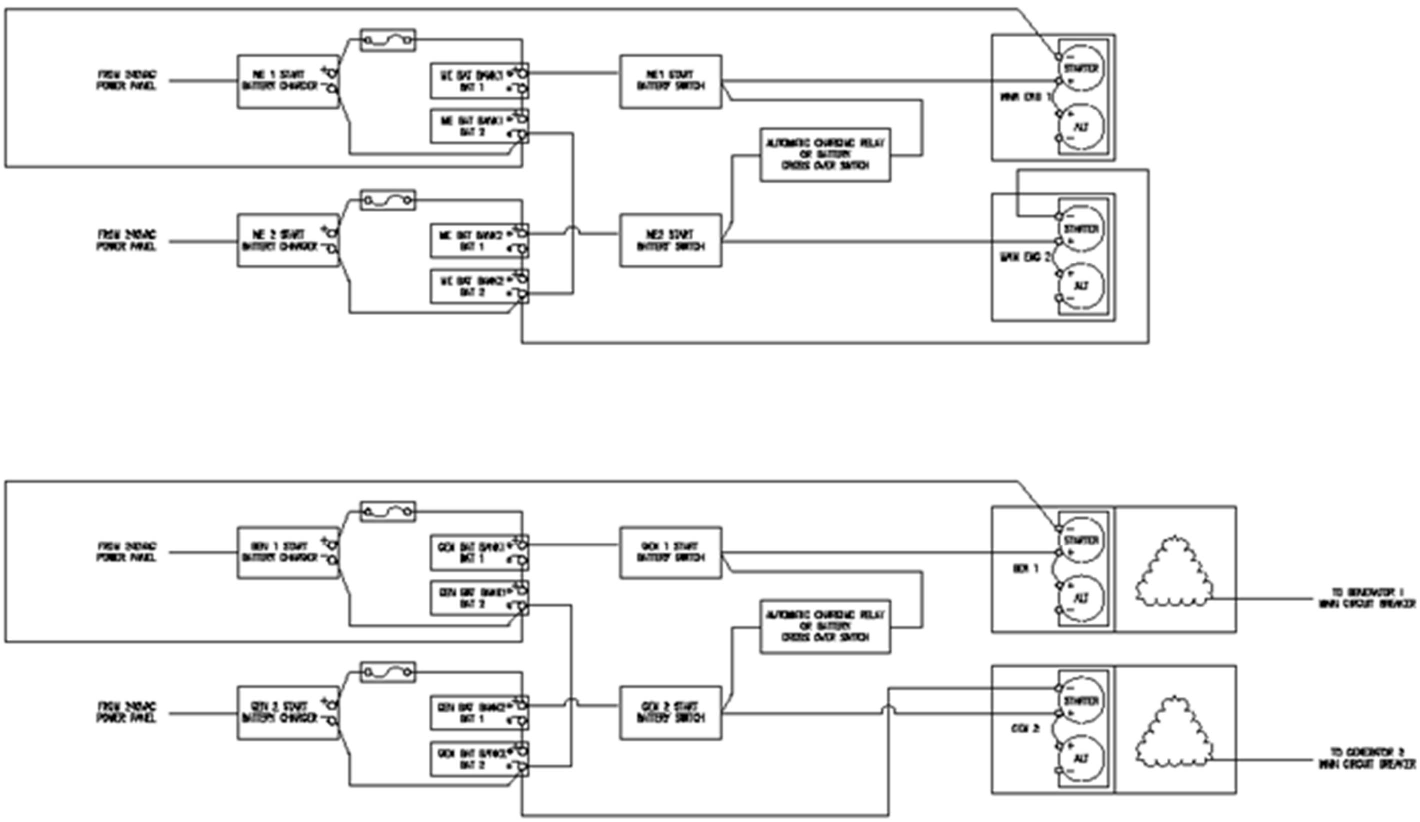


Figure 313-1 Notional Engine and Generator Battery Charging Diagram

SECTION 314 POWER CONVERSION EQUIPMENT

<u>314.1 General</u>

314.1.1 Transformers and AC/DC converters shall have KVA ratings sufficient to support 115% of the maximum demand load plus the spare circuits. Transformers shall be dry type.

<u>314.2 Performance</u>

314.2.1 Power conversion equipment with similar power characteristics shall be from the same manufacturer.

<u>314.3 Battery Charger</u>

314.3.1 The battery charger for each starting system shall be powered from the 240VAC distribution system. It shall be a fully automatic, multi-stage charger designed to maintain its associated battery bank in a fully charged and equalized state without requiring manual intervention.

SECTION 320 GENERAL REQUIREMENTS FOR ELECTRIC POWER DISTRIBUTION SYSTEMS

<u>320.1 Receptacles</u>

320.1.1 Receptacles shall be installed as required to permit the use of portable tools without more than a 50 foot extension cord.

320.1.2 Receptacles shall be installed throughout accommodation, electronic and office spaces in locations designated in the EPLA for electric equipment use.

320.1.3 Each machinery space shall be fed by separate feeders from the switchboard within or adjacent to the space.

SECTION 324 SWITCHBOARDS & POWER PANELS

324.1 General

324.1.1 Programmable controllers shall be compliant with IEC 61131-3

324.2 Ship Service Switchboard

324.2.1 A dead-front switchboard shall be provided, built, designed and tested to UL 891. Except that destructive tests do not need to be performed.

324.2.2 Circuit breakers shall meet the requirements of UL 489 SA.

324.2.3 Switchboard buses shall be copper. Bus connections shall have silver surfaced contacts. Bare bus bars shall meet the minimum spacing requirements of one (1) inch.

324.2.4 Terminals in switchboards and panelboards shall be located so that it will not be necessary to reach across uninsulated line bus to make connections or replace fuses.

324.2.5 Switchboard instruments shall be flush mounted. The instruments shall display the monitored parameter in an analog format for instantaneous recognition of trend information. The instrument's scale shall be marked to indicate the rated value of the parameter being measured. Instruments shall have sensitivities which are twice the sensitivity of the associated controls; for example, 5% sensitivity instruments shall be used with 10% sensitivity controls.

324.2.5.1 An ammeter shall be provided on the switchboard capable of measuring the ships ampere draw while on shore power.

324.2.6 Switches for instrument transfer or control functions shall be of the rotary type, equipped with clearly marked escutcheon plates to indicate function and position. Switches shall be mounted on and operated from the front of the switchboard. Switches shall have detents to securely hold the switch in the selected position.

324.2.7 Indicator lights shall be flush mounted and shall incorporate long life LED. Bulb size and base shall be standard throughout the switchboard.

324.2.7.1 Indicator lights shall be provided to indicate when the ship service generator motorized breakers are in the tripped position.

324.2.8 A diagrammatic wiring diagram of each switchboard section shall be framed and mounted on the inside of the rear hinged access cover for the appropriate section. The mounting shall include a protective clear coating which shall be acid resistant. In addition to showing point-to-point wiring, the diagrams shall also depict the system's setup, control and power sources.

324.2.9 Circuit breakers for the ship service generator shall be interlocked with the manual and automatic synchronization systems so as to prohibit being closed onto an energized bus without first being synchronized. The circuitry for the manual and automatic synchronization systems shall be independent to the extent that a failure on either system will not render the other synchronization system inoperable.

324.2.10 Where the Class Rules are silent, interrupting devices, control, instrumentation, and metering and protective and regulating equipment assemblies shall be in accordance with IEEE-STD-C37.20.1.

324.2.11 Any manual or automatic bus transfer switch shall be as compact as possible and shall be designed specifically for an ungrounded marine power system.

324.3 Power Distribution Panels

324.3.1 Power and lighting distribution panels shall be installed as required by system design. These panels shall be provided with a minimum of one spare circuit breaker for each five active circuit breakers or fraction thereof. The spares shall be of the size and type most frequently used within the panel.

324.3.2 All motors regardless of horsepower shall have overload protection independent of the feeder breaker.

324.3.3 Power distribution panels designed for galley equipment shall be readily accessible and located in close proximity to the galley.

SECTION 330 LIGHTING SYSTEMS

330.1 General

330.1.1 Lights shall be light-emitting diode (LED).

330.1.2 Illumination levels in all compartments shall meet the requirements of ABS Crew Habitability on Workboats Section 6/Table 1. Any supporting human factors requirements shall comply with ASTM F1166.

330.2 Performance

330.2.1 Lighting distribution in any compartment which has multiple power sources shall be arranged such that the failure of one circuit does not leave any area without light.

330.2.2 Floodlights shall be provided for the working deck.

330.3 General Illumination

330.3.1 The flag, hull illuminating, and exterior lights shall be switch controllable from the pilothouse.

330.3.2 Switches for exterior lighting shall be located in the pilothouse and be protected from inadvertent operation.

330.3.3 Exterior lighting shall be provided with an electrical means to de-ice and prevent icing.

SECTION 402 SECURITY REQUIREMENTS

402.1 GENERAL

402.1.1 This section applies to boat equipment containing either software or firmware, including embedded software and firmware in equipment primarily described by other sections.

402.1.2 Outdated or superseded versions of software and firmware shall be removed prior to, or immediately after, installation of new versions.

402.1.3 Prior to acceptance testing, software and firmware, including what is embedded in equipment, shall be verified to match the version recommended by the OEM. Non-field programmable devices and field-programmable devices with analog or discrete signal interfaces are exempt from these verification requirements.

402.1.4 Equipment within the HM&E Boundary shall not be connected to the Navigation or Communication Boundaries.

402.2 ACCESS CONTROL

402.2.1 If equipment has an account management or login function, the Contractor shall reset to the OEM default username/password. Any temporary or test accounts shall be deleted, except as otherwise directed by the USCG.

402.2.2 Wireless Technology

402.2.2.1 Wireless functions shall be disabled, either physically or via software, unless otherwise directed by USCG.

402.2.3 Physical Data Interfaces

402.2.3.1 Physical data interfaces (e.g., communications ports such as USB, SD cards, etc.) that are not necessary to satisfy the requirements herein shall be removed of disabled, except as otherwise directed by the USCG.

402.2.3.2 Physical data interfaces shall be secured within enclosures that are lockable or that require tools to open, except as otherwise directed by the USCG.

402.2.4 Logical Data Interfaces

402.2.4.1 Functions, ports, protocols, and services that are not necessary to satisfy the requirements shall be removed or disabled, except as otherwise directed by the USCG.

402.3 EXTERNAL DEVICE CONNECTIONS

402.3.1 The Contractor shall only connect OEM-approved devices to the OEM equipment that contains software or firmware.

402.3.2 The connected device shall be dedicated for that specific equipment/task and shall not be used for any other purpose or connect to any external networks or systems.

402.4 INTERFACE AND TECHNICAL DOCUMENTATION

402.4.1 System/Sub-System and Interface Design Descriptions shall be developed, maintained, documented, and provided. These describe system/subsystem wide design and architecture and contain a topology diagram noting the interfaces, connections and type of connection, boundaries, and physical locations of the boat equipment. These shall be formatted in accordance with IEEE 12207, Systems and Software Engineering – Software Life Cycle Processes.

402.4.2 Equipment configuration settings shall be developed, maintained, documented, and provided in a Firmware Support Manual.

402.4.2.1 For equipment on the boat, including special tools to support the boat, the Firmware Support Manual shall contain:

402.4.2.1.1 Installation, removal, configuration, reset, and recovery procedures;

402.4.2.1.2 Firmware and software version numbers;

402.4.2.1.3 Field-programmable device firmware and software;

402.4.2.1.4 Adjustable parameter settings;

402.4.2.1.5 Network settings and addresses;

402.4.2.1.6 Login procedures and elevated-privilege modes;

402.4.2.1.7 Data logging, storage capacity, and audit functions;

402.4.2.1.8 Data ports and protocols and how they are enabled/disabled;

402.4.2.1.9 Wireless functions and how they are enabled/disabled;

402.4.2.1.10 List of required maintenance tools; and

402.4.2.1.11 Software license information.

402.5 VULNERABILITY AND RESOLUTION REPORT

402.5.1 The Contractor shall screen provided software and firmware against the latest vulnerability information in the National Vulnerability Database (NVD) and Common Vulnerabilities and Exposures (CVE). The Contractor shall address any listed vulnerability and inform the USCG of the vulnerability and resolution.

SECTION 404 RADIO FREQUENCY, TRANSMISSION LINES, AND CABLES

404.1 GENERAL

404.1.1 This section sets forth the general requirements for radio frequency transmission lines and cabling, and provides requirements for installation. For all other cable types or where this section is silent, electrical/electronic cabling shall meet the requirements of Specification Section 301, Wiring and Electrical Cable, and shall be labeled in accordance with Specification Section 305.

404.1.2 Cable connectors not inside a watertight enclosure shall be made with IP 67 connectors.

404.1.3 Cables associated with electronic equipment shall be continuous runs. There shall be no splices. The only exception is for quick disconnects associated with removable items.

404.2 INSTALLATION REQUIREMENTS

404.2.1 Radio frequency transmission lines shall:

404.2.1.1 Be protected from mechanical abuse and heat damage.

404.2.1.2 Have no physical or electrical interference with equipment, cables, or other radio frequency transmission lines.

404.2.1.3 Have an entrance that prevents the introduction of moisture and dirt.

404.2.1.4 Be installed such that characteristic impedance of each line is not materially changed.

404.2.1.5 Have a loop of excess cable to allow removal of equipment from consoles for accessing the back of the associated items by turning the items over without disconnecting any conductors.

404.2.1.6 Have stuffing tubes or multi-cable transits for passing through decks or bulkheads to maintain their watertight, airtight, fume-tight, or light-tight integrity of the structure. Stuffing tubes shall be ASTM F1836M-15 type. Multi-cable transits shall be ABS- or UL-certified.

404.2.2 Cables near fluid routing shall have drip-proof shields or other barriers installed to protect from leak damage.

404.3 COAXIAL CABLE

404.3.1 Coaxial cable types shall be appropriate for the application. Standard methods shown in IEEE STD 45 shall be used for:

404.3.1.1 Entry of coaxial cable to accessories, equipment, and wiring boxes;

404.3.1.2 Passing coaxial cable through bulkheads;

404.3.1.3 Protection of cable against heat, condensation, and mechanical damage; and

404.3.1.4 Supporting and securing cable to decks and bulkheads.

404.3.2 The inside bend radius of the coaxial cable shall be greater than 10 times the cable diameter, except when the cable is subject to repeated flexure, in which case the inside bend radius shall be greater than 20 times the cable diameter.

404.3.3 Terminal boxes, branch boxes, or other forms of standard electric wiring equipment shall not be used to terminate or connect the coaxial cable.

SECTION 405 ANTENNAS

<u>405.1</u> <u>GENERAL</u>

405.1.1 Antennas shall be arranged for satisfactory system performance across all frequencies and shall meet the EMI/RFI reduction requirements.

405.1.2 Antennas shall be physically separated to reduce electrical interaction and to avoid physical contact due to antenna deflection caused by ice loading, wind, sea conditions, or other adverse conditions.

405.1.3 Antenna mounts shall be constructed to support the antenna without failure while subjected to the HSC-L operating environment.

SECTION 406 GROUNDING AND BONDING

<u>406.1</u> <u>GENERAL</u>

406.1.1 Grounding, bonding, and shielding of equipment and cable shall meet the requirements of MIL-STD-1310 to ensure safety and electromagnetic compatibility.

406.1.1.1 RF transmission lines and cables shall be electrically bonded and be routed within the structure to protect against EMI. Type I or II bond straps shall be used where required.

406.1.1.2 Cables that are routed topside or in exposed locations shall be shielded in accordance with the requirements of MIL-STD-1310G, either by

406.1.1.2.1 Use of shielded cables; or

406.1.1.2.2 Use of single or multi-cable conduit.

SECTION 407 EMI AND RADIO FREQUENCY INTERFERENCE (RFI) REDUCTION

<u>436.1</u> <u>GENERAL</u>

407.1.1 EMI and RFI shall be minimized in accordance with OEM recommendations and allow the operation of all equipment simultaneously without disrupting other equipment.

407.1.2 Safety rails, lifelines, and lanyards shall not generate EMI.

407.1.3 The Contractor shall consider the impacts to crewmembers when designing the placement of equipment that emits electromagnetic radiation and shall identify hazardous areas.

407.1.2.1 Hazardous areas shall be marked using applicable marking:

407.1.2.2 Attach Type 1 identification marker, (NSN: 7690-01-377-5893);

407.1.2.3 Attach Type 2 identification marker, (NSN: 7690-01-377-5896);

407.1.2.4 Attach Type 3 identification marker, (NSN: 7690-01-377-5895);

407.1.2.5 Attach Type 4 warning sign, (NSN: 7690-01-377-5899); and

407.1.2.6 Attach Type 5 warning sign, (NSN: 7690-01-377-5374).

SECTION 410 REQUIREMENTS FOR CONTROL STATIONS AND DATA DISPLAY SYSTEMS

410.1 GENERAL

410.1.1 Controls and gauges shall be within sight of the seated operator, clearly identifiable, accessible, and operable during daytime, nighttime, and cold weather operations, as specified in the Control Criticality Table (Specification Table 410-1).

410.1.2 Neither polarized nor tinted display surfaces shall be used.

410.1.3 Displays shall be configured to minimize glare and reflection.

410.1.4 Displays shall have easily legible alphanumeric characters.

410.1.5 Displays shall be flush mounted to the console surface unless approved by the USCG.

410.1.6 Touch screens shall be avoided if other options are reasonably available.

410.1.6.1 Touch screens shall be fully functional when wet. Displays involving touch screen shall have finger rails, or similar support, at a minimum, along the top horizontal and each vertical side of the display to allow for stabilization of the operator's hand while using the thumb to operate specific buttons during operations.

410.1.6.2 A touch screen shall not be the sole input means. Full functionality shall be provided without the use of a touch screen.

410.2 LOCATION OF CONTROLS, DISPLAYS, AND EQUIPMENT

410.2.1 ASTM F1166, Section 6, Section 8, and Specification Table 410-1 shall be used to establish efficient location of controls, gauges, displays, and equipment.

410.2.2 Headset and microphone console-connected wires shall be at least three inches away from the helm and propulsion controls when worn or operated by the helmsman.

410.2.3 Controls that require adjustment while underway shall be provided with finger rails, or similar support, to allow for stabilization of the operator's hand while using the thumb to operate specific buttons during operations.

410.3 ANTHROPOMETRY REQUIREMENTS

410.3.1 Controls, displays, and equipment identified on the Control Criticality Table (Table 410-1) shall meet the following definitions:

410.3.1.1 "Primary Viewable" (PV) shall be within 35 degrees left and right of center and 25 degrees above and 35 degrees below the horizontal line of sight for the seated positions.

410.3.1.2 "Secondary Viewable" (SV) shall be within 60 degrees left and right of center and 50 degrees above and below the horizontal line of sight for the seated positions.

410.3.1.3 "Reachable" (R) shall be within the Functional Reach (extended), Vertical Arm Reach (sitting) for the 5th percentile female to 95th percentile male in accordance with ASTM F1166, Figures 50 and 51, for seated operations. For standing operations, reach shall be within the Forward Functional Reach (back of shoulder to fingertips) and Maximum Overhead Gripping Reach for the North American 5th percentile female to 95th percentile male in accordance with ASTM F1166, Tables 16 and 17.

410.3.1.4 The viewing angle of the console shall be in accordance with ASTM F1166, Fig 20.

Table 410-1: Control Criticality Table – Pilothouse

TABLE UNDER DEVELOPMENT

SECTION 421 NON-ELECTRONIC NAVIGATION SYSTEM

421.1 MAGNETIC COMPASS

421.1.1 The Contractor shall provide permanent magnetic compass(es) that shall:

421.1.1.1 Meet ISO 25862;

421.1.1.2 Have a dampened display that is at least 4-1/2 inch in diameter;

421.1.1.3 Be capable of compensating for deviation; and

421.1.1.4 Be provided with lighting dimmable to 0 for night operations.

421.1.2 The compass shall be installed in the Bridge Console, forward of the primary helm station, and within easy view of the helmsman while underway.

421.2 Clinometers

421.2.1 Heeling and trimming Clinometers meeting standard CID-A-A-59308 shall be provided and installed on the fore and aft axis and athwart ships in the pilothouse.

421.3 Day Shapes

421.3.1 Day shapes shall be provided in accordance with COMDTINST M16672 2D "Navigation Rules, International – Inland" for each of the operation scenarios described in SECTION 422.1.1

SECTION 422 ELECTRICAL NAVIGATION AIDS

422.1 NAVIGATION LIGHTS

422.1.1 The HSC-L shall have LED navigation lights in accordance with the USCG Navigation Rules and Regulation handbook; International Rules for Prevention of Collisions at Sea (72 COLREGS) Annex I and Inland Navigation Rules Annex I requirements corresponding with underway operations, anchoring, towing astern, less than 200 meters, more than 200 meters and towing alongside

422.1.1.1 The Navigation lights shall be powered by a single circuit breaker and switched using a single toggle or rotary type selector switch for the different navigational configurations.

422.2 Searchlight

422.2.1 The HSC-L shall have mounted searchlight(s) able to rotate 360 degrees on the horizontal axis.

422.2.1.1 The searchlight shall be Imtra part number (DHR 230) with Imtra controller part number (DHPAN2019NF) or equal.

422.2.1.2 The searchlight shall be integrated into the HSC-L electrical system and not use batteries for control power.

422.3 Law Enforcement Light

422.3.1 A blue strobe Law Enforcement Light shall be installed on the mast, located to provide maximum visibility to other vessels.

SECTION 423 ELECTRONIC NAVIGATION SYSTEMS

423.1 GENERAL

423.1.1 The Contractor shall provide an electronic navigation system using the USCG standardized SINS equipment listed below:

423.1.1.1 12-inch Multifunctional Display (MFD) Raymarine Axiom Pro 12 RVX with RealVision 3D, 1kW CHIRP Sonar (E70372);

423.1.1.1.1 Qty (1) Master MFD provided with two micro-SD cards;

423.1.1.1.2 Qty (2) Slave MFDs provided with one micro-SD card;

423.1.1.1.3 Micro-SD cards to be R70838 or equal (formatted in FAT32 or NTFS, and Speed Class 10 or UHS Speed Class 1);

423.1.2 Transducer options;

423.1.2.1 SS60 Depth & Temp Stainless Steel Through Hull (SS 60, 12 Degree, CP370 Connector), A80568;

423.1.2.2 SS60 Depth & Temp Stainless Steel Through Hull (SS 60, Zero Degree, CP370 Connector), A80569;

423.1.2.3 SS60 Depth & Temp Stainless Steel Through Hull (SS 60, 20Degree, CP370 Connector), A80570;

423.1.3 Heading Sensor (EV-1, E70096);

423.1.4 Global Navigation Satellite System (GNSS) Receiver (RS150, E70310);

423.1.5 Qty (2) i70S remote information display (E70327-S2);

423.1.6 Radome 4KW 24” HD (E92143);

423.1.7 Qty 3 National Marine Electronics Association (NMEA) 0183 to 2000 Converter (A80455);

423.1.8 RayNet Network Switch (RNS-5, A80731); and

423.1.9 Remote Grip/Pointer (RMK-10, A80438).

423.1.10 SONAR module (CP370, E70207)

423.1.11 Weather Station (200WX, A0755)

423.1.12 The components shall be integrated using NMEA 2000 protocol to the maximum extent to provide navigational functionality.

423.1.13 The selected transducer shall be of sufficient length (including adapter[s] as needed) to reach at least two MFDs.

SECTION 425 AUTOPILOT

<u>425.1 GENERAL</u>

425.1.1 The HSC_L shall have an autopilot system with the following features:

425.1.1.1 Automatic heading control;

425.1.1.2 Adjustable steering response; and

425.1.1.3 Automatic heading adjustment based on pitch, roll, and yaw.

425.1.1.4 The autopilot shall not interface with SINS-2 or AIS-2.

425.1.1.5 The autopilot shall have an independent heading sensor and display.

SECTION 436 ALARMS

<u>436.1 GENERAL</u>

436.1.1 All systems requiring alarms shall have their alarm signal integrated into and indicated by the Machinery Control and Monitoring System see Specification SECTION 202.

SECTION 441 EXTERIOR COMMUNICATIONS SYSTEM

<u>441.1 GENERAL</u>

441.1.1 The Contractor shall provide and install (in accordance with the OEM’s requirements) the following communication systems:

441.1.1.1 At least one Motorola APX-8500 Tactical Radio Systems;

441.1.1.2 At least one Standard Horizon GX-6000 DSC Class D;

441.1.1.3 Communication systems installation shall be modular (with connector terminated cables) to the maximum extent practicable to facilitate maintenance and future radio upgrades.

441.1.2 The transceivers for the radios shall be mounted to be readily accessible to load frequencies and the code plugs. Transceivers shall be located within the Pilothouse.

441.1.3 Each radio shall perform independently and have a dedicated antenna(s) through which it can transmit and receive. Radios shall not interfere with each other.

441.1.4 Each radio shall have a dedicated IP 65 water-resistant speaker that is clearly audible during all operating conditions. Each speaker shall have a volume control. Each speaker shall have a visual indicator of transmission if available.

441.1.5 Each radio shall have a dedicated water-resistant palm microphone (at least IP 65).

441.1.6 For each radio, the Contractor shall furnish all programming hardware, cables, adapters, and special tools needed for testing and alignment.

<u>441.2</u> <u>TACTICAL RADIO SYSTEM</u>

441.2.1 The Tactical Radio System equipment shall be configured from the following list of components:

441.2.1.1 P25 Multiband Mobile Radio (Transceiver), inclusive of Mount Kit, Model Number: APX-8500;

441.2.1.2 P25 Multiband Mobile Radio Remote-Mounted Control Unit, inclusive of Mounting Kit, Model Number: O7 Control Head;

441.2.1.3 Multiband Radio Marine-Grade Remote Speaker, PN: HSN4038; and

441.2.1.4 Multiband Radio Marine-Grade Remote Microphone, PN: HMN1089.

441.2.2 The tactical radio system's triband antenna system shall:

441.2.2.1 Be a marine ground plane independent antenna;

441.2.2.2 Be capable of full transmit and receive coverage across every frequency band enabled in the radio, 136-174, 380-520, and 762-870 MHz bands; and

441.2.2.3 Have a maximum voltage standing wave ratio < 2.5:1 at every frequency enabled in the tactical radio system shall be provided.

441.2.3 Components needed to deliver a fully integrated and operable tactical radio system.

441.2.4 The following equipment listed below shall be provided with each boat:

441.2.4.1 One Multiband Radio Programming Cable, PNL HKN6184;

441.2.4.2 One Multiband Radio Data Interface Cable, PNL HKN6172; and

441.2.4.3 One Multiband Radio Key Variable Loader (KVL) Key Loading Cable/Adapter, PN: HKN6182.

<u>441.3</u> <u>COMMERCIAL MARINE VHF-FM/DSC</u>

441.3.1 The Commercial Marine VHF FM/DSC Radio System equipment shall be configured from the following list of components:

441.3.1.1 VHF-FM band (156 MHz – 162 MHz), dedicated antenna;

441.3.1.2 DSC Marine Radio (Transceiver Class D) With Bracket Mount Kit, PN GX-6000; Kit also includes:

441.3.1.2.1 Microphone; and

441.3.1.2.2 Radio Programming Cables, Programming and Firmware;

441.3.1.3 DSC Marine Radio Remote Control Unit (Transceiver Class D) with Bracket Mount Kit, PN SSM-70H; Kit also includes

441.3.1.3.1 Microphone; and

441.3.1.3.2 Transceiver to Remote Control Unit Cable;

441.3.1.4 Transceiver/Remote Control Unit Remote Radio Speaker, PN MLS-410SP-B;

441.3.1.4.1 Transceiver Radio flush mount kit, PN MMB-84; and

441.3.2 The Commercial Marine VHF FM/DSC Radio System shall be integrated and displayed in the SINS-2 MFDs.

SECTION 443 LOUD HAILER

443.1 GENERAL

443.1.1 The boat shall have a system capable of emitting automatic audible signals and external voice commands in all operating environments. An additional dedicated GX 6000 for this application may be provided.

443.1.2 The system shall provide sound signals to meet the requirements of the USCG Navigation Rules and Regulations Handbook; International Rules for Prevention of Collisions at Sea (72 COLREGS) Annex III; and Inland Navigation Rules Annex III for sound signaling.

443.1.3 The system shall have a volume control.

443.1.4 The system shall be independent of other communication systems.

443.1.5 The speaker shall be an IP 67 and mounted in accordance with the OEMs recommendations.

443.1.6 The system shall have listen-back capability.

443.1.7 The system shall have a siren/yelp feature that is activated by means other than the handheld microphone.

443.1.8 The system shall be configured with a remote water-resistant microphone (IP 65 or greater).

443.1.9 A horn signal shall be operated by a dedicated momentary switch accessible from each helm location.

SECTION 455 AUTOMATIC IDENTIFICATION SYSTEMS

455.1 GENERAL

455.1.1 The boat shall have a USCG Second Generation Automatic Identification System (AIS) using equipment listed below:

455.1.1.1 Raymarine AIS5000 Transponder Bundle (Transponder, GPS antenna, Power/Control/RF Coax cables, Mounting Bracket, Junction Box, and Manual), Raymarine Part Number E70529;

455.1.1.2 VHF Antenna, Raymarine Part Number A80590; and

455.1.1.3 DeviceNet to STng Cable, Raymarine Part Number A06045.

455.1.2 Components to deliver a fully integrated and operable AIS shall be provided. The components shall be integrated using NMEA 2000 protocol to the maximum extent. The AIS with SINS shall be integrated.

SECTION 457 ELECTRO-OPTICAL AND INFRARED SURVEILLANCE SYSTEM

457.1 ELECTRO-OPTICAL (EO) AND INFRARED (IR) CAMERA SYSTEM

457.1.1 A marine grade Electro-Optical (EO)/Infrared (IR) camera system shall be installed. It shall:

457.1.1.1 Be integrated to display on the SINS MFD(s);

457.1.1.2 Have thermal/infrared and visible light/EO camera(s);

457.1.1.3 Have a minimum 4X zoom function;

457.1.1.4 Be capable of detecting a 30-foot vessel at a range greater than 2 NM in total darkness;

457.1.1.5 Be provided with remote-controlled pan (360º) , tilt (+/- 90º), and home control; and

457.1.1.6 Have image stabilization.

457.1.2 The EO/IR camera system shall be capable of monitoring visibility both fore and aft to aid in navigation, identify targets of interest, and monitor vessels being towed astern.

457.2 MONITORING CAMERA SYSTEM

457.2.1 A marine grade visible and IR camera system that allows the crew to monitor interior compartments and exterior deck spaces shall be installed.

457.2.2 The following interior compartments and exterior spaces shall have the following quantities of camera(s) with unobstructed view of the compartment/space:

457.2.2.1 Machinery Space(s) – one port and one starboard;

457.2.2.2 Forward Deck Space – one centerline;

457.2.2.3 Aft Deck Space – one port and one starboard;

457.2.3 The output video of each camera shall be viewable using dedicated display.

SECTION 505 GENERAL REQUIREMENTS FOR PIPING SYSTEMS

505.1 GENERAL

505.1.1 This Section specifies additional general requirements for the design, construction, material, fabrication, arrangement, installation, testing, and cleaning of piping systems and piping components. Requirements peculiar or supplemental to a specific system are specified in the section governing that system. Where supplemental or differing requirements are specified in the section governing the system, the system section requirements take precedence.

505.1.2 Design pressures shall not be less than maximum allowable working pressures. Normal, abnormal, and emergency operation and casualty conditions shall be considered in determining the maximum allowable working pressure.

505.1.3 Normal, abnormal, emergency, and casualty conditions shall be considered in determining the maximum system temperature. The design temperature shall equal or exceed the maximum system temperature.

505.1.4 Piping systems and components shall be free of vibration outside of design limits

505.2 DESIGN AND SELECTION OF COMPONENTS

505.2.1 Material selections shall be in accordance with the diagrams provided.

505.2.2 Unless otherwise specified, components shall meet the specifications of ABS MVR.

505.2.3 The specified component pressure and temperature rating listed in referenced specifications or standards shall be equal to or greater than the system design pressure and temperature.

505.2.3.1 Joint Bolting. Selection of bolting lengths shall be in accordance with ASTM F704.

505.3 PRESSURE RELIEF

505.3.1 Relief valves on non-fired pressure vessels shall take into account maximum filling rate AND the possibility of fire in the compartment and be sized in accordance with ASME BPVC Section VIII.

505.3.2 Where relief valve discharge piping is provided, it shall be arranged and sized so that the backpressure does not cause unsatisfactory relief valve operation, either from a stability or capacity standpoint. Back pressure build-up shall be based on the maximum capacity of the installed relief valve rather than the minimum required relief valve flow rate based on source flow.

505.3.3 The open ends of relief valve discharge piping shall be located so that discharge does not damage machinery or equipment or endanger personnel.

505.4 PRESSURE, TEMPERATURE, LEVEL AND FLOW CONTROLS

505.4.1 Orifices

505.4.1.1 Orifices shall be installed in a flanged joint and constructed in accordance with USCG Drawing YD-505-001.

505.4.1.2 Orifices provided with equipment and machinery shall be installed in accordance with their specifications or the manufacturer's standard recommendations.

505.4.1.3 Where orifices are installed in constant flowing copper nickel piping systems, piping material downstream of each orifice shall be 70-30 copper-nickel for at least ten(10) pipe diameters.

505.4.1.4 Where required, Orifices shall have a cavitation index K of 1.0 or greater calculated as follows:

$$K = [\frac{(H_u - H_v)(1 - (D_o/D_i)^2)}{H_u - H_d}] - 1$$

Where:

K = cavitation index

Hu = static head upstream of the orifice, ft.

Hd = static head downstream of the orifice at 10 pipe diameters, ft.

Hv = absolute fluid vapor pressure, ft.

Do = orifice diameter, in.

Di = pipe inside diameter, in.

Where the cavitation index is found to be less than or equal to 1, the condition shall be corrected by installing a multistage orifice.

505.4.5 Control valves shall be evaluated for cavitation. The valve shall be designed and configured such that the downstream restriction does not affect valve performance other than for the intended purpose of correcting cavitation using the following formula for cavitation index:

$$K = \frac{(P2 - Pv)}{(P1 - P2)}$$

Where:

K = cavitation index

P2 = absolute pressure downstream of the valve

P1 = absolute pressure upstream of the valve

Pv = absolute fluid vapor pressure.

The cavitation index shall be in accordance with the manufacturer's data for avoidance of incipient cavitation. Installation of cavitation trim, downstream orifice plate(s) are permitted means to bring the cavitation index into compliance, provided that they do not affect the operation of the valve.

505.5 VALVES

505.5.1 Valves shall be installed in accordance ASTM F1166 Section 12, Valve placement, Orientation, and Location.

505.5.2 Globe and angle valves shall be arranged with the system pressure or vacuum under the disc so that in the shut position, the pressure or vacuum is not exerted on the bonnet joint or stem packing

505.6 PIPE SUPPORTING ELEMENTS

505.6.1 Supports shall be installed on or adjacent to concentrated weights in the piping system to preclude contact with adjacent pipe, equipment, and structure under loading or under working of the ship.

505.6.2 Design and installation of rigid pipe hangers shall be in accordance with ASTM F708.

505.7 PIPING STRUCTURAL REQUIREMENTS

505.7.1 Piping systems shall have the flexibility to prevent overstressing of piping materials or supports, leakage of joints and unacceptable distortion of connected equipment.

505.7.2 Flexibility shall be incorporated into system design by changes of direction in the piping through the use of bends, loops, and offsets. Use of expansion joints shall be minimized.

505.8 ARRANGEMENT

505.8.1 Piping shall not obstruct or interfere with the operation of doors, hatches, manholes, scuttles, or openings covered by portable plates, nor obstruct or interfere with access to access panels, access doors, inspection manholes, hand holes, sight glasses, drain plugs or test cocks used for inspection and maintenance of machinery and equipment.

505.8.2 Connections to pumps, tanks, and equipment shall have take down joints to facilitate maintenance.

505.8.3 Local high and low points shall be avoided in the arrangement of piping. Where necessary for the proper functioning of the system and its connected machinery and equipment, vents, and drains shall be installed at the high and low points.

505.8.4 Waster pieces shall be installed in the overboard discharges and other locations where joints between ferrous and non-ferrous materials are required. A short sleeve type waster piece, with dimensions in accordance with Coast Guard Drawing FL-4823-84 for the pipe size required shall be installed.

505.8.5 Overboard discharges shall be fit with a swing check and cutout valve. Unless otherwise specified, the cutout valve shall be at the shell of the ship with the swing check valve located as close as practicable to the cutout valve and installed in a fore and aft direction.

505.9 COMPARTMENTAL

505.9.1 Piping within tanks shall be of a material which will not cause contamination of the tank contents, and the material for this piping shall be suitable for both the tank and pipe contents.

505.9.2 Piping cleanouts and takedown joints shall not be located in the galley area.

505.10 INTERFACE TO SHIP STRUCTURE

505.10.1 The penetration of decks and bulkheads shall be minimized. Bulkheads and decks shall generally be penetrated close to boundaries of compartments.

505.10.2 Piping penetrations through watertight bulkheads and decks shall be installed type approved.

505.10.3 Single pipe penetrations through watertight bulkheads may be installed using ABS-approved Roxtec model SPM, or equal, sleeves/seals for piping not located in bilge areas, or below deckplate. Roxtec or equal products are prohibited in hydrocarbon fluid systems (e.g., hydraulic oil, fuel oil, etc).

505.11 JOINTS

505.11.1 Takedown joints in piping containing combustible fluids shall be located as remotely as practicable from hot surfaces.

505.11.2 Raised face flanges shall not be used against bronze or other relatively low strength composition valves, fittings or flanges.

505.12 FIRE HAZARD REDUCTION

505.12.1 Flammable fluid piping shall not be located within 18 inches of hot surfaces.

505.12.2 Piping with hot surfaces shall not be located within 18 inches of tanks containing flammable fluid.

505.12.3 In machinery spaces and in other operating spaces that contain both machinery and electric equipment, combustible or flammable fluid piping shall be kept as far from electrical equipment as practicable.

505.13 TAILPIPES

505.13.1 Bellmouths shall be installed on tailpipes and provide an area not less than 1-1/2 times the inside area of the tailpipe and shall terminate in a flanged antivortex bellmouth.

505.13.2 Tailpipe location with respect to adjacent plating or other components shall ensure a free suction area around the open-end periphery not less than 1- 1/2 times the inside area of the tailpipe.

505.13.3 Each tailpipe shall have a minimum of one valve between the tailpipe and the piping system.

505.13.4 Tailpipes shall terminate ½ the tailpipe nominal pipe size from the tank bottom except for the following:

505.13.4.1 Fuel tank stripping and overflow: 3/4".

505.13.4.2 Fuel service to engine: 10% of tank height.

505.13.4.3 Fuel return from engine to top of tank.

505.13.4.4 Sewage collection, holding, and transfer: 1.0x tailpipe nominal pipe size.

505.14 PIPE INSULATION

505.14.1 Piping systems shall be insulated and lagged in accordance with ASTM F683 except for prescribed modifications.

505.14.2 Piping in the temperature range of –20 °F to 125 °F shall use elastomeric cellular insulation. The requirements of ASTM F683 5.4 shall not apply.

505.14.3 Calcium silicate and expanded perlite insulation shall not be used.

505.15 PIPING AND VALVES

505.15.1 Piping arrangement shall not obstruct or interfere with operation, inspection, or maintenance of other equipment on the boat.

505.15.2 Valves shall be suitable for intended application and shall be located to allow for operations and maintenance and be reachable by personnel in PPE.

505.15.3 Valves for on/off applications shall be stainless steel flanged, socket weld, butt-weld, or threaded ball valves with stops in the open and close positions. Butt-welded valves, fittings, and flanges shall not be used for sizes smaller than 1 ¼ inches.

505.15.4 Multi-position valves, when installed, shall clearly indicate their intended positions

505.15.5 Piping conveying fluids that is subject to constant or frequent flow shall be protected from erosion by the following mitigations:

505.15.6 Sizing pumps, including flow and head characteristics to match system and attached equipment design flow requirements.

505.15.7 Valve handles shall be rated for the intended application and corrosion resistant.

505.15.8 Avoiding the use of tees or other fittings which cause a sudden change in direction.

505.15.9 Threaded pipe, valve, and fitting installations of dissimilar metals in raw water systems shall not be made

505.16 HOSE AND HOSE ASSEMBLIES

505.16.1 Installed hoses and hose assemblies shall comply with SAE J1942 for the intended service.

505.16.2 Hoses may be installed between two points to provide flexibility, or to isolate or control vibrations. Hoses shall not be subject to torsional deflection (twisting) under normal operating conditions. Their lengths shall be limited to that necessary to provide flexibility and for proper operation of the machinery. Hoses may be used in every class of piping subject to pressure, temperature, and fluid service limitations as specified by the manufacturers and fire hazardous zone requirements.

505.16.3 Flexible piping connections shall absorb the maximum designed excursions of resiliently mounted equipment without overstressing the hose, attached piping, or the equipment components to which the hose is attached. Flexible hose assembly installations shall meet the requirements of SAE J1273. Hose length shall not exceed 30 inches, except where required to address equipment movement.

505.16.4 In potable water systems, braided, stainless steel jacketed, flexible piping may be used between the cut-off valve and each potable water system end user. The flexible piping shall be National Sanitation Foundation (NSF) approved for drinking water.

505.16.5 Fuel, lube oil, and hydraulic systems shall be stainless steel and have either crimp-on or re-useable end fittings except for the fuel fill hose and fuel tank vent hose. End fittings shall be pressure rated equal to or greater than the mating hose and compatible with applicable connection points of the system.

505.16.6 Hoses in the raw water and bilge systems shall use hose barb fittings with two stainless steel hose clamps (if practicable). For raw water hoses in sizes exceeding that covered by SAE J1942, use thermoid bellowsflex 7910, or equal.

505.16.7 Hoses for vital systems shall meet the regulatory requirements of 46 CFR Part 56. Vital systems are those identified by 46 CFR Part 56.07-5(f).

505.16.8 Hose assemblies shall be hydrostatically tested to 150% of the system service pressure. Tests shall be reported in the hose assembly testing report.

505.16.9 A metal tag shall be affixed to each hose upon installation.

505.16.10 Tags shall have the following information:

505.16.10.1 Item (hose or flexible joint) serial number;

505.16.10.2 Hydrostatic test pressure;

505.16.10.3 Hydrostatic test date [DD/MM/YY] or “NA” if not tested; and

505.16.10.4 Service life date (replacement date) [QTR "Q"/YY]. References to calendar year quarter shall include the letter "Q" after the quarter number to avoid confusion with months. The 4th quarter of 1999 would be represented by "4Q99." Service life date is the install date plus 5 years for installed fire systems and exterior hoses and 12 years for fuel, lube oil, jacket water, raw water, and hydraulic systems, bilge water, and other hoses.

505.16.11 Tags shall be attached using materials that will not damage hoses or flexible joints. If metal strapping causes chaffing, nylon cable ties may be used. Nylon cable ties shall not be used on hoses that have operating temperature over 180°F.

505.17 EXPANSION JOINTS

505.17.1 Non-Metallic Flexible Expansion joint design shall conform to the requirements of ASTM F1123.

505.17.2 Flexible piping connections shall absorb the maximum anticipated excursions of resiliently mounted equipment, without overstressing the expansion joint, attached piping, or the equipment components to which the expansion joint is attached.

505.17.3 Expansion joints shall not be used in hydraulic systems.

505.17.4 All rubber expansion joints used for fluid systems (excluding diesel air intake and exhaust, and heating, ventilation, and air conditioning [HVAC]) shall meet the requirements of ASTM F1123.

505.17.5 The outside cover of expansion joints shall be neoprene.

505.17.6 For lube oil service and seawater systems, the inner sleeve of expansion joints shall be nitrile-butadiene (BUNA-N).

505.17.7 Each installed expansion joint shall be tagged with the information required for hoses and shall be included in the hose assembly testing report.

505.18 MACHINERY AND PIPING SYSTEM DRAINAGE

505.18.1 Drain plugs shall be installed at low points in all piping systems or equipment that may entrap fluid, including components and parts of equipment. All drains shall be accessible for operations and maintenance.

505.18.2 Each discharge and drain, except engine exhaust, shall have a flanged through-hull penetration connecting to a 316 stainless steel ball valve and 316 stainless steel check valve.

505.18.3 Discharges above the water line shall be fitted with drip edges to prevent drainage from staining or corroding the hull.

SECTION 506 OVERFLOW, AIR ESCAPES, AND SOUNDING

506.1 GENERAL

506.1.1 Overflows, air escapes, and sounding tubes for potable water, fuel oil, and black/graywater tanks shall be constructed of stainless steel. Other tanks shall use 90-10 copper nickel. Overflows and air escapes shall terminate in the weather and be protected from the weather and water ingress.

506.2 OVERFLOWS AND AIR ESCAPES

506.2.1 All tanks served by pumps shall be provided with air escapes and overflows.

506.2.2 Overflows and air escapes shall be combined wherever practicable. Air escapes shall be provided and installed in all tanks having filling or suction connections. Piping shall be sloped to ensure gravity drainage of the piping back to the tank.

506.2.3 The size of the overflow pipe from any tank shall not be less than 125% of the size of either the transfer or filling piping to the tank, whichever is larger.

506.2.4 Overflow pipe wall thickness shall be in accordance with ABS MVR 4-6-4/9.5.6.

506.2.5 Overflow pipe overboard discharges shall be in accordance with ABS MVR 4-6-4/9.5.3.

506.2.6 The sewage holding tank overflow shall terminate in a dedicated hull penetration with a lockable gag scupper valve.

506.2.7 The sewage holding tank overflow cross sectional area shall not be less than the combined cross-sectional area of all soil and waste lines which enter the tank.

506.2.8 Fuel oil tank overflows shall be in accordance with ABS MVR 4-6-4/9.5.5.

506.2.9 Air escapes shall be sized to limit the air velocity to not more than 25 ft/s when the tank or compartment is being filled or drained at its maximum design rate and in accordance with ABS MVR 4-6-4/9.3.3.

506.2.10 Air escape pipe height and wall thickness shall be in accordance with ABS MVR 4-6-4/9.3.2.

506.2.11 Air escapes for different services may be combined, providing requirements within this Section are met, except that air escapes from tanks carrying different liquids shall not be combined. If air escapes are combined, a drop-out section shall be installed in each air escape to permit testing the tanks individually. If air escapes are joined together, or to a header, the area of common air escape or header shall be not less than that required to limit the air velocity to 25 ft/s when the tanks are filled or drained at maximum designed rates.

506.2.12 Air escapes from tanks or compartments carrying flammable, combustible, or toxic vapors shall terminate clear of ventilation intakes, openings into the vessel, and sources of ignition such as combustion exhaust gas outlets.

506.2.13 The fuel oil tank air escapes shall terminate in return bends and be protected in accordance with ABS MVR 4-6-4/9.3.5.

506.2.14 Air escapes from fuel tanks shall be independent of all other air escapes.

506.2.15 Sewage holding tank air escapes shall be sized such that the seal of any plumbing trap will not be subjected to a pressure differential of more than 0.25 kPa (1 inch of water) when all sewage pumps are discharging at their maximum anticipated flow rates, but in no case shall be less than 2-inch NPS.

506.2.16 Air escapes from sewage holding tanks shall be independent of all other air escapes.

506.3 SOUNDING TUBES

506.3.1 Sounding tubes shall be provided and installed for all fuel tanks, potable water tanks, . Sounding tubes shall not be installed in sewage holding tanks or greywater tanks.

506.3.2 The upper ends of sounding tubes for fuel tanks shall terminate on the weather deck. The sounding tube caps for all oil tanks shall be designed so that any accumulated air pressure in the tube will be gradually released and equalized before the cap or plug is completely unscrewed.

506.3.3 Potable water tank sounding tubes shall extend at least 18 inches above the tank top, be provided with caps that can be padlocked closed and have individual sounding rods that can be permanently stowed in their respective tubes.

506.3.4 The upper ends of other sounding tubes which terminate above the full load waterline shall be located, where possible, on open decks or in passageways. These ends shall be fitted with a valve and a cap and chain except where they may form an obstruction, in which case flush-type plug fittings shall be provided.

506.3.5 Where a sounding fitting must be located in a compartment, its location shall be readily accessible and shall not interfere with the function of the compartment. Sounding fittings shall not be located in locked or carpeted compartments, or where oil from a tank being filled could accidentally be discharged from an open fitting onto a hot surface or on electrical or electronic equipment.

506.3.6 Flush sounding fittings shall not be located where water might normally be on the deck. A wrench for opening the sounding tube cap or plug shall be mounted on the bulkhead adjacent to the sounding tube.

506.3.7 Sounding tubes shall measure fluid in a tank or compartment at its lowest level. A sounding tube of not less than 1-1/2 inch NPS shall be installed. Such tubes may be run on an angle or they may be bent to a radius of not less than 10 feet. Sounding tube wall thickness shall be in accordance with ABS MVR 4-6-4/11.3.3.

506.3.8 A striking plate shall be provided at the bottom of each sounding tube or welded to the tank bottom in way of the sounding tube to protect the tank plating from damage by the sounding device.

<u>506.4 ELECTRONIC TANK LEVEL INDICATION</u>

506.4.1 Black and Grey Water tanks shall have a radar type tank level indication device for local and remote reading of the tank level. Vega Pulse 6x shall be used.

506.4.2 Radar type tank level indication devices shall be installed on fuel oil tanks, and potable water tanks. Where a remote tank level indicating device is provided for determining the level in a tank containing flammable liquid, the failure of the device shall not result in the release of the content of the tank through the device.

506.4.3 TLI standpipes shall be constructed of non-ferrous material

506.4.4 TLI standpipes shall be separate from manual sounding tubes.

SECTION 507 MACHINERY AND PIPING DESIGNATION AND MARKING

<u>507.1 GENERAL INFORMATION.</u>

507.1.1 Information and identification labels shall be permanently installed. Equipment identification labels shall include the following information:

507.1.1.1 Equipment Nomenclature

507.1.1.2 Serial number

507.1.1.3 Manufacturer name

507.1.1.4 Equipment component location/identification number

507.1.1.5 Critical ratings, as applicable to the equipment.

507.1.2 Identification labels installed on machinery or equipment shall be metallic. Information labels installed in the following instances shall also be metallic and of corrosion-resistant material, such as CRES or anodized aluminum:

507.1.2.1 On decks or other surfaces subject to abrasion

507.1.2.2 Labels for piping system valves and fittings

507.1.2.3 Wherever a label is attached by wire

507.1.2.4 Labels installed on surfaces exceeding 126°F.

507.1.3 Materials.

507.1.3.1 Identification label plates installed on machinery or equipment shall be metallic.

507.1.3.2 New identification label plates, information label plates, and piping system/HVAC system label plates shall comply with MIL-DTL-15024, except as noted herein. Equipment identification labels shall be etched, stamped or engraved to a depth which will withstand wear throughout the life of the equipment.

507.1.3.3 Interior Piping System/HVAC System label plates shall be Type A (etched), B (engraved), or C (stamped) stainless steel per ASTM A240, type 316L, finish 2B or Type H (photosensitive) aluminum per ASTM B209 Alloy 1100. Where subject to a severe environment such as foot traffic, use Type A (etched), B (engraved), or C (stamped) stainless steel per ASTM A240, type 316L, finish 2B.

507.1.3.4 Piping System/HVAC System label plates in the weather shall use Type A (etched), B (engraved), or C (stamped) stainless steel per ASTM A240, type 316L, finish 2B.

507.2 PIPING SYSTEM LABLE PLATE SCHEME

507.2.1 Designation. Piping system components are assigned a three-part designation as follows:

507.2.1.1 System designation (See Table 507-1)

507.2.1.2 Component identification (See Table 507-2)

507.2.1.3 Basic location number (See 507.2.3)

507.2.1.4 An example of the three (3)-part designation is as follows: FM-V-1-25-3, which indicates a valve is installed in the firemain system on the Main Deck at Frame 25 and it is the second valve starboard of centerline.

Table 507-1. Piping System Designation Letters.

SYSTEM	DESIGNATION
Auxiliary Freshwater Cooling (Notes 8)	AFW
Auxiliary Seawater Cooling (Notes 1 and 4)	ASW
Ballast	BAL
Chilled Water (Note 5)	CW
Combustion Air Exhaust	CE
Combustion Air Supply	CS
Condensate	CN
Deballast	DBAL
Firemain	FM
FM-200 (Heptafluoropropane) Fire Extinguishing	HFP
Fuel Service (Note 2)	FS
Fuel Stripping	FST
Fuel Transfer and Filling	FT
Hot Water/Hydronic Heating (Note 5)	HW
Hydraulics (Note 6)	HYDR
Jacket Water (Note 2)	JW

Lubricating Oil Fill, Transfer, and Purification	LO
Lubrication Oil Service (Notes 2)	LOS
Lubrication Oil Service, Reduction Gear (Note 3)	RLO
Main Drainage (Bilge)	MD
Overflows	TV
Potable Water	PW
Potable Water, Cold	CPW
Potable Water, Hot	HPW
Sounding Tubes	ST
Vacuum Sewage Collecting, Holding, and Transfer (Note 7)	VCHT
Vent (Tanks and Voids)	TV
Waste Drains (Plumbing/Grey Water and Interior Deck Drains)	PWD
Water Mist Fire Extinguishing (Machinery Spaces)	WM
Weather Deck Drain	WDD

1 - Includes seawater cooling for propulsion, ship service generators, auxiliary machinery, auxiliaries, and miscellaneous systems and equipment when centralized seawater cooling is provided.

2 - Generic system designation. The system designation shall be preceded with P for propulsion and G for generator applications when systems are separate and distinct from one another.

3 - System designation shall be used only for those propulsion plants (typically diesel or gas turbine) where the reduction gear lubricating oil service system is separate and distinct from the propulsion unit lubricating oil service system.

4 - That portion of the central seawater cooling system common to seawater demands shall be designated SW. Those portions of the central seawater cooling system associated with auxiliary machinery, auxiliaries, and miscellaneous systems and equipment shall be designated ASW.5 - Chilled water and hot water designations. The system designation shall be followed with an S for supply and an R for return side of the system.

6 - Hydraulic system designations. The system designation shall be preceded with a SG for steering gear, CR for crane, AW for anchor windlass. For other hydraulic systems use the designator HYDR and specify the system as part of the label plate description.

7 - For a non-vacuum flush system remove the “V” from the system designation.

8 - Includes freshwater cooling for propulsion, ship service generators, auxiliary machinery, auxiliaries, and miscellaneous systems and equipment when centralized freshwater cooling is provided.

507.2.2 Piping Component Identification Letters. The piping component identification letters used in the valve/component label plate scheme shall be in accordance with Table 507-2.

Table 507-2. Piping Component Identification Letters.

Component	Designation
Valve	V
Differential Pressure Switch	DP
Float Switch	FS
Gauge	GA
Liquid Level Switch	LS
Orifice	OR

Pressure Switch	PS
Pressure Transmitter	PT
Pulsation Damper	PD
Salinity Cell	SC
Special Fitting such as Strainer, Filter, Flexible Connection	F
Thermometer	TH

507.2.3 Basic Location Number. The basic location number shall consist of a series of three numbers, separated by hyphens, designating in the following sequence the vertical, longitudinal, and transverse location of an item in the cutter/vessel. Assignment of basic location numbers for label plates shall be determined as follows.

507.2.3.1 A2.4.1 First Number. The first number shall indicate the vertical level at which the component is located, by deck or platform number.

507.2.3.2 Machinery spaces which have multiple decks/levels within the space or varying heights within a single deck/level shall use projected continuous deck levels when determining the vertical level numbering for the basic location numbers. Machinery spaces shall include Machinery Spaces and any other compartment with the compartment designation ending in “E”.

507.2.3.3 Second Number. The second number shall indicate the longitudinal location of the component by frame number. If the component is not on a frame, the next forward frame number shall be used. If the next forward frame is in the compartment forward of the compartment with the component, the next frame number aft of the component shall be used.

507.2.3.4 Third Number. The third number shall indicate the transverse location by assigning consecutive odd numbers for starboard locations and consecutive even numbers for port locations. The numeral "1" shall indicate the lowest centerline (or centermost starboard) component.

507.2.3.5 Numbering Selection. Consecutive odd or even numbers shall be assigned to components as they would be observed as being above, from forward to next aft frame, and then outboard of the preceding component. This means starting at the lowest center most location in the space/deck and working in an upward direction then aft to the next frame and then moving outboard either to the starboard or port side.

507.2.4 Piping Systems Color Coding. Piping system components shall be color coded in accordance with Table 507.2-3. Piping system color coding not specified in Table 507-3 shall be painted in accordance with COMDTINST M10360.3 (series) Coatings and Color Manual.

Table 507-3. Color Code Requirements.

Fluid	Color	Fed Std 595 Color Number and Chip	Extent
Chilled Water	Striped Light Blue/ Dark Green	15200/14062	Note 1
Firemain (including root valves)	Red	11105	Note 2
Fresh water,	Light Blue	15200	Note 1
Fuel	Yellow	13538	Note 1
Hydraulic	Orange	12246	Note 1
Jacket Water/Waste Heat	Striped Light Blue/ Black	15200/17038	Note 1
Lube Oil	Striped Black/Yellow	17038/13538	Note 1

Potable Water	Dark Blue	15044	Note 1
Seawater Included Main Drainage and Waste Drainage	Dark Green	14062	Note 1
Sewage	Gold	17043	Note 1
Water Mist Fire Extinguishing	Striped Red/Dark Blue	11105/15044	Note 3
1 - Color code only valve handwheels (including the spokes) and levers on valves not exposed to the weather. Valves and handwheels exposed to the weather at the immediate shipboard shore connection shall have label plates or plain language markings that clearly delineate the service for each connection. 2 - All fire plugs and handwheels including associated components (strainer, wye, gate, applicators, and hose racks) shall be color-coded.3 - Color code all handwheels including the spokes, all levers, all pushbuttons including the enclosure and cover, and SOPV enclosures. - General Info: A - Handwheels and levers are preferred to be coated with plastisol per Commercial Item Description A-A-59464A. Valve handwheels and operating levers in lieu of plastisol may be painted with brush or spray using enamel, Master Painters Institute Reference #9 where surface temperature does not exceed 82°C (180°F), but should not be applied in handwheels or levers where they could become immersed, such as in tanks and bilges. B - If necessary, thin enamel or clean equipment using paint thinner/solvent per MIL-PRF-680C.			

507.2.4 Other Piping System Requirements

507.2.4.1 Identification. In addition to special colors required for particular systems, all piping systems shall be marked for identification. This marking shall be the system's functional name, pressure (only if the fluid has major pressure differences for similar functions), destination where feasible, and directional flow arrow. The direction of flow arrow shall be an arrow 3 inches long pointing away from the lettering (for reversible flow, point an arrow away from each end of the lettering).

507.2.4.2 Application. The marking shall be applied with paint and use of stencils. The marking shall generally be black in color, but white for dark colored piping. The markings shall be applied in conspicuous locations, preferably near control valves.

507.2.4.3 As an alternative on the inside of the ship, markings can be applied using preprinted retro-reflective labels equal to what is specified in the 3M U.S. Navy Shipboard Damage Control/Luminous Label Catalog. Marking of piping located in tanks, voids, cofferdams, bilges and similar unmanned spaces shall be by label plate curved to fit the piping and attached in place by full circumferential wiring or banding. The label plate shall meet the material requirements of 507.2.2.3.

507.2.4.4 Spacing. Markings shall be spaced not more than 4.5m (15 ft) apart, measured along the run of the pipe, unless piping is concealed behind sheathing, in which case the markings shall be spaced not more than 1.5 m (5 ft) apart.

507.2.4.5 Multiple Markings. Piping passing through spaces including unmanned spaces such as tanks, voids, cofferdams and bilges shall be marked at least once in each space; piping in machinery spaces shall be marked at least twice, once near entry and once near exit. Systems serving propulsion plants and systems conveying flammable or toxic fluids shall be marked at least twice in each space. Where piping is behind protective battens, attached label plates shall list all pipes behind the batten. One label plate shall be provided to list all pipes installed in a pipe tunnel or similar space.

507.2.4.6 Deck Plates. The method for marking deck plates for sounding pipes shall be as shown on drawing NAVSHIPS No. 810-1385848. Where compliance with drawing NAVSHIPS No. 810-1385848 would result in letters less than 3.2 mm (1/8 inch) size, an approved 3.2 mm (1/8 inch) thick CRES label plate inscribed with the required information shall be welded to the deck adjacent to the corresponding sounding tube deck box.

507.2.4.7 Other Fittings. Deck plates for sounding tubes shall be marked with the tank name and number. For remote operating stations, labels shall contain the functional service of the valve being remotely operated. Labels for relief valves and temperature control valves shall also indicate the set pressure or the set temperature.

507.2.4.8 Installed Instruments. Labels shall be provided for all installed instruments. Inscriptions shall indicate the purpose or use of the instruments and shall bear the designation of the related component or system.

507.2.4.9 Hidden Valves. Where a valve is installed below a detachable floor plate or grating, or is similarly hidden, a marking shall be installed both on the valve and on the portable access plate.

507.3 HVAC SYSTEM LABEL PLATE DESIGNATIONS.

507.3.1 Where a label is provided in accordance with the following requirements and is located so that it serves the purpose of another label, only one label is required. Labels shall be located on fan casings, closure housings or handwheels, coil casings, or adjacent structure.

507.3.2 Label plates shall be installed for components of ventilating and air conditioning systems in accordance with Table 507-4. These plates are in addition to identification or information plates installed by equipment manufacturers and the closure classification label plates for damage control.

Table 507-4 – HVAC System Label Plate Designations

COMPONENT (ITEM TO BE IDENTIFIED)	DESIGNATIONS REQUIRED	EXAMPLES
Fan	Location number and type of systems (supply, exhaust, or recirc.) and compartment number where controller and remote master switch are located	Fan 2-28-2 supply Contr. Loc. 2-28-0-L Remote Master Switch Loc. 4-30-1-E
Fan coil assemblies	Location number of compartment served. Thermostat location of compartment numbers where controller and remote master switch are located	Fan coil assembly 2-28-2 Serving 2-28-0-L, 2-48-0-L 2-48-1-L Thermostat 2-28-2 Controller 2-28-0-L Remote Master Switch Loc. 3-48-1-P
Fan coil assemblies Controller and Remote Master Switch	Location number of component served. Compartments served including those that may receive blowout air, recirc systems that receive replenishment air, and compartment where remote master switch or controller is located (See Notes 7 and 8)	Fan 2-28-2 Supply Supply 1-18-2-C, 2-25-2-L Replen. Recirc. Systems 2-30-1, 2-32-2 Remote Master Switch Loc. 4-30-1-E
Watertight closure (on	Location number of closure	Clos 2-35-1

discharge side of supply systems or suction side of exhaust systems)	and compartment secured by the duct (See Notes 1 and 6)	Supply 3-27-1-L, 4-29-1-E
Watertight closures (if above service is through a preceding or subsequent closure which controls ventilation to more compartments)	Location number of closure and the preceding or subsequent closure (See Notes 1 and 6)	Clos 2-35-1 Prior 1-37-1 Next 2-34-3
Weather closure for natural ventilation ducts	Location number of closure and compartment served by duct	Clos 01-1-10-L Exh 2-10-1-L
Closures on the discharge side of exhaust systems and on suction side of supply systems, including weather closures	Location number of closure and ventilating fan served	Clos 01-12-2 Supply Sys 2-16-2
Trunks and ducts passing thru a compartment, unless identified by other labels in that compartment	System location number (See Notes 1 and 2)	Supply Sys. 2-121-1
Terminals, where necessary to identify their character	Type of terminal (supply, exhaust, return, or replenishment)	Supply
Duct coils (reheating and Cooling)	Location number of coil and compartment served Thermostat location Contactor location if duct coil is electrical.	Heating coil 2-27-2 Serving 2-59-2-L, 2-68-2-B Thermostat 2-68-2 Contactor 2-28-2
Duct coils (preheating)	Location number of coil and ventilating system served Location of contactor	Heating coil 02-80-1 Supply Sys 01-79-1 Contactor 02-83-1
Unit heaters, unit coolers, and gravity coils	Location number of component Location of thermostat	Unit Htr 01-73-1 Thermostat 01-74-2
Thermostat	Location number of thermostat and cooling coil, heater coil, damper fan coil assembly controller	Thermostat 01-31-2 Controlling Cooling coil 01-51-2 and Heating coil 01-52-2
Electric contactor for electric heater	Location number of contactor and electric heater controlled	Contactor 01-31-2 Controlling heater 01-52-2
Weather openings to natural ventilation ducts	Component and compartments served	Intake Nat. Sup. 2-70-4-E
Weather openings to supply or exhaust systems	Component and system served	Intake Supply Sys 2-16-2

<table>
<tr><td>Disposable filters (final or prefilter or both)</td><td>Location number of filter and compartment served (See Note 3)</td><td>Disposable filter 2-59-2
Prefilter 2-58-2
Serving 3-124-1-E
(See Note 3)</td></tr>
<tr><td>Cleanable filters</td><td>Location number of filter and amount and size required (See Note 4)</td><td>Filter 3-82-1
3-Size-12AF</td></tr>
<tr><td colspan="3">1 - Duct system designation may be combined with the closure designation.
2 - Ducts and trunks penetrating compartment boundaries shall be labeled unless easily identified by labeled system components nearby.
3 - If equipped with a gauge, include following note on the label, "Replace when gauge indicates dirty filters".
4 - If equipped with a gauge, include following note on the label, "Clean when gauge indicates dirty filter".
5 - Closure labels shall be arranged to be plainly visible when the closure is closed or opened. Each remote operation station for a closure shall be provided with a label plate which is a duplicate of that installed at the closure.
6 - Where a remote switch is specified, the controller label plate shall denote the compartment in which the remote master switch is located and vice versa; the remote switch label plate shall denote the compartment in which the controller is located.
7 - Controllers and remote master switches and label plates shall be installed warning of dual power sources.</td></tr>
</table>

507.4 INSTRUMENTS

507.4.1 Label plates shall be provided for installed indicating and recording instruments. Inscriptions shall indicate the purpose or use of the instrument and shall bear the designation of the related component or system wherever appropriate. Label plates shall be installed below the instrument and on the board supporting the instrument unless otherwise approved.

SECTION 512 HEATING, VENTILATION AND AIR CONDITIONING

512.1 GENERAL

512.1.1 The objective of the HVAC system is to provide efficient and effective heating, cooling, and ventilation to all of the compartments on the vessel, as well as to establish the required air supply and exhaust to ensure crew safety and support equipment requirements. It will consist of fan coil units in the living spaces, workshop, and pilot house as well as ventilation systems for the machinery spaces to maintain the design temperatures identified in the diagram through the whole range of operating conditions identified in Section 070. Replenishment air for the fan coil units will be supplied by a dedicated ventilation supply system, and exhaust will be handled naturally (pilot house and engine room) or through mechanical ventilation. Heating and cooling will be provided by a 2-pipe thermal fluid system with 45°F chilled water in cooling season and 180°F hot water in heating season. Heating of conditioned spaces will be via the fan coil units, while duct heaters, convection heaters, and unit heaters will provide the balance of the heating capacity. Supply air for recirculating system replenishment and for the auxiliary machinery space will be preheated.

512.2 SYSTEM REQUIREMENTS

512.2.1 Weather Openings. Weather openings shall be designed and located to prevent accumulation of ice and/or snow in all operating conditions down to the design temperature, shipping seawater, driving rain, or spray. Resistance to entrance of water may be either inherent by reason of location or design or obtained by use of water excluding ventilators. Ventilator throat heights shall meet the requirements derived from heel angles, down flooding angles and trim lines required to meet damage control requirements without the use of closing apparatuses where possible.

512.2.2 Supply system intakes shall be located to prevent recirculation of air from combustion air discharge openings, exhaust air, and noxious or toxic fumes and smoke. Supply system intakes shall not be in the vicinity of sewage, gray water, or flammable liquid tank vents and shall be located so that it is not possible to ingest fumes from such vents during any foreseeable condition of operation.

512.2.3 Hazardous Spaces. Compartments containing A2L refrigerants shall be compliant with ASHRAE Standard 15, including but not limited to the use of refrigerant monitoring devices and airflow alarms.

512.2.4 Routing of Duct. Installation of duct sections, terminals, cooling coils, and fan coil units over electrical and electronic equipment is not permitted.

512.2.5 System Test and Balance Requirements

512.2.6 System testing and balancing shall be in accordance with ASHRAE Standard 151 and SOW Section 512. At the conclusion of balancing, balancing dampers shall be drilled and permanently riveted in place.

512.3 SYSTEM VELOCITIES

512.3.1 The following table lists the maximum design air velocities for ventilation systems and their components.

TABLE 512-1 - Air Velocities of Systems and Their Components

Feature	Maximum Velocity (ft/min)
Adjustable blast terminals, machinery space	3,500
Adjustable blast terminals, ventilated spaces	2,500
Air filters (face velocity)	900
Ducts and bellmouth terminals (throat velocity), mechanical ventilation or recirculation (rectangular)	3,000
Ducts and bellmouth terminals (throat velocity), mechanical ventilation or recirculation (round)[2]	3,500
Ducts, natural ventilation	2,500
Exhaust commercial grille (face velocity), other	1,000
Expanding cone terminals	3,000
Heating coils, duct (face velocity)	1,800
Natural ventilation openings through non watertight structure[1]	1,000
Weather openings, intake, face velocity	2,000
Weather openings, exhaust, face velocity	2,500
Demisters	2,000

512.4 EQUIPMENT REQUIREMENTS

512.4.1 Filters. Disposable Air filters having a MERV rating of 8 shall be provided for all preheaters as well as cooling and dehumidifying equipment. Ventilation supply systems containing preheaters shall have a filter installed upstream of the preheater.

512.4.2 Filters shall be accessible and easily removable. Access doors to any compartment or plenum that contains a filter shall not be a bolted manhole cover. Air filter assemblies shall ensure a tight seal between the cells and the filter frames. Filters shall be installed so that they are protected from the weather and give heating and cooling coils maximum protection. Air filter frames installed in built up systems shall be galvanized steel or stainless steel. Frames shall be airtight.

512.4.3 Screens. Weather terminals and ventilation exhaust inlets shall be provided with galvanized steel, aluminum, or CRES screens, ½” mesh with 0.063” wire diameter.

512.4.4 Terminals

512.4.4.1 Supply Terminals. Supply terminals shall be adjustable blast type E (Juniper JE-102E or equal) for ventilation systems, diffusing type for air conditioning systems, and expanding cone for replenishment air branches. Blast terminals, Type E, shall be constructed of CRES 304.

512.4.4.2 Exhaust Terminals. Bellmouth exhaust terminals shall be used in all locations, except where side duct openings would more efficiently serve to exhaust heat from equipment with internal exhaust blowers.

512.4.4.3 Location of Terminals. Terminals shall be arranged to prevent water from dripping, splashing, or being blown on electric or electronic equipment; to prevent short circuiting of air between supply and exhaust terminals, or between terminals and compartment accesses; to prevent dead spaces; and to exhaust heated air or fumes directly from the sources. Compartment air distribution shall be arranged for, as near as practicable, a full sweep of the room.

512.4.4.4 Supply and exhaust terminals shall be installed in the overhead, except as follows:

512.4.4.5 Replenishment air terminals shall be directed into and terminated not more than 300 mm (12 inches) from the fan coil unit intakes or directly connected to the return for ducted systems.

512.4.4.6 Except for main machinery spaces, the exhaust terminals in spaces containing hydraulic oil reservoirs shall be located close to the reservoir breather pipe.

512.5 CONSTRUCTION REQUIREMENTS

512.5.1 Unless otherwise specified, duct construction shall be in accordance with SMACNA HVAC Duct Construction Standard and ASHRAE Handbooks. All joints in watertight duct must be welded. Gasketed watertight flanges shall be used for takedown and attachment to equipment.

512.5.2 Entrance Requirements. Installation of fans, cooling coils, heaters, and supply system take-offs shall ensure full and unstratified airflow.

512.5.3 Materials. Non-watertight and watertight ductwork shall be fabricated from one of the following:

512.5.3.1 Aluminum, ASTM B209, Alloy 5052.

512.5.3.2 ASTM A527 galvanized steel.

512.5.3.3 Hot rolled steel to ASTM A568, galvanized after fabrication to equal ASTM A527 where ductwork is 1/8 inch thick or less.

512.5.3.4 CRES (Stainless) duct shall be fabricated from ASTM A240 Grade 316L (S31603).

512.5.3.5 Flange bolting in exterior and moisture-prone locations or in CRES duct shall be CRES 316, ASTM F593G (bolts) and ASTM F594G (nuts).

512.5.3.6 Flange bolting for galvanized steel ducting in interior locations shall be zinc plated steel, MIL-S-1222, SAE J429/J995 Grade 5 or greater, or ASTM A193/A194 Grades 2H, B7 and B16. ASTM F593G may be substituted at all locations calling for zinc plated steel bolting.

512.5.4 Duct Shapes. Where rectangular ducts are used, the ratio of the large to small dimension shall not exceed 3.5.

512.5.5 Spirally Wound Ducts. Spirally wound galvanized steel or aluminum duct and fittings may be used for round or flat oval non-watertight ductwork.

512.5.6 Corrosion Protection. All HVAC system components subject to seawater spray, driving rain, or condensation including supply and exhaust system fan rooms and intake/discharge plenums shall be given corrosion protection as follows:

512.5.7 Duct sections expected to collect water from one of the above sources shall be constructed from CRES 316L.

512.5.8 Aluminum and galvanized steel surfaces shall be protected inside and out in accordance with Section 631.

512.5.9 Corner Radii. Where corner radii are required on rectangular duct, the connecting duct shall be faired in if the reduction in cross sectional area is 5 percent or greater.

512.5.10 Gaskets. Gaskets shall be used for flanged joints. Gaskets for flanged joints shall be rubber, cloth inserted, except that rubber, silicon gasket material 3.0 mm (1/8 inch) thick with a temperature rating of 218 deg. C (425 deg. F), MIL-A-A-59588 Class 2-B, 40 durometer, may be used for duct heater flanges, watertight duct flanges, and access plates.

512.5.11 Flanges. Flanges or flange corners protruding below the headroom level shall be padded or provided with a headroom flange to prevent injury to personnel. Watertight flanges shall be bolted together with a flat, continuous face, and fully welded to the duct.

512.5.12 Watertight Ducts. Welded construction shall be used for watertight ductwork. Take down joints and equipment connections shall be flanged.

512.5.13 Minimum Thickness. The minimum thickness of materials for ducts shall be as follows:

TABLE 512-2 - Sheet for Fabricated Ductwork

DIAMETER OR LONGER SIDE (INCHES)	NONWATERTIGHT		WATERTIGHT	
	GALVANIZED STEEL (INCH)	ALUMINUM (INCH)	GALVANIZED STEEL (INCH)	ALUMINUM (INCH)
Up to 6	.018	.025	.075	.106
6.5 to 12	.030	.040	.100	.140
Above 12	.036	.050	.118	.160

TABLE 512-3 - Welded or Seamless Tubing

TUBING SIZE (INCHES)	NONWATERTIGHT	WATERTIGHT
	ALUMINUM (INCH)	ALUMINUM (INCH)
2 to 6	.035	.106

6.5 to 12	.050	.140

TABLE 512-4 - Spirally Wound Duct (Nonwatertight)

Diameter (Inches)	Steel (Inch)	Aluminum (Inch)
Up to 8	.018	.025
Over 8	.030	.032

512.5.14 Blind Rivets. Aluminum blind rivets may be used in non-watertight aluminum ductwork.

512.5.15 Screws. The use of sheet metal screws is prohibited. Tapping Screws, ASME B18.6.3-2010, may be used where metal thickness allows a thread engagement equal to a standard nut of the same screw size and where the thread cutting screw does not have to be removed.

512.5.16 Ducting and Equipment Access Openings. Cleaning and inspection openings shall be provided to access all sections of the system.

512.5.16.1 On each side of duct heaters

512.5.16.2 On the air inlet side of centrifugal fans, vane turns and splitters.

512.5.16.3 At the impeller end of axial fans.

512.5.16.4 In exhaust ducts serving the machinery rooms so that all interior areas of the ducts from the exhaust inlet to the weather can be reached and cleaned.

512.5.17 Duct Hangars. Hangers for ductwork and heaters shall properly and adequately support the equipment for all static and dynamic conditions outlined in Section 070. This includes ramming of ice.

512.5.18 Flexible Duct Connections. Flexible joints shall be used on all fan duct connections and shall be flanged rubber spools. The spool material shall be at least 3/16 inch thick and at least 2 inches long in the direction of airflow. The joint shall be able to deflect sufficiently to allow free travel of equipment.

512.5.19 Balancing Provisions. System balance shall be achieved by orifice plates and/or dampers. Where dampers are used for balancing, they must be permanently affixed by riveting following balancing. Dampers must be present at every supply and exhaust terminal

512.5.20 Deck Penetrations. Ventilation ducts that pierce decks shall be watertight for a distance of 3 inches above the deck. This shall be achieved by a section of watertight pipe, or a spool welded to the deck.

512.6 CONTROL REQUIREMENTS

512.6.1 Fan motor controllers for the engine room shall be located outside the compartment in the ATON Workshop.

512.6.2 Chilled Water Cooling Coils.

512.6.2.1 Recirculating Systems. Chilled water flow through duct cooling coils, fan coil units, air handlers shall be controlled by three-way modulating control valves.

512.6.2.2 Preheater Thermostats. Preheaters shall be controlled by an enforced zero voltage firing thyristor (SCR) power controller.

512.6.2.3 Thermostat for preheater thyristor controls shall be a proportional controller. The sensing element shall be a duct mounted iron-constantan type “J” thermocouple downstream of the pre-heater. Thermostat shall be set for a constant preheat air delivery temperature.

512.6.2.4 Hot water preheaters shall use PID controllers to control a 3-way v-ball valve or direct acting globe valve.

512.6.3 Recirculating, Reheat, and Space Heat Thermostats. The thermostats shall be located in the return air stream from the space served, wherever practical. Room thermostats shall be mounted approximately 1.5 m (5 ft) above the deck in a location which is representative of the average temperature of the space. Thermostats shall be located away from sources of heat or stagnant air pockets.

512.6.4 Fan and Damper Control Interfaces and Interlocks. Controls of auxiliary equipment, such as cooling coils, electric duct heaters, or battery charging units, shall be interlocked with the fan controller of the fan serving or associated with that equipment or device such that when fan is secured the auxiliary equipment or device is secure.

512.6.5 Automatic closures shall be installed in the supply and exhaust ducting serving compartments protected by fixed firefighting systems. Fans and closures shall be interlocked to secure airflow before fixed firefighting systems are discharged.

512.7 DAMAGE CONTROL REQUIREMENTS

512.7.1 Duct Penetrations. Where ducts penetrate bulkheads and decks required to be watertight for intact or damage stability, they shall be fitted with watertight closures.

512.7.2 Watertight Closures are required as follows:

512.7.2.1 At watertight bulkhead penetrations.

512.7.2.2 Where watertight closures are required at duct penetrations of compartments, they shall be located as close to the penetration as is practicable.

512.7.3 Fire Tight Integrity. Requirements of NVIC 9-97 and MSC PRG.GEN-04 shall apply.

512.7.4 Fire Dampers. Fire dampers shall be Coast Guard Approved Type 164.139.

512.7.5 Non-manned spaces other than voids, tanks, and exterior lockers shall be provided with ventilation. Ventilation may be omitted for interior spaces without equipment or transmission loads and a deck area of less than 10 ft2.

512.7.6 The spaces containing batteries or diesel stowage shall be vent

SECTION 514 HVAC CHILLED AND HOT WATER SYSTEMS

514.1 HVAC CHILLED AND HEATING WATER SYSTEM REQUIREMENTS

514.1.1 The HVAC Chilled and Heating Water System will supply heating and cooling thermal fluid (in this section, “water” is to be interpreted as thermal fluid) in a 2-pipe configuration to HVAC services including but not limited to: duct coils, fan coil units, unit heaters, and convection heaters. Heating water will be supplied at 180°F by an Aldrich L-17W fired heater through a dedicated hot water pump, Cooling water will be supplied at 43°F and 3 gpm/ton by two Dometic 48,000 btu/h chillers through a dedicated chilled water pump. A common expansion tank will set the system pressure upstream of the fired heater and chillers as well as handle the thermal expansion and contraction of the fluid due to the variation in operating conditions. Each service will be piped in parallel branches off the main, and will contain a wye strainer, three-way modulating or two-position zone valve depending on the service, flowmeter, service valves and balancing valve(s). Fan coil units will have a dedicated balancing valve for heating and cooling mode, while preheaters, unit heaters, and convection heaters will be fitted with a single balancing valve and be secured in cooling season. When in port, heating will be provided by an immersion heater using ship’s electric power via the shore tie.

514.1.2 The system shall be as shown on the drawings.

514.2 THERMAL FLUIDS

514.2.1 The working fluid for hydronic heating systems shall be a mixture of water and glycol such as CAT ELC. If the working fluid is to be used in air intake preheaters or any other services exposed to the cutter design service temperature, the ratio of water and glycol serving that system shall have a freezing point no higher than -55 degrees F.

514.2.2 The thermal fluid shall be pre-mixed in a tank that contains not less than 200% of the system volume. The tank need not be dedicated to the thermal fluid system and may be combined with the diesel engine jacket water replenishment tank.

514.2.3 The filling system shall have an automatic shut off to ensure the system is not overfilled. Automatic shut off for the system shall be triggered via level indication or pressure indication in expansion tanks.

514.2.4 Where the expansion tanks are air over water, the filling system shall be a pump drawing suction from the fill tank. The cut-in and cut-out commands shall be via level controls allowing 2-minute pump run time, and the required volume shall be added to the required acceptance volume.

514.2.5 Where the expansion tanks are bladder-type, the filling system shall be a pump drawing suction from the fill tank The cut-in and cut-out commands shall be via pressure controls, with the cut-in pressure being the pressure required to fill the system and the cut-out pressure being the system cold fill pressure allowing for the full thermal expansion of the system without opening the relief. The pre-fill of the expansion tank between these two pressures shall be added to the acceptance volume when sizing the tank. This volume shall be based on a minimum pump run time of 2 minutes. Alternately, a hydropneumatics charging tank sized for a 2-minute run time may be installed, with a pressure regulating valve set at the set at the cold fill pressure feeding the expansion tank.

514.2.6 The injection point of the filling system to each system shall be to the expansion tank or suction side of the pump.

514.2.7 The fill pump shall be sized to have a max fill time of 1 hour.

514.3 SYSTEM DESIGN

514.3.1 The smallest size pipe used shall be 1/2-inch NPS.

514.3.2 Maximum velocities in piping shall not exceed the following:

514.3.2.1 Freshwater above 140°F: $3\sqrt{D}$ up to 2.5 ft/sec.

514.3.2.2 Freshwater in chilled water service: $5\sqrt{D}$ up to 8 ft/sec.

514.3.2.3 Freshwater solutions above 140°F with propylene glycol or ethylene glycol: $3\sqrt{D}$ up to 3 ft/sec.

514.3.2.4 Freshwater solutions in chilled water service with propylene glycol or ethylene glycol: $5\sqrt{D}$ up to 8 ft/sec.

514.3.2.5 Where V is the velocity in feet per minute and D is the actual inside diameter of the pipe or tubing in inches.

514.3.3 Each heat exchanger shall be isolatable from the secondary circuit to allow for maintenance and cleaning as needed, without affecting system operation.

514.3.4 20-mesh line strainers shall be installed in supply piping to equipment and components where protection is not provided by integral strainers.

514.3.5 Hydronic heating operated equipment and automatic controls shall be provided with cutout valves and union fittings (or flanges). Union (or flange) end valves shall be used.

514.3.6 The hydronic heating piping installation for each heating consumer shall include supply and return isolation cutout valves, a 20-mesh wye strainer, a 3-way zone valve, and a constant flow fitting in the return piping downstream of the bypass tie-in. A venturi-style mechanical flowmeter shall be installed local to the coil to read flow rate at the coil without including bypass flow.

514.3.7 Balancing shall be accomplished by an automatic flow control device, Hays commercial marine Mesurflo or equal.

514.3.8 Automatic balancing valves shall be installed such that no configuration of the mixing and/or manual bypass valves may bypass the flow control valve.

514.3.9 Large services including air handling units and all ducted heating coils shall utilize a modulating, fail-in-place 3-way v-ball valve for zone control. The v-ball valve shall be sized such that its head loss is 50% of that of the entire branch, excluding the balancing valve. Modulating valves shall be configured for an equal percentage characteristic curve.

514.3.10 Small stand-alone services such as unit heaters and convection heaters may utilize a 3-way 2-position solenoid or motor operated valve, or a 3-way modulating valve.

514.3.11 Thermal fluid consumers shall be fitted with cutout valves and take down joints arranged such that any component can be removed without disrupting system operations.

514.3.11.1 Circulator pumps shall be cast iron, Bell & Gossett Series PL or NRF or equal. A separate circulator pump shall be provided for heating and cooling modes, sized for the required flow rates.

514.3.11.2 The circulator pump shall be capable of supplying flow as required in Section 512 with a minimum pressure drop of 20 psid across the constant flow fitting in the most hydraulically remote loop.

514.3.11.3 The pumps shall have a recirculation line installed between the pump supply and return within a loop to protect the pump from total shut-off. The recirculation line shall be sized to pass 5% of the heating water flow, or that volume which is required by the pump manufacturer, whichever is greater.

514.3.12 Compression or bladder type expansion tanks shall be provided for the system. Each expansion tank shall be sized to accommodate the thermal expansion of the total volume of fluid in its portion of the system over the entire temperature gradient possible. The inlet and outlet connections of each tank shall be connected to the hydronic system via locked open cutout valves. Filling and make-up to the expansion tanks shall be from the dedicated fill system.

514.3.13 Expansion tanks shall be installed upstream of the fired heaters. Pumps shall be installed downstream of the fired heaters.

514.3.14 The hydronic heating system shall be provided with permanently installed flow meters for each supply to the main. Each meter shall have a measurement capacity of 30 to 120 percent of the design flow rate and shall read in gal/min. Meter assemblies shall be provided with flanged or union ends.

514.3.15 Drains shall be provided at every system low point and shall be capable of draining the entire hydronic heating system. Drains shall be provided with a ball or gate valve and a cap or plug. The hydronic heating system shall drain to the contaminated waste water tank or portable storage containers. If portable storage container(s) are provided, the drain end shall be fitted with a garden hose thread fitting. If drained to the contaminated waste water system, draining shall be via an air gap and open funnel.

514.3.16 Air vents shall be provided at system local high points to ensure that the system does not become air bound.

SECTION 516 REFRIGERATION SYSTEMS

516.1 GENERAL

516.1.1 HVAC chilled water will be a mixture of water and glycol provided at 45°F by two Dometic MCGX 48,000 btu/h chillers operating at 3.0 gpm/ton or equal. The chillers will be staged in a lead-lag configuration to provide capacity control. Refrigeration of food stores will be provided by a standalone reefer/chiller unit.

516.2 OPERATION.

516.2.1 Pressure Relieving Systems. Where required by ASHRAE Standard 15, each refrigerant circuit shall be provided with a pressure relieving system. Each pressure relieving system shall consist of a high-pressure rupture disc assembly, a refrigerant relief valve, a low-pressure rupture disc, and associated piping. Refrigerant overboard discharge shall be piped to a main weather deck. The discharge shall terminate in a safe location where refrigerant cannot be picked up by HVAC systems or where water can be trapped and collect against a rupture disc. Piping arrangement shall prevent seawater corrosion of the rupture discs. Refrigerant relief piping from multiple units in the same compartment may be combined. The required capacity for refrigerant relief piping shall be the sum of all connected relief valve capacities.

516.2.2 Relief valves. Relief valves and refrigerant relief piping shall be provided and sized in accordance with ASHRAE Standard 15. Relief valves shall be brass or bronze body with corrosion resistant internal parts.

<u>516.3</u> <u>WATER CHILLER PLANTS</u>

516.3.1 Quantity. The chilled water plant shall consist of two chillers. The chiller shall be a Dometic MCGX or equal, minimum capacity rating 48,000 btu/h at 3.0 gpm per ton, 45F leaving water temperature, and condenser entering conditions as described herein.

516.3.2 Plant controls.

516.3.2.1 Control centers - The water chiller plant shall be provided with means for local control and monitoring at the plant. Items such as pressure and temperature gauges which have piping or capillary tubing shall be mounted at the front of the cabinet on an adjacent gauge board. Controls for analog refrigeration condensing units shall be mounted on the front control cabinet. Control centers shall provide for sequential lead-lag control of chiller units. The “lead” and “lag” chillers shall be selectable at the control center.

516.3.2.2 Motor control centers and/or lead-lag controls shall include a front-mounted LOCAL-OFF-AUTOMATIC selector switch with visual indicators (lights) for operating the unit. This switch shall provide for operating the unit locally independent of the master controllers having the unit and its auxiliaries secured or having the unit operate in conjunction with the master controllers.

<u>516.4</u> <u>SELF-CONTAINED EQUIPMENT</u>

516.4.1 Self-contained equipment shall be defined to include reach-in chill and freeze boxes (internal gross volume of less than 100 cubic feet each), ice machines, beverage dispensers, compressed air driers, and similar small refrigerated units. New self-contained equipment shall use refrigerants that are hydrofluorocarbon (HFC) or HFC/HFO (hydrofluoroolefin) blends, are non-ozone depleting, have a Global Warming Potential < 2500, and are classified as Safety Group A1 or A2L in ASHRAE 34. In addition, refrigerants must be listed in 40 CFR 82 as an acceptable refrigerant under the Environmental Protection Agency Significant New Alternatives Policy Program (SNAP).

516.4.2 Refrigerators and freezers shall be of commercial marine quality meeting National Science Foundation (NSF) Standards for Food Service, Standard 7 (Food Service Refrigerators and Storage Freezers). Food service equipment cited by ASTM designation shall be in full compliance with any supplemental requirements defined in this specification.

SECTION 517 FIRED HEATERS AND OTHER HEAT SOURCES

<u>517.1</u> <u>GENERAL</u>

517.1.1 This section contains the requirements for the cutter heating plant, which will consist of an Aldrich L-17W fired heater to provide 180°F hot water at approximately 12 gpm while underway and a Chromalox 50kW electric circulation heater while in port and connected to the shore tie. The heating plant provides heat to vessel HVAC heating equipment.

517.1.2 Electrical power and ship service air provided to the heating plant, as required, shall be arranged so that the failure of a fired heater, electric power supply panel or air supply pipe will not disrupt operation of other fired heaters or electric heaters.

517.1.3 Temperature and pressure shall not exceed the design values of equipment served.

517.2 FIRED HEATERS

517.2.1 Design Standards – Fired heaters shall be certified to ASME Boiler and Pressure Vessel Code Section IV.

517.2.2 The fired heater shall be an Aldrich Model L-17W oil-fired vertical firetube unit operating at 180F (cut-in) to 185F (cut-out).

517.2.3 The system shall be compliant with ASME CSD-1 and UL 726.

517.2.4 Controls.

517.2.4.1 Fired heaters shall be interlocked with the watermist fire protection system serving the compartment.

517.2.4.2 Fired Heaters shall be equipped with a "first-out" safety alarm and indicating system which will shutdown the unit in the event any condition exists which could cause damage to the equipment, sound an audible alarm, and give a visual indication for the initial cause for shutdown. The alarms shall sound locally and in the Pilot House.

517.2.4.3 Hydronic heating coils and heat exchangers shall be as shown on the drawings.

517.3 ALTERNATE HEAT SOURCE.

517.3.1 An immersion heater shall be provided in parallel with the fired heater(s) to supply heat when in port and connected to shore power. The immersion heater shall be a Chromalox NWH-18-050P-E4 240V-3P50kW circulation heater or equal.

517.3.2 The immersion heater shall be controlled by an enforced zero voltage firing thyristor (SCR) power controller. The thermostat for the thyristor controls shall be a proportional controller. The sensing element shall be mounted in the pipe 5 diameters downstream of the heater. Thermostat shall be set for a constant water delivery temperature.

517.3.3 The immersion heater shall be equipped with safety controls in accordance with ASME CSD-1 including but not limited to a secondary manual reset temperature switch.

SECTION 520 SEAWATER SERVICE

520.1 GENERAL

520.1.1 A 3 HP AMPCO ZC2 series pump provides 40 GPM at 38 PSI to the 90-10 copper-nickel piping system, which utilizes either silbraze or press-connect fittings. The system supplies two 1-inch rigid hose reels located fore and aft of the pilothouse and a 3/4-inch connection at the working deck for deck washdown.

520.1.2 The system shall be a dry type capable of being charged from the Pilothouse or the local pump control with no other actions required. The seawater service system shall consist of one power driven pump and shall service two hose stations.

520.2 PUMP

520.2.1 The pump shall be an Ampco Model ZC2 1.5 x 1.25 – 5.375” impeller with a 3 HP, 3500 rpm motor producing 40 gpm at 38 psi, or equal. Pump construction shall be nickel aluminum bronze or duplex 2205 stainless steel. A pressure gauge shall be installed in the pump suction and discharge.

<u>520.3 PIPING</u>

520.3.1 Arrangements - The piping shall be pitched to a common drain. The drain will gravity drain the entire system without the removal of components. The system is intended to remain dry when not in operation.

<u>520.4 HOSE STATIONS</u>

520.4.1 Hose stations shall be located as shown on the drawings.

<u>504.1 PORTABLE PUMP</u> – SECTION UNDER DEVELOPMENT

SECTION 528 PLUMBING AND DECK DRAINS

<u>528.1 GENERAL</u>

528.1.1 All plumbing drains and interior deck drains will direct water to sewage/graywater tanks with a 500-gallon capacity. Condensate from AC cooling units will be routed through water seal traps to a 7.5-gallon condensate sump tank and discharged overboard at least 12 inches above DLWL.

528.1.2 Gravity drains from galley sink, wash basins and shower shall be lead through water seal traps to a Sewage/Gray water holding tank. Gravity drain piping shall be installed in accordance with the International Plumbing Code as modified herein, with a minimum slope of 1/4 inch per foot. Waste water sump tank and pump systems shall be installed, see Section 593 for waste water sump tank and discharge pump.

<u>528.2 PLUMBING AND DECK DRAINS</u>

528.2.1 Plumbing drains consist of sewage drains and gray water drains as defined in Section 593.

528.2.2 For requirements on vacuum flush toilets and other sanitary fixtures, see Section 593.

528.2.3 Gravity drains shall be a minimum of 1 1/4 inches NPS.

528.2.4 The design of gravity plumbing drains shall provide drainage under conditions of list to either side under service conditions. Sewage/gray water holding tank mounting and piping connections shall be made so that the piping can be disconnected and the tank removed for cleaning without destruction of the foundation or any piping.

528.2.5 Air conditioning cooling unit(s) condensate shall be lead through water seal traps to condensate sump tank(s). Each condensate sump tank shall have a 7.5-gallon capacity and be made of aluminum or polyethylene. Each sump tank shall be cleanable and vented to the atmosphere.

528.2.5.1 Pump(s) and piping components, shall be provided to allow for automatic discharge of air conditioning unit(s) condensate overboard.

<u>528.3 TRAPS AND CLEAN-OUTS</u>

528.3.1 Each fixture trap shall have a readily accessible means for cleaning. No fixture shall be double trapped. Traps shall be installed in a fore-and-aft direction. Each trap shall have a water seal of not less than two (2) inches nor more than four (4) inches.

528.3.2 Clean-outs. Clean-out fittings shall be installed in accessible locations to permit cleaning drain pipes. Clean-outs shall be installed at each change of direction greater than 90 degrees. Clean-outs shall be the same nominal size as the pipe served. Clean-outs for drains shall be installed with a minimum clearance of 12 inches for rodding.

<u>528.4 PLUMBING VENTS</u>

528.4.1 Protection of trap seals from siphonage or back pressure shall be accomplished by air admittance valve.

SECTION 529 DRAINAGE AND BALLASTING SYSTEMS

529.1 GENERAL

529.1.1 The purpose of the Bilge system is to provide the capability of emergency dewatering in the case of flooding. The system shall also be capable of pumping out bilge contents to a shoreside facility. The emergency dewatering shall be accomplished via Rule 4000 bilge pumps. A portable pump will be utilized for pumping bilge water to shore facilities.

529.1.2 This section contains additional requirements for drainage and ballasting systems.

529.2 DRAINAGE SYSTEM

529.2.1 Pumps shall be capable of local control

529.2.2 Each bilge pump shall be located so that it takes suction from the lowest practical point of the bilge space.

529.2.3 Switches shall be in one location within the Pilothouse arranged for easy operation while minimizing the risk of accidental operation. The switches shall provide for a manual on/off and automatic operation.

529.2.4 The Pilothouse shall drain water either to a bilge pump location or directly overboard through a check valve.

529.3 BALLAST SYSTEM - SECTION UNDER DEVELOPMENT

529.3.1 The forepeak Ballast Tank shall be filled and un-filled pier side using pier resources.

SECTION 530 AUXILIARY FRESHWATER COOLING

530.1 GENERAL

530.1.1 The freshwater cooling system utilizes Alfa Laval T-6 PFG, or equal, to freshwater heat exchangers and circulates freshwater to cool auxiliary equipment. The freshwater piping consists of schedule 10 stainless steel piping gr304l or 316l, stainless steel buttwelds, socketweld or press connect fittings, and stainless steel valves. Seawater is drawn from keel cooler pockets serving as sea chests, pumped through 90/10 copper nickel piping with bronze silbraze or copper nickel welded fittings to the heat exchanger then discharged over the side.

530.2 HEAT EXCHANGERS

530.2.1 Piping shall be arranged to allow the heat exchanger to be isolated and serviced without disturbance to the connected piping systems. Where this is not possible, take down joints shall be provided between the heat exchanger and cutout valves.

530.2.2 Heat exchangers fitted with an internal strainer shall be mounted such that the strainer can be removed without interference or introducing the strained materials back into the cooler during removal. The strainers shall be located such that personnel can access and clean the baskets while standing on deck plates without the use of ladders or stools.

530.2.3 Each heat exchanger shall accomplish temperature regulation via a 3-way temperature regulating valve configured to bypass water around the heat exchanger. Means of completely isolating flow to the heat exchanger when the equipment is not calling for cooling is required and must be accomplished without personnel intervention unless otherwise approved by USCG.

530.2.4 Plate type heat exchangers for seawater service shall be constructed using titanium plates in accordance with ASTM B265. This standard applies for seawater plate type heat exchangers, both propulsion and auxiliary systems.

530.2.5 Seawater cooled plate heat exchangers nozzles that are in contact with seawater shall be titanium. The seawater inlet and outlet piping serving the heat exchanger shall be galvanically isolated from the titanium plates with non-metallic hoses or non-conductive rubber flex couplings at each fluid connection.

530.2.6 Plate type heat exchangers for closed-loop freshwater service shall be constructed using stainless steel plates. This standard applies for freshwater plate type heat exchangers, both propulsion and auxiliary systems.

530.3 FRESHWATER COOLING SUPPLY REQUIREMENTS

530.3.1 General Requirements of Freshwater Cooling Systems

530.3.1.1 Where a fixed orifice plate is required, it shall be installed downstream of the heat exchanger and shall not be directly connected to the heat exchanger flange.

530.3.1.2 High points in each cooling loop shall be fitted with automatic vent valves or a deaerator system to purge air from the system.

530.3.2 Head Tanks

530.3.2.1 For each loop, a head tank (air over water compression or bladder type) shall be provided. Each head tank shall be sized to accommodate the thermal expansion of the total volume of fluid of the system over the entire temperature gradient possible, to include ambient space temperatures.

530.3.2.2 Head tanks shall be sized in accordance with ASHRAE Handbook HVAC Systems and Equipment to accommodate the system fluid expansion from minimum to maximum fluid temperatures.

530.3.2.3 The tank location shall be accessible for routine inspection.

530.3.2.4 Systems utilizing compression type head tanks shall be provided with alarms to indicate low water level.

530.3.3 Additives.

530.3.3.1 Additives shall be specifically accounted for in system design, as reduced fluid heat transfer coefficients relative to water will reduce system cooling capacity. Other additives are subject to USCG approval. Additives shall not be added with a dedicated hard piped connection to the system.

530.3.3.2 The coolant shall be a corrosion inhibiting solution that provides protection against freezing at temperatures down to -35°F and in alignment with manufacturers recommendations.

SECTION 533 POTABLE WATER SYSTEMS

533.1 GENERAL

533.1.1 The Potable Water Systems stores and distributes hot and cold potable water to plumbing fixtures and a window washing system. Two potable water tanks each with a capacity of 250 gallons are configured to receive potable water from a shore tie filling line. The Potable Water System includes a 20-gallon pressurized tank and pump that is capable of the delivery of water at a minimum pressure of 20 psi at shower heads and as recommended by equipment vendors, with the system operating at design capacity. One instantaneous water heater is sized to provide 120°F water to sinks and shower.

533.1.2 The freshwater systems shall be constructed in accordance with ABYC H-23, except as modified herein.

533.1.3 Compression tanks. Unless otherwise specified, compression tanks shall be of commercial marine grade. Each tank shall be fitted with a gauge glass, relief valve, one (1) combined ANSI flanged inlet and outlet connection, cutout and drain valves, air charging connection, a connection for pressure switches (as required), a pressure gauge and other fittings necessary for each complete system. Tanks shall be in accordance with ASME Boiler and Pressure Vessel Code, Section VIII, Division 1.

533.1.4 Exterior fresh water piping, including hose valves, shall be equipped with a cutout valve and drain valve. Cutout and drain valves shall be installed within a heated compartment so that exterior fresh water piping can be drained in freezing temperatures

533.1.5 Potable water components and installation shall be certified lead free and compliant with ANSI/NSF 61 or ANSI/NSF 372.

533.2 POTABLE WATER STORAGE

533.2.1 Storage tanks.

533.2.1.1 There shall be two (2) potable water tanks and they shall be approximately equal capacity. They are non-integral tanks made of stiffened composite or aluminum material.

533.2.1.2 Sinks and instant hot water feeds shall have replaceable water filter(s).

533.2.1.3 Only sealing compounds that conform to SAE AMS8660 and NSF/ANSI Standard 61 shall be used.

533.2.1.4 Potable water outlets, connections, and vents shall be provided with protection against siphonage and backflow of contaminants.

533.2.1.5 Each potable water storage tank shall have a tank level sensing system.

533.3 SERVICE PUMP

533.3.1 Pump shall be Par-Max model Q402J-115S-3A or equal.

533.3.2 Pump shall have a disconnect switch nearby and shall be equipped with a pressure switch that starts the pump at 20 psi and stops the pump at 40 psi.

533.4 PRESSURE TANK

533.4.1 Pressure tank shall have an FDA approved vinyl bladder or polypropylene liner within a welded steel shell. Tank shall be sized to a minimum 1.5 minute pump cycle time to prevent short cycling. Pressure tank shall be pressurized 2-4 psi below minimum cut-in pressure Amtrol Model WX-202, or equal, meets these requirements.

533.5 WATER HEATER

533.5.1 Water heater shall be 16kw instantaneous hot water unit, Hubbell model MTX016-3T or equal.

SECTION 541 FUEL SYSTEMS

541.1 GENERAL

541.1.1 The fuel oil system is constructed of Schedule 10 316L stainless steel pipe with 316L stainless steel fittings and valves. A booster pump takes suction on a header connected to each fuel oil tank and discharges to a common header supplying each piece of fired equipment in the Machinery Room. A return line allows fired equipment to send fuel back to the fuel oil tanks.

541.1.2 The fuel system shall be designed in compliance with ABYC standard Section H-33.

541.2 FUEL FILLING SYSTEM

541.2.1 The filling system shall be designed such that piping and components are sized to receive a minimum of 100 gal/min at the deck filling connections.

541.2.2 The deck filling connection shall be 2-1/2 inches and located as shown on the drawings and shall terminate in a blind flange. A containment coaming shall be provided around the deck connection and be drained to an overflow tank via a threaded plug.

541.2.3 The fuel tank combined vents and overflow shall be collocated with the deck filling connection in the common containment coaming area.

541.2.4 The containment area shall have a drain plug as low as possible in the side for draining seawater/rain water.

541.2.5 The drain plug shall be installed in the drain hole to the overflow tank when not in use and shall be fitted with a keeper chain.

541.3 FUEL SERVICE SYSTEM

541.3.1 Fuel service system shall supply fuel to fired equipment and return excess fuel to the storage tanks. Piping and components shall be sized to provide for operation of the propulsion engines and generators from any tank. The engine fuel service system shall take suction from a header cross-connected to all tanks through independent tailpipes. The fuel supply line to each engine shall be provided with equipment as necessary to maintain fuel oil pressure at the engine manufacturer's requirements.

541.3.2 A Fass model IND250G pump, or equal, shall be installed as a booster pump to supply fuel to diesel engines.

541.3.3 A quick closing valve and cutout valve shall be installed at the tank boundary connection with the booster pump suction pipe. The quick closing valve shall be capable of being operated locally and remotely.

541.3.4 A cutout valve shall be provided in the pump discharge header between the connections of each engine.

541.4 TANKS

541.4.1 Each fuel tank shall have a tailpipe for fill/transfer and a tailpipe for stripping.

541.4.2 Tank liquid level indicating systems readings shall be displayed both in the Pilothouse and in the Engine Room.

541.4.3 A high-level audible and visual alarm shall be provided for each tank, set to actuate at 95% of tank capacity. A low-level audible and visual alarm shall be provided for each tank, set to actuate at 25% of tank capacity. Alarms shall actuate both locally and in the Pilothouse and shall be able to be silenced through acknowledgment of the alarm.

541.4.4 An overflow tank with a minimum capacity of 50 gallons shall be installed.

541.4.5 A high-level audible and visual alarm shall be provided for the overflow tank, set to actuate at 10% of tank capacity. Alarms shall actuate both locally and in the Pilothouse and shall be able to be silenced through acknowledgment of the alarm.

541.4.6 TANK STRIPPING

541.4.6.1 Each tank shall have a stripping suction line, hard piped in the tank, connected to a common stripping header to support use of a portable pump operating at a minimum of 10 gpm.

541.4.6.2 The common stripping header line shall be fitted with a ball valve at the tank boundaries and at the portable pump connection.

SECTION 555 FIRE EXTINGUISHING SYSTEM

555.1 PORTABLE FIRE EXTINGUISHERS

555.1.1 Portable fire extinguishers shall be installed in accordance with and meet the requirements of ABYC A-4, Fire Fighting Equipment. The number and location of portable fire extinguishers shall be in accordance with ABYC A-4 Table 2, except as modified herein. At least one portable extinguisher shall be USCG Type 20-B.

555.1.2 Fire extinguishers shall be mounted so that they are secure but readily accessible with a quick and positive release for immediate use. Fire extinguishers shall be located so that they do not hinder crew movement and operation.

555.1.3 Fire extinguisher mounting brackets shall be marine type, heavy-duty, made of corrosion-resistant metal and appropriately sized to the extinguisher.

555.1.4 All portable fire extinguishers shall have an inspection tag installed on the device.

555.2 FIXED FIRE SUPPRESSION SYSTEM

555.2.1 The fixed firefighting system serving the machinery space shall be a Marioff MAU50 pre-engineered water mist system, or equal. The system shall be installed in accordance with the type-approved Design, Installation, Operation, and Maintenance manual (DIOM).

555.2.2 The fixed fire suppression system shall have at least two manual actuation stations: one in the main egress route outside the engine room spaces, and another in a conspicuous location in the pilothouse.

555.2.2.1 The fire suppression system(s) shall not be capable of automatic discharge.

555.2.2.2 Emergency stop stations shall be provided for ventilation systems in accordance with IEEE-STD-45 and 46 CFR, Subchapter J. Where fire extinguishing systems are provided for compartments with mechanical ventilation, an interlock shall be provided to stop the applicable ventilation fan and sound an alarm, as required by the particular fire extinguishing system

555.2.2.3 Upon activation, automatic switches shall: Shut down engines, gensets or other equipment that processes or consumes diesel in the protected space.

555.2.2.4 Energize electrical warning indicators, and audible/visual alarms; and

555.2.2.5 Secure ventilation fans serving the protected space and closing any ventilation dampers

SECTION 556 HYDRAULIC OIL SYSTEMS

556.1 GENERAL

556.1.1 Cranes and exposed deck machinery shall use an environmentally acceptable lubricant (EAL) in accordance with EPA 800-R-11-002.

556.1.2 The contamination level for hydraulic systems shall be at least ISO 4406 code 20/18/15 or in accordance with manufacturer specifications, whichever is more stringent.

556.1.2.1 Cleanliness shall be achieved and maintained in accordance with ISO 4406. Cleaning by mechanical means, followed by dry air blast, shall be used where cleaning of pumps, motors, rams, valves and other components and accessories become necessary.

556.2 SYSTEM ARRANGEMENTS

556.2.1 IA hydraulic system shall be used for powering the crane.

556.2.1.1 Hydraulic circuits shall be situated below the working deck.

556.2.1.2 Hydraulic piping deck penetrations shall be within 12 inches of the equipment served.

556.2.1.3 Piping installed in the weather shall be protected from accidental impact damage.

556.2.1.4 Hydraulic system components shall be provided with a drip pan or deck coaming to contain hydraulic oil incidental to use, leakage and maintenance of the units.

556.2.1.5 Isolation valves shall be installed as close as practical to each piece of equipment and each hydraulic pump.

556.3 SYSTEM PERFORMANCE

556.3.1 The temperature of hydraulic fluid shall be maintained within the operating limits of the crane OEMs. Systems shall support continuous service at full flow and at least three (3) hours of service at no flow conditions without overheating.

556.4 HYDRAULIC OIL REQUIREMENTS

556.4.1 All hydraulic systems shall use the same oil.

556.4.2 Stowage shall be provided onboard for five (5) gallons of hydraulic oil.

556.5 HYDRAULIC PUMP

556.5.1 Hydraulic pumps shall be marine grade in accordance with SAE J1776-2014.

556.5.2 The net positive suction head available under expected conditions of operation shall meet the pump manufacturer's recommendations.

556.5.3 Hydraulic pumps shall be driven by electric motors. Pumps shall be capable of continuously variable output from zero to full power and shall be configured to start unloaded.

556.6 PIPING SYSTEMS

556.6.1 Hydraulic piping systems shall include connections to allow flushing in accordance with ISO 12669.

556.6.2 Fittings installed to connect rigid piping shall be welded.

556.6.3 Piping may utilize tubing and swaged marine fittings in lieu of welded fittings . Where tubing fittings are used, threaded tubing connections shall be SAE J1453 straight thread with "O" ring seal.

556.6.4 Where equipment is furnished with other than SAE J1453 interfaces, SAE J514 37 degree flare fittings may be used.

556.6.5 Tapered pipe threads and 45 degree flare fittings shall not be used.

556.6.6 Take-down joints and connections to equipment or components shall be flat-face O-ring seal union type, or SAE type 4-bolt flanges with O-ring seal shall be used.

556.7 VALVES

556.7.1 Adjustable valves shall be lockable at service adjustment.

556.7.2 Manifolded valves shall be individually replaceable. Valves within junction blocks shall be removable cartridge type only; junction blocks with non-replaceable valve body elements are not permitted.

556.7.3 Pressure control valves (including unloading, backpressure, and sequence valves) and check valves shall be damped.

556.7.4 Relief valves shall begin opening at 110% of operating pressure and be set by a flow test of at least 80% of expected flow. The relieving capacity shall not be less than full pump flow with a maximum pressure rise in the system of not more than 10% of the relief valve setting.

556.7.5 When relief valves are set, tags shall be attached to the valves which include the following information:

556.7.5.1 Date set

556.7.5.2 Name of activity and person conducting or witnessing test

556.7.5.3 Setting and test conditions

556.7.5.4 Test stand: Flow rate used for setting pressure, i.e., (3450 lb/in2 at 20 gal/min)

556.7.5.5 Hand pump: Cracking pressure and describe test conditions

556.7.6 Spool Requirements

556.7.6.1 Control valve spools for deck equipment shall be high strength stainless steel, and the spools shall be in a totally encapsulated/flooded assembly.

556.7.6.2 The thermal growth characteristics and resulting change in leakage characteristics shall be considered in the selection.

556.7.6.3 Control valve leakage under start-up and service thermal conditions, shall be considered in the selection of control valve lapping tolerances.

556.7.6.4 Control valves with through flow bypass shall have mastering chamfers on both working and bypass spool edges.

556.8 FILTERS AND STRAINERS

556.8.1 Filtration shall be provided as indicated below:

556.8.1.1 Strainers in pump suction lines from reservoirs.

556.8.1.2 Filters in the discharge lines of pumps except:

556.8.1.2.1 Filters in the discharge of hand pumps are not required.

556.8.1.2.2 Filters in the discharge lines of pumps in reversible flow service are not required, provided that alternative systematic filtration of the circulating fluid acceptable to the pump and equipment manufacturers and suitable for the service intended.

556.8.1.3 Pumps or motors with case drains shall employ either case drain or return line filters.

556.8.1.4 Filters in return lines for systems serving more than one compartment or more than five actuators.

556.8.1.5 Filters in supply lines to contaminate sensitive components (e.g. servo valves and pilot control valves) unless such components are in close proximity to the main supply line filter.

556.8.2 Strainer Selection. Suction line strainers shall be cleanable wire mesh strainers and shall be removable from the reservoir without disassembly. The elements shall be 50 mesh or finer if fluid returning to the reservoir is filtered by a return line filter, or 100 mesh or finer if fluid returning to the reservoir is not subject to return line filtration. Suction line strainers with a mesh finer than 200 shall not be used.

556.8.3 Filter Selection

556.8.3.1 Filters shall be selected in accordance with SAE J2333 and applicable NFPA standards.

556.8.3.2 Filter flow rate capacity shall be based upon peak flow rates and fluid viscosities at both normal and cold-iron start up condition.

556.8.3.3 Filter housings in low pressure, case, and drain line services shall have a bypass relief valve. Filter housings for use in pressure filtration shall not be equipped with a filter bypass.

556.8.3.4 Filters shall be equipped with an indicator or differential pressure gauge to indicate when servicing is required. Mechanical pop-up indicators shall not be used in applications where the pressure loss across the filter may vary significantly due to changes in fluid temperature or flow rate.

556.8.3.5 Filters shall be readily accessible and filter elements shall be removable for servicing without disconnecting the attached pipe or dismounting the filter housing from its foundation. Filter assemblies shall be arranged for viewing of the element indicator or differential pressure gauge by watchstanding personnel.

556.8.3.6 Filters, bowls, and housings shall be equipped with drains or other devices which will permit changing of elements with minimum spillage of fluid. If this is accomplished by drain ports, the installation shall be such that a container can be used to catch fluid from the drain port.

556.8.4 Filtration elements internal to valve or manifold bodies shall not be used. Where such filters are normally supplied by the equipment manufacturer for protection of contaminate sensitive elements, external filtration shall be provided.

556.9 RESERVOIRS

556.9.1 Reservoirs shall have quick disconnect fittings and cutout valves for attachment of the portable filtration cart suction and discharge lines for fluid transfer and service.

556.9.1.1 Reservoirs shall be equipped with drain valve configured for gravity drainage via hose to portable containers.

556.9.1.2 Reservoirs shall only be filled via filter cart.

556.9.1.3 Quick-disconnect fittings shall be provided with attached plastic plug or cap to minimize contamination of the fitting.

556.9.2 Hydraulic system reservoir shall have an additional 15 percent additional capacity over the calculated minimum reservoir volume.

556.9.3 Reservoirs shall have an exterior, magnetic float type tank level indicating system with continuous reading colored magnetic flag indicators similar to ASTM F2044.

556.9.4 Information plates shall be installed adjacent to each fluid filling opening identifying the system, fluid and filtration level.

556.10 TEST FITTINGS.

556.10.1 Test fittings shall be provided in the supply and return lines of each circuit.

556.11 CONTROLS AND MONITORING

556.11.1 Control systems shall not be wirelessly accessible.

556.11.1.1 Systems shall have equipment for wireless access physically removed.

556.11.1.2 Control systems shall be physically secured with a locking enclosure.

556.11.2 A panel shall be provided in the same compartment as the hydraulic pump and include the following:

556.11.2.1 Pressure and temperature gauges necessary for the operation of the unit.

556.11.2.2 On/Off control and status indication of reservoir heaters.

556.12 ALARMS

556.12.1 The following alarm conditions shall be monitored. A summary alarm shall indicate in the Pilothouse and individual alarms on a HPU Control Panel.

556.12.1.1 Hydraulic oil high temperature

556.12.1.2 Reservoir low level

556.12.1.3 Circuit high pressure

556.12.1.4 Hydraulic oil cooler high differential pressure

556.13 INSTRUMENTATION

556.13.1 Pressure transducers shall be installed in hydraulic pump outlets and in return lines.

556.13.2 Thermal transducers shall be installed in hydraulic reservoirs and return lines.

556.13.3 Thermal and pressure transducers shall display on local control stations and support remote indication.

556.14 SPECIAL TOOLS.

556.14.1 Special tools necessary for maintenance shall be provided.

556.15 EAL VACUUM DEHYDRATION CART

556.15.1 A combination filter and water removal cart shall be provided for maintenance and decontamination of reservoirs using the selected EAL and marked with its oil type.

556.15.2 A cart and cart storage shall be provided in the same space as the centralized HPU. The cart may be permanently mounted.

556.15.3 The vacuum dehydration system shall meet the following salient characteristics:

556.15.3.1 Compatible with the selected EAL.

556.15.3.2 Filtration: Eight (8) microns or less.

556.15.3.3 Water Removal: 100% free, 100% emulsified, dissolved < 50 ppm.

556.15.3.4 Particulate Removal: ISO cleanliness less than 15/13/12.

556.15.3.5 Minimum flow rate of four (4) gal/min.

556.15.4 The vacuum dehydration cart shall be operational with 115 VAC, 60 Hz supplied power.

556.15.5 The vacuum dehydration cart shall have dimensions that are able to pass through watertight doors and hatches as specified herein.

556.15.6 Quick disconnect hoses at least ten (10) feet in length shall be provided for supply and return. The quick disconnect fittings shall be common to the fittings installed on the reservoirs. Storage for the hoses shall be provided on the cart. An additional suction hose with suction fitting for drums shall be provided and stowed.

SECTION 561 STEERING SYSTEMS

561.1 GENERAL

561.1.1 The steering system provides a means for the crew to turn and steer the vessel with either wheel steering on the bridge, or via a follow-up lever in an aft steering station. The helmsman's movement of the wheel or lever is relayed and amplified via a power unit and hydraulic cylinders to accordingly turn the rudders. The rudders can be turned a maximum of 35 degrees to port or starboard, and a feedback system relays the rudders' position back to the helmsman with station mounted indicators.

561.1.2 Steering system shall be a power assisted hydraulic system capable of actuation from the pilothouse forward and aft consoles. Wheel steering shall be provided in the forward helmsman console. Aft steering station shall include an electric follow-up lever type steering mounted on the console.

561.1.3 Steering system shall move the rudder from 35 degrees port to 35 degrees starboard in no more than six seconds and hold the rudders at any angle, with the boat going ahead or astern at maximum boat speed.

561.1.4 Rudders shall be inter-connected with a mechanical tie-rod linkage with adjustable spherical rod ends. The linkage shall be arranged to permit adjustment of rudder toe in and out. Mechanical stops shall be fitted against each tiller arm to limit rudder movement beyond 36 degrees port and starboard. Actuator(s) shall be arranged to permit adjustment of rudder(s) neutral position. Stops shall be welded in place after final alignment of the rudders and tie rod. Rod end adjusters shall have double nuts to prevent movement after final alignment.

561.1.5 The port and starboard rudder steering gear shall have permanent witness marks that indicate rudder position. Marks shall indicate the neutral rudder angle and each 5 degree increment of rotation up to 35 degrees from the neutral rudder angle. The neutral rudder angle (zero) and each ten degree increment shall be labeled with 1/4 inch numbers. The neutral angle shall include toeing. The rudder amidships angle (rudders aligned with the centerline) shall be scribed on both rudder stocks, tillers and steering gears at assembly and marked with centerline symbol in 1/4 inch letters.

561.1.6 Hydraulic steering system shall be completely separate from the hydraulic buoy handling system.

561.1.7 Tiller actuation shall be by hydraulic cylinder(s). A single actuator shall consist of a balanced trunnion mounted cylinder.

561.2 POWER STEERING HYDRAULIC SYSTEM

561.2.1 Hydraulic steering system shall be comprised of two circuits: a hand operated manual circuit that is the control element and a power circuit that is the working element.

561.2.2 Hand operated manual circuit shall consist of the helm unit and the cylinder(s). Helm unit shall be a self follow-up bi-directional axial piston pump pumping fluid to the steering cylinder(s) to move the rudders in the appropriate direction.

561.2.3 A change over valve shall be installed to protect the manual circuit from over pressure, assist in air purging and provide for automatic transfer to manual emergency steering upon failure of the power system.

561.2.4 Power circuit system shall consist of one hydraulic pump, inline filter, and power cylinder(s). The pump output flow shall be metered and controlled to assure sufficient flow. After being diverted through the sequence-relief valve, fluid from the power pump shall enter the control valve manifold, where it shall further distribute to the proper side of the steering cylinder. The fluid shall pass through a filter before it returns to the reservoir to be recirculated.

561.2.5 Hydraulic fluid for both circuits shall be stored in one reservoir, and the helm pump may have a separate header tank.

SECTION 573 CROSS DECK WINCHES AND TIE DOWN FITTINGS

573.1 CROSS DECK WINCHES

573.1.1 The vessel shall have two cross deck winches located at the working deck as shown on the drawings. Suitable winch is Warn DC3000 24V PN 85252 with two layers of line (50 feet), or equal.

573.1.2 The winch shall use Class II high-strength synthetic cordage recommended for the application be the manufacturer with a safety factor of 5:1 based on the rated minimum breaking strength. Suitable line is Samson Amsteel Blue 3/8 diameter.

573.1.3 The winch shall be equipped with a roller fairlead to maintain required fleet angle to the drum.

573.2 TIE DOWN FITTINGS

573.2.1 Burchell Type 14 Inconel tie down fittings shall be provided on the working deck to secure buoys. (Burchell Professional Group, Inc. www.burchellpg.net)

573.2.2 Griping tie downs which interface with the Burchell fittings shall be provided.

SECTION 581 ANCHOR

581.1 GENERAL

581.1.1 The purpose of the anchoring system is to provide a means of stowing, deploying and retrieving one anchor.

581.1.2 Anchor arrangements shall be such that the anchor or anchor chain does not foul or damage the hull, hull appendages or equipment when weighing, dropping, or riding at anchor

581.1.3 The anchor and anchor rode shall be stowed on the foredeck.

581.1.4 The anchor rode shall be installed on a reel protected from the weather.

581.2 ANCHOR

581.2.1 A 300 ft anchor rode consisting of a combination of line and chain attached to the anchor shall be provided and configured as follows:

581.2.1.1 One (1) Fortress FX-55 or equal anchor.

581.2.1.2 A minimum of 25 ft. of not less than 1/2 inch BBB galvanized chain with a minimum Working Load Limit (WLL) of 4500 pounds shackled to the anchor.

581.2.1.3 Double braided nylon line with a minimum circumference of 2-¾ inches with and a Minimum Breaking Strength (MBS) of 19,000 lbs inches spliced with a thimble to the chain.

581.2.2 Line end terminations shall be greater than 95% efficient as recommended by the cordage manufacturer.

581.2.3 Fittings used in the anchor rode shall be galvanized and have a WLL exceeding the WLL of the chain.

581.2.4 A spare Fortress FX-55 or equal anchor shall be provided onboard.

581.3 WINDLASS

581.3.1 Anchor deployment and recovery shall utilize a Vertical Windlass with a wildcat capable of hauling both rope and chain. The windlass shall have an inhaul rate of not less than 30 ft/min under rated load.

581.3.2 The anchor windlass shall be constructed of stainless steel.

581.3.3 The anchor windlass foundation and fasteners shall be designed for not less than three times the rated capacity of the windlass.

581.3.4 Means shall be provided to manually operate the anchor windlass.

581.3.5 The anchor windlass shall be electrically powered and have a load rating not less than 600 lbs.

581.3.6 The anchor rode shall be arranged to be routed through an open centerline chock and secured to a bitt when deployed.

SECTION 582 MOORING AND TOWING

582.1 MOORING LINES

582.1.1 The vessel shall have four (4) 3 ¼-inch circumference Samson RP-12 Ultra Blue or Cortland Supertuf Plus or equal lines. These mooring lines shall be 90 feet in length with an eye splice in each end.

582.1.2 Mooring Lines shall be stowed in bins.

582.2 HULL FITTINGS

582.2.1 Hull fittings such as chocks and bitts shall be provided to enable mooring and towing operations. Chocks not mounted in bulwarks may be open.

582.2.2 Fittings shall be cast steel or welded steel and have a Safe Working Load exceeding the MBS of the higher of the mooring, towing or anchoring line which may use the fitting.

582.2.3 Chocks shall have a minimum of opening of nominally 3 inches tall at the centerline and 6 inches wide at the centerline.

582.2.4 Bitts are to be 6 inches in size and be of double type welded construction designed for welding to the deck.

582.2.5 Bitts and Chocks shall be arranged as shown on the drawings .

582.2.6 Supporting structure shall meet manufacturer specifications and the requirements of ABS MVR 3-5-1.

SECTION 583 LIFE SAVING APPLIANCES

583.1 LIFE RINGS

583.1.1 Three (3) each of 24-inch diameter life rings shall be installed. The rings shall meet 46 CFR 160.05 and SOLAS Life Saving Appliance requirements and shall be stowed in approved, float free cradle devices.

583.1.1.1 Two (2) life rings shall be located on the pilothouse level (1 Port/1 Starboard), and one (1) life ring shall be located on the aft Working Deck

583.1.1.2 The rings shall be installed vertically in open racks and shall be capable of being rapidly lifted free and easily cast loose. In case of submergence, the rings shall float free.

583.1.2 The rings shall be marked with two (2) inch wide reflective tape into four (4) quadrants. The tape shall wrap around the floatation section of the ring and shall be visible from any direction. The ring quadrants shall be marked with two (2) inch black decal lettering or stencils with the words “U.S.”, “COAST”, and “GUARD” (left to right, clockwise) with the boat name or number on the bottom quadrant.

583.1.3 Accessories

583.1.3.1 A rescue line throw bag shall be stowed adjacent to each life ring, so that it floats free with the ring. The CRES snap hook from the throw bag shall be attached to the life ring but shall be easily detachable.

583.1.3.2 A floating electric marker light unit shall be stowed adjacent to each life ring, so that it floats free with the ring. A short lanyard of 3/8 inch double braid construction and CRES snap hooks shall be provided and attached to the marker light. The CRES snap hook shall be attached to the life ring but shall be easily detachable.

583.1.3.3 A cleat shall be provided in the vicinity of the life ring for securing the line from the throw bag, after it is thrown.

583.2 LIFE RAFTS

583.2.1 At least one life raft shall be provided. These shall be USCG approved inflatable life rafts and shall be stowed in white hard-side containers and secured on the main deck or 01 deck. The life rafts shall be stowed in separate cradle foundations, capable of being launched by gravity. The life rafts shall be stowed in accordance with manufacturers' recommendations.

583.2.2 Each life raft shall have a capacity of 125% of the vessel's complement.

583.2.3 Life rafts shall be accessible for manual release.

583.2.4 The racks shall include neoprene rubber strips as an interface to support the raft canister. The racks shall have a system or feature, operable by one (1) person, to push the raft out of the rack against at least a 15-degree adverse list.

583.2.5 Life rafts shall not become entrapped within the vessel structure when deployed against at least a 15-degree adverse list.

583.2.6 Life rafts shall be secured in the racks with 5/16 inch diameter wire rope or equivalent high strength cordage (Kevlar) based rigging. The tension rigging shall utilize the grooves in the raft container for compression straps. The system shall be designed to retain the raft against a storm loading force of not less than 3375 lbs.

583.2.7 The rigging shall include a hydrostatic release unit (HRU), provision for easy manual release, and a severable lashing for emergency release. The rigging shall have an aluminum plate strong-back to spread the wire ropes or strapping apart to secure the load. The rigging shall include threaded U-bolt, threaded clevis bolt or threaded turnbuckle provisions for at least two (2) inches of adjustment to the rigging length to secure the raft in the frame. Nuts on bolts shall be backed with locking nuts.

583.2.8 The HRU release button shall be exposed for operation. The unit shall be shackled to ships structure; it shall not be bolted to structure. The unit shall be suspended in the retention rigging and shall not be installed in such a manner that the HRU is side loaded or bent in any manner.

583.2.9 A severable lashing, consisting of two (2) master links or shackles approximately three (3) inches apart, bound together with five (5) complete turns of ¼ inch nylon or polyester cord, shall be provided. The lashing shall be finished with a clove hitch and two (2) half hitches. The cordage ends shall be seared. The lashing shall be arranged to avoid chaffing.

583.2.10 The sea painter from the life raft container shall be secured in accordance with manufacturer's recommendations.

583.2.11 A painter cleat shall be provided for each raft on the deck edge or railing adjacent to raft stowage, for control of the painter to a deployed raft.

SECTION 589 CRANES – SECTION UNDER DEVELOPMENT

589.1 GENERAL

589.1.1 A hydraulic, deck mounted crane is required for buoy operations with a safe working load of approximately 4,500 lbs. The crane shall be Appleton Marine Knuckle Boom Crane KB21-21 or equal.

589.1.1.1 The crane manufacturer shall hold a Monogram License from American Petroleum Institute (API).

589.1.1.2 The crane shall meet ABS and API Specification 2C requirements.

589.1.2 The crane shall be located in accordance with the drawings.

589.1.3 Besides buoy handling the crane shall be capable of handling debris and general goods. It will not be used to lift personnel.

SECTION 593 ENVIRONMENTAL POLLUTION CONTROL SYSTEMS

593.1 SEWAGE SYSTEM

593.1.1 The purpose of sewage system is to efficiently collect, hold, and transfer sewage generated onboard to offshore locations. It is also capable of discharging sewage to a shoreside facility using the JABSCO 18690-000 pump at 10 gpm, or equal. The Dometic VG4 Vacuum Generator, or equal, operates on 24V DC power to generate vacuum pressure, enabling effective waste transport from the toilet to the sewage/graywater holding tank. Allowable piping materials are carbon steel, stainless steel, or 90-10 copper nickel.

593.1.2 Arrangement and location of sewage collection tanks, pumps, and processing equipment shall be as shown on the drawings.

593.1.3 The sewage/gray water holding tank shall receive waste from the vacuum sewage collection system and gray water system.

593.1.4 The sewage system shall consist of two (2) vacuum operated water closets, a holding tank, vacuum pump, shore discharge pump, and necessary wiring, piping and controls.

593.1.5 Water closets shall be installed as shown on the drawings. The 24-volt DC vacuum pump Dometic VG 4 shall draw waste from the bowl to the Sewage water holding tank. Flushing shall be with potable water. Holding tank shall be stainless steel or GRP with gel coat inside, with a minimum capacity of 500 gallons.

593.1.6 Emptying the tank shall be accomplished with a 115-volt AC macerator/pump, Jabsco Model 18690-0000 or equal, through a deck fitting. Pump shall be capable of at least 10 gpm at 20 feet of discharge head and shall have a shutoff head of at least 30 feet. The tank shall be tested to 5 psi or 1-1/2 times the total head of water from the lowest point in the tank to the highest point in the tank's vent, whichever is greater.

593.1.7 System shall have sewage level indicating lights at 1/3 full, 2/3 full and 3/4 full and shall be marked as such. The level indicating lights shall be mounted on the forward pilothouse console and shall be powered by 24-volt DC. Means of verifying tank level shall be provided at the tank.

593.1.8 Piping and connections shall be in accordance with Section 505. Connections shall not be compatible with the potable water system or deck washdown system.

593.1.9 The sewage system shall include a 50-foot collapsible hose (such as Goodyear Collapsible Discharge Hose) with appropriate connections for transferring waste to a shore-side sewage facility. One end of the hose shall be equipped with a screw or cam-lock fitting for attachment to the deck discharge fitting. The opposite end shall have a complementary fitting, allowing both ends to be connected together when the hose is not in use. Additionally, a connection shall be provided to enable the deck washdown pump to flush the waste hose into the shore-side facility.

593.1.10 Each water closet shall have a means of isolation to allow for maintenance.

SECTION 600 OUTFITTING

600.1 GENERAL

600.1.1 The Contractor shall provide the items and quantities identified in Specification Table 600-1 to 600-4 with each boat.

600.1.2 Stowed items shall be located and arranged to support effective use by the crew.

Table 600-1: General Outfit

Description	Quantity	Additional Information	Location	Stowage
Ensign, National (16”x24”)	1 Ea	NSN 8345-00-245-2040	Mast	See Location
Ensign, US Coast Guard (15”x24”)	1 Ea	NSN 8345-00-242-0275	Mast	See Location
HSC-L Boat Information Book	1 Ea	In accordance with SOW Section 086	Pilothouse	See Location
Hearing Protectors, Ear Muff Style (Pair)	2 Ea		Outside entrance to engine room	See Location
Hearing Protectors, Foam Insert Type, Disposable, Corded	1 Box		Workshop	See Location
Flashlight, Non-Explosive	2 Ea		One outside entrance to engine room, one at entrance to Pilothouse	See Location
Blankets, Wool	4 Ea	NSN 7210-00-082-5668	Habitability Storage	See Location
Pillows	4 Ea	NSN 7210-01-015-5190	Habitability Storage	See Location
Binoculars, 7x50 Marine Waterproof	1 Ea		Pilothouse	See Location
Push-Button Portable Signal Horn, Air, with 8 oz Propellant	1 Ea		Pilothouse, navigation stowage	See Location
Spare 8 oz Propellant for the Portable Signal Horn	1 Ea		Pilothouse, navigation stowage	See Location
Handi-Wipes	2 Pkg	To fit dispensers, see Specification Section 644	Head and Galley	See Location
Medium Duty Wipes	1 Box	NSN 7920-01-448-7053	Engine Room	See Location
Paper Towels	1 Roll	To fit holder, see Specification Section 644	Near sink	See Location
Toilet Paper	1 Roll	To fit holder, see Specification Section 644	Head, in water resistant holder	See Location
Covers (Console, seat, towline reel, and P-6 pump container)	1 Boat Set	See Specification Section 613		
Anchor	1 Ea	See Specification Section 581		
Anchor Rode	1 Ea	See Specification Section 581		

Table 600-2: Navigation Equipment Outfit

Description	Quantity	Additional Information	Location	Stowage
Navigation Rules and Regulations Handbook	1 Ea	International Regulations for Preventing Collisions at Sea (72 COLREGS) and the U.S. Inland Navigation Rules (33 CFR 83), as amended; their Annexes; and some associated provisions	Navigation Station	See Location
Parallel Plotter	1 Ea	Weems and Plath #120 or equal	Navigation Station	See Location
Dividers	1 Ea	Weems and Plath #176 or equal	Navigation Station	See Location
Slide Rule, Nautical	1 Ea	Weems and Plath #105 or equal	Navigation Station	See Location
Search Pattern Wheel	1 Ea	Weems and Plath #113 or equal	Navigation Station	See Location
Pencils, #2	5 Ea		Navigation Station	See Location
Pencil Sharpener	1 Ea		Navigation Station	See Location
Anemometer, Hand Held Electronic	1 Ea		Navigation Station	See Location
Stop Watch	1 Ea	NSN 6645-00-126-0286	Navigation Station	See Location
Penlight with Red Lens	1 Ea		Navigation Station	See Location

Table 600-3: Rescue and Survival (R&S) Equipment Outfit

Description	Quantity	Additional Information	Location	Stowage
24” Ring Buoys, Lifesaving	3 Ea	NSN 4220-00-275-3156	IAW arrangement	See Location
Light, Marker, Distress	4 Ea	FML-7 or equal.	With Ring Buoys	See Location
Snap Hook for Marker Light Lanyard	4 Ea	Lifesaving Systems P/N 365 or equal	With Ring Buoys	See Location
Throw Bag, 75’ of Line	4 Ea	Lifesaving Systems Model 237P or equal	With Ring Buoys	See Location
Snap Hook for Throw Bag	4 Ea	Lifesaving Systems P/N 365 or equal	With Ring Buoys	See Location
Pyrotechnics Storage Container	1 Ea	Lifesaving Systems P/N 527 or equal		Watertight
Personal Flotation Devices, Type I, Adult	12 Ea	Stearns Merchant Mate I or equal		Open
Personal Flotation Devices, Type I, Youth	6 Ea	Stearns Merchant Mate I or equal		Open
Damage Control Kit Bag	1 Ea	NSN 8465-01-117-8699 Rescue	Interior readily accessible with space for full bag; contents of bag GFE	Watertight
Knife with Case	1 Ea	NSN 4220-00-141-1909	Adjacent to aft towing bitt	Open
USCG Boat Response Aid Kit	2 Ea	USCG Boat Response Aid Kit NSN 6545-01-646-2623, ITEM # 80-0353 or equal	Interior, medical assistance location	See Location

Table 600 4: Mooring and Towing Equipment Outfit

Description	Quantity	Additional Information	Location	Stowage
Mooring Lines, 3-1/4" circ. DBN x 90 ft	4 Ea	Samson RP-12 Ultra Blue or Cortland Supertuf Plus or equal	Aft deck locker	
Heaving Lines, 100 ft	3 Ea	Life Saving Systems Model 227-H or equal	Aft deck, secured to handrails	
Fenders, Cylindrical	4 Ea	Min. 11" dia x 41" long pneumatic with 10 ft x 1/2" dia. DBN tag lines	Deck lockers	
Fenders, Spherical	4 Ea	14" diameter with 10 ft x 1/2" dia. DBN tag lines	Deck lockers	
Telescoping Boat Hooks, 12 ft	2 Ea		Exterior, readily accessible from side decks	
Shackle (1 ¼") Galvanized Steel	1 Ea			
Nylite Shackle, Size 4 (Orange)	1 Ea	American Group: (800) 227-7673 or equal		
Folding Grapnel	1 Ea	P/N G8008-MKII/ Bosun Supplies or equal		
Drogue, Stubbies	1 Ea	Stubbies Upholstery & Canvas P/N-47MLB Drogue or equal		
Drogue, Bag Flotation	1 Ea	Stubbies Upholstery & Canvas P/N- 47FLOATDROGUEBAG or equal		
Drogue Tending line, 1 ½" circ. DBN x 200 ft, hardware to include: 316 Stainless Steel 5/16" swivel, 316 Stainless Steel 5/16" screw pin shackle, and 1/2" plastic thimble spliced into the line at one end.	1 Ea	200 ft length of line to include length of spliced-in thimble		
Snap Hook	1 Ea	Life Saving Systems Model #367 Snap Hook and #377 Snap Hook Clip or equal		
Snap Hook Tending Line, 2" circ. black, DBN line x 15 ft	1 Ea	Line spliced onto the snap hook at one end		
Towing Line, 5" circ., 1200 ft, DBN	1 Ea	TBD		
Towing Line, 3 ¼" circ., 900 ft, DBN	1 Ea	TBD		
Towing Gear Stowage Case with Shackle (1 ¼"), Nylite Shackle, Folding Grapnel	1 Ea	Pelican Case 1550 P/N 527 or equal		

SECTION 602 HULL, MECHANICAL DESIGNATION, AND MARKING

602.1 BOAT IDENTIFICATION PLATE

602.1.1 The boat shall be provided with a manufacturer's identification plate, mounted in accordance with the Boat Management Manual, COMDTINST 16114.4 (series). The plate shall identify boat and manufacturer information in accordance with 33 CFR 181. The USCG boat number will be provided at the time an individual boat is ordered.

602.1.2 The identification plate shall be a minimum of 4 by 6 inches and be made of metal with either engraved or raised molded permanent letters. The plate shall be a minimum of 1/8 inches thick. The label plate shall be mounted in a readily viewable location inside the Pilothouse.

602.2 HULL MARKINGS

602.2.1 The boat shall include USCG visual identification in accordance with Coast Guard Drawing No. FL-2804-021 per instructions for vessel length 50' to 64'. Markings shall include the Coast Guard stripe and emblem on the port and starboard bow, the words "U.S. COAST GUARD" on the port and starboard aft of the stripe, the boat number port and starboard forward of the stripe, and the hull number above the station name on the stern.

602.2.2 Painted on letters are not acceptable. Lettering color shall be white.

602.2.3 Travel lift sling positions shall be permanently marked on the exterior of the boat in locations that can be easily seen from the working deck.

602.3 LABEL PLATES

602.3.1 Each control, switch, gauge, and valve shall be provided with a durable and permanently attached label plate to indicate its function.

602.3.2 Each radio, microphone, and speaker shall be labeled with the radio information (e.g., VHF/UHF/700/800, HF).

602.3.3 Each tank deck connection shall be labelled in accordance with ABYC standards including fuel fill, potable water, and sewage.

602.3.4 Label plates for differential pressure gauges shall state a maximum allowable differential pressure before cleaning of strainers is required.

602.3.5 All labels shall be a minimum of 12-point font.

602.3.6 The entrance to the engine room shall have a warning sign indicating the presence of a noise hazard. The sign shall be labeled "DANGER, NOISE HAZARD AREA – HEARING PROTECTION MUST BE WORN."

602.3.7 All permanent notices and signs shall be mounted and sized in accordance with ABYC T-5, Safety Signs and Labels. The Contractor may propose certain notices and signs to be provided in the BIB rather than being mounted as approved by the USCG.

SECTION 603 DRAFT MARKS

603.1 GENERAL

603.1.1 Draft marks shall be in accordance with Coast Guard Drawing No. FL-2804-021.

SECTION 604 LOCKS, KEYS, AND TAGS

604.1 DEFINITIONS

604.1.1 Master key - A key that will operate two or more locks that are keyed differently from each other.

604.2 GENERAL

604.2.1 Locks and latch sets shall be marine grade hardware.

604.2.2 Locks shall have a unique serial number stamped or engraved on the outer surface. Keys shall be stamped or engraved with the serial number of the lock served.

604.2.3 All external-facing doors with access to the interior of the boat shall be provided with a hasp-type mechanism that may be secured using a padlock.

604.2.4 All external-facing hatches with access to the interior of the boat shall be provided with non-integrated means to secure the hatch in a closed position. Hatches shall not be lockable from the interior of the boat

604.2.5 All potable water fills and sounding tubes shall be provided with a hasp-type mechanism that may be secured using a padlock.

604.2.6 Hasps and staples shall be fitted so that when the door is opened or closed, padlocks cannot be caught between the door and the door frame.

604.2.7 Each lock shall be provided with two (2) keys. Two (2) master keys shall be provided for opening all locks except crew lockers and berth lockers.

SECTION 612 RAILS, STANCHIONS AND LIFELINES

612.1 DEFINITIONS

612.1.1 Lifelines – Refers to protective devices in the weather fabricated of flexible line or cables supported by stanchions allowing for flexibility or easy removal to gain access.

612.1.2 Liferails - Refers to protective devices fabricated of rigid pipe or tubing wherever there is danger of crew members falling overboard or to a lower level of 24 inches or more including along deck edges, waterborne mission walkways, and around antenna platforms.

612.1.3 Grabrods - Refers to bars installed near ladders where they will aid personnel in ascending, descending, or stepping from the same.

612.1.4 Handrail - Refers to protective devices fabricated of rigid pipe or tubing on a bulkhead adjacent to ladders, ladderways, and walkways or ramps to steady movement along the same.

612.1.5 Stanchions – Vertical supports for lifelines or liferails.

612.2 GENERAL

612.2.1 A recommended arrangement for liferails and lifelines is depicted in the design package drawing XXXX General Arrangement .

612.2.2 Stanchions and liferails shall be galvanically compatible with the deck or structure materials to which they are fastened.

612.2.3 Lifelines shall be white.

612.2.4 Unguarded openings between adjacent sections of liferails and lifelines or between the ends of the rails, lines, and adjacent structure shall be no more than five (5) inches.

612.2.5 Removable lifelines shall be used for the working deck area, starting even with the aft edge of the pilothouse.

612.2.6 Removable lifelines and stanchions shall not be installed on the deck edges forward of the aft edge of the crane pedestal.

612.2.7 The transverse area between the termination of the lifelines and the crane pedestal shall be protected by a removable wire or chain.

612.2.8 Liferail and lifeline stanchions shall be constructed per ABS rules 3-2-17. Braced stanchions shall be provided as necessary to withstand the pull of the lifeline. End stanchions shall have support braces.

612.2.9 Handrails shall be provided around platforms, deck plates, gratings, walkways, passageways, switchboards, hot surfaces, and moving parts of machinery. Handrails shall be located so as not to interfere with the opening and closing of hatches.

612.2.10 Steel shall be used for handrails, stanchions, and grabrods in machinery spaces except for those attached to switchboards. Handrails shall be provided on bulkheads, in ladderways, and by ramps and ladders rising over 24 inches.

SECTION 622 DECK COVERING

622.1 GENERAL

622.1.1 This section contains requirements for ladders and other climbing aids, handrails, floor plates, staging, gratings, and service platforms.

622.1.2 Unless otherwise specified, inclined and vertical ladder dimensions shall be designed to ASTM F1166 using the requirements for stairs.

622.1.3 Vertical ladders, stringers, and supports shall support a 500 lb. static load for every 78 inches of the height of the ladder.

622.1.4 Ladders, handrails, staging, and service platforms shall not interfere with the opening and closing of hatches, scuttles, doors, manholes, hinged gratings, or equipment.

622.1.5 Floor plates and platforms shall be capable of supporting a load of 144 pounds per square foot. Gratings shall be capable of supporting a load of 100 pounds per square foot.

622.1.6 Ladders shall be removable, bolted in place via padeyes and fitted with handrail or grabs.

622.1.7 All decking system and latching mechanisms shall:

622.1.8 Be constructed from corrosion-resistant materials; and

622.1.9 Be designed to securely hold in the open and closed positions in all operating environments.

622.2 REMOVABLE DECKING SYSTEM

622.2.1 A removable decking system may be provided in lieu of hatches to provide a flat walking surface and to allow access to the machinery, tanks, and bilges below the decking for inspection and maintenance while at the dock or underway.

622.3 MACHINERY SPACE DECKING

622.3.1 Machinery space decking systems shall allow the crew members access to and around machinery and bilge below without having to step on hull plating and structure.

622.3.2 Deck panels shall include quick-acting latching mechanisms that interlock mechanically with the framework in a manner that holds the panel in place securely.

622.3.3 Deck panels requiring lifting for regular maintenance or inspection, particularly underway, shall be hinged on the appropriate side to facilitate the maintenance or inspection activity.

622.3.4 Deck panels shall be able to be removed or replaced by one crew member without the use of tools.

622.3.5 The working surface of the deck panel shall have a slip-resistant surface and shall be free of protrusions and other features that could present a tripping hazard in the installed configuration.

SECTION 625 FIXED PORTLIGHTS AND WINDOWS

625.1 GENERAL

625.1.1 Windows and portlights shall be designed in accordance with ABS MVR Sections 3-2-17 and 3-6-1.

625.1.2 All windows/portlights shall be marine grade windows.

625.1.3 Light transmission levels through pilothouse windows and windows in pilothouse doors shall comply with the requirements of ABYC H-3 Exterior Windows, Windshields, Hatches, Doors, Portlights, and Glazing Material.

625.1.4 Each forward-facing window (windshield) shall have a wiper system with a window washing system. The wiper system shall:

625.1.5 Clear at least 60 percent of the window and accommodate visibility for 5th percentile female to 95th percentile male while underway at cruising speed at performance weight condition.

625.1.6 Have at least three speeds (intermittent, low, and high) and be self-parking.

625.1.7 Windows in the pilothouse shall be electrically heated for defrosting and defogging.

625.1.8 Windows that can be opened shall:

625.1.8.1 Be able to be secured in the open and closed position.

625.1.8.2 Not impede access to the side decks of the boat.

625.1.8.3 To the extent practicable, be located to avoid taking on spray when the craft is underway.

625.2 WINDOWS AND PORTLIGHTS IN DOORS

625.2.1 Windows and portlights installed in doors shall be fixed, unheated, and made of heat-treated and shatterproof glass.

625.2.2 Doors to the weather shall have windows installed.

625.2.3 The following interior doors shall be fitted with portlights: watertight doors, and door to the workshop.

625.2.4 Stairs leading directly to the pilothouse shall have means to make them light tight to prevent spillage into the pilothouse

625.3 WINDOW MATERIALS

625.3.1 Forward-facing windows shall be constructed of monolithic tempered and laminated glass, or multi-ply tempered and laminated glass.

625.3.2 Window thickness shall be at a minimum in accordance with TS 625 – Boat Windows, as shown in Table 625-1.

Table 625-1: Boat Window Minimum Thickness by Material Type

Material	Minimum Thickness, US (in)
Monolithic Tempered Glass	1/4
Multi-Ply Laminated & Tempered Glass	0.30
Polycarbonate (PC)	5/16
Poly (methyl) methacrylate (PMMA)	5/16

SECTION 631 PAINTING

631.1 GENERAL

631.1.1 This section provides additional requirements for coatings consisting of surface preparation, painting, preservation, and corrosion control.

631.1.2 Coatings, color schemes, and corrosion control for all exterior and interior areas of the boat shall comply with the Black Hull requirements given in the COMDTINST M10360.3D, SFLC Standard Specification 6310, CGTO PG-85-00-60-S.

631.1.3 Underwater hull coating systems shall use International Intershield 163 Inerta 160, or equal.

631.1.4 The Contractor shall have each coating system inspected by a certified AMPP CIP Level 2, Certified Coatings Inspector or equivalent and submit an inspection report on the coating system preparation and application

631.1.5 Prior to installing insulation, piping surfaces with expected surface temperatures below 200°F shall be coated as follows:

631.1.5.1 The surface preparation shall be SSPC-SP 3.

631.1.5.2 The coating system shall be one coat of high build epoxy at 5.0 to 6.0 mils Dry Film Thickness (DFT) qualified for use under MIL-PRF-23236 or MIL-PRF-24647, followed by one coat of Silicone Alkyd, Type II C1 at 2.0 to 3.0 mils DFT qualified for use under MIL-PRF-24635.

631.1.5.3 The top-coat color shall correspond the piping contents in accordance with the COMDINST M10360.3D USCG Colors and Coatings Manual.

631.2 FAYING SURFACE COATING

631.2.1 All faying surfaces, including the hull area in contact with the fender and the bottom of lockers or similar spaces that may contain wet items such as lines or fenders, etc., shall be coated for corrosion resistance.

631.2.2 Joints and crevices that may trap water shall be sealed with one part polyurethane caulking compound or be coated as a faying surface.

631.3 ANTI-SWEAT TREATMENT COATING

631.3.1 Anti-Sweat Treatment Coating System, if applied, shall be applied as follows:

631.3.2 The surface preparation shall be by abrasive blast to bare metal with clean, fine aluminum oxide, garnet, or equivalent inert material conforming to CID A-A-59316, Type I or Type IV to achieve a profile of 1.5 to 3.5 mils.

631.3.3 Ceramic Insulation Coatings shall be:

631.3.4 White water based acrylic coating;

631.3.5 Meeting IMO and SOLAS requirements; and

631.3.6 Low volatile organic compounds (i.e., less than 0.07 lbs per gallon).

631.4 COATING SYSTEM:

631.4.1 First coat: MIL-PRF-23236 or MIL-PRF-24647 high build epoxy, black in color (SAE AMS-STD-595, ID # 17038) at 5.0 to 6.0 mils Dry Film Thickness (DFT);

631.4.2 Second coat: Ceramic Insulation Coating, 20.0-22.0 mils Dry Film Thickness (DFT);

631.4.3 Third coat: Ceramic Insulation Coating, 20.0-22.0 mils Dry Film Thickness (DFT); and

631.4.4 Fourth coat: Ceramic Insulation Coating, 20.0-22.0 mils Dry Film Thickness (DFT).

SECTION 633 CATHODIC PROTECTION

633.1 GENERAL

633.1.1 The boat's hull shall be protected from corrosion with a sacrificial anode cathodic protection (CP) system designed and installed in accordance with ABS Guidance Notes on Cathodic Protection of Ships and ABYC E-2.

633.2 SACRIFICIAL ANODE CATHODIC PROTECTION SYSTEM

633.2.1 Anodes shall be in accordance with ABYC E-2 and ABS guidance.

633.2.2 Sacrificial cathodic protection systems installed on the boat shall provide a minimum service life of four years.

633.2.3 Sacrificial anodes shall be installed in a manner to facilitate ease of replacement and not require special tools or training.

633.2.4 The boat shall be equipped with a hull electrical potential monitoring system which measures potential difference between the underwater hull and anodes vs. a permanently installed external electrode fixed to the hull.

SECTION 635 THERMAL, ACOUSTIC AND FIRE PROTECTION INSULATION

635.1 THERMAL, ACOUSTIC AND FIRE PROTECTION INSULATION TREATMENT

635.1.1 Insulation materials shall be U.S. Coast Guard approved. (https://cgmix.uscg.mil/equipment/equipmentsearch.aspx) or meet the applicable standards listed herein.

635.1.2 Blown-in insulation shall not be used.

635.1.3 All fire insulation shall be rated for unrestricted use.

635.2 APPLICATION AND INSTALLATION

635.2.1 Insulation on bulkheads shall extend from 6 inches above the deck to the overhead. Where insulation is subject to damage, a continuously welded Z-bar coaming shall be installed at the bottom edge of the insulation. Doors and hatches shall meet the fire and thermal insulation requirements of the bulkheads and decks in which they are installed.

635.2.2 Fasteners shall not crush or otherwise reduce the insulation value of the insulation.

635.2.3 Insulation shall be installed so that movement of the enclosed structure, such as by vibration, will not damage the insulation.

635.2.4 Insulation shall be segmented and overlapped to provide a tight fit without voids around webs and flanges of beams and stiffeners and other fittings.

635.2.5 Thermal and acoustic insulation shall not be installed in bilge regions.

635.2.6 Sheathing shall be installed wherever necessary to protect the insulation from mechanical or fluid damage. A surface treatment or covering shall be installed on sheathing where necessary for the protection of personnel from cuts or abrasion or from hot surfaces

635.3 STRUCTURAL FIRE PROTECTION

635.3.1 Boundaries of the engine room with other spaces shall be treated with a minimum of A-30 Structural Fire Protection (SFP) insulation.

635.4 THERMAL INSULATION

635.4.1 Thermal insulation shall be provided on compartment boundaries, including webs and flanges of beams and stiffeners. The following additions and exemptions apply to the thermal insulation requirements:

635.4.2 Weather doors and doors between conditioned and non-conditioned spaces shall be insulated to the same level as the boundary in which they are installed to the maximum extent possible. Insulation shall extend 12 inches beyond the area requiring insulation, including abutting uninsulated boundaries and stiffener webs and flanges.

635.4.3 Vapor barrier and Lagging - Vapor barrier treatment shall be applied to all thermal insulation to prevent penetration of moisture and shall remain intact and continuous over the protected area by means of sealed joints and edges. Lagging shall be installed as part of the vapor barrier treatment over thermal insulation.

635.5 ANTI-SWEAT TREATMENT

635.5.1 Anti-sweat treatment shall be applied to the warm side of uninsulated boundaries and surfaces and closures susceptible to condensation, including webs and flanges of beams and stiffeners. Anti-sweat treatment shall only be used on surfaces unsuitable for thermal insulation.

635.5.2 Anti-sweat treatment shall not be applied to any part of a door, hatch or scuttle operating mechanism, or contact surfaces.

635.5.3 Anti-sweat treatments are considered coatings and shall be in accordance with Specification Section 631.

635.6 ACOUSTIC INSULATION

635.6.1 Acoustic insulation shall be installed on compartment surfaces to meet the ship space airborne noise criteria.

635.6.2 Perforated acoustic insulation shall not be installed in machinery spaces where such perforations could act as collectors of combustible material.

635.6.3 Spray on acoustic treatments and damping compounds are considered coatings and shall be in accordance with the vessel specification and be U.S. Coast Guard approved (https://cgmix.uscg.mil/equipment/equipmentsearch.aspx).

635.6.4 If acoustic insulation is required for any area for which thermal insulation is required, the acoustic treatment shall be applied to the plane surfaces and thermal insulation shall be applied to beams and stiffeners.

635.6.5 Where both fire protection and thermal or acoustic insulation are required, and the fire insulation does not meet the thermal or acoustic specifications, additional fire insulation may be installed to satisfy the requirements if permitted by the certificate. All other combinations of fire protection with thermal or acoustic insulation shall require approval by the U.S. Coast Guard technical authority.

SECTION 640 LIVING SPACES

640.1 GENERAL

640.1.1 Space arrangement plans are provided in the design package.

640.1.2 Clearance shall be provided around furniture to ensure the full use of drawers, lockers and writing shelves. Articles of equipment such as coat hooks and light switches shall be located to ensure their utility.

640.1.3 Berths shall be installed oriented generally fore and aft.

640.1.4 A minimum clear area of 77-1/2 inches long by 27-1/2 inches wide

640.1.5 Padding with waterproof covers and a means to secure them to the berth;

640.1.6 A red/white LED reading light; and

640.1.7 Steps or a ladder shall be provided to access berths as required.

640.1.8 Berths may be multi-purpose and used for seating.

640.2 FURNITURE AND FURNISHINGS

640.2.1 Deck and bulkhead mounted furnishings shall be properly secured to prevent movement. Movable components such as drawers, sliding shelves, and hinged mechanisms shall be captive in any normal operating position to prevent movement and shall be sufficiently snug to prevent rattle of parts during vessel vibration.

640.2.2 Unless otherwise approved, materials used for furnishing specified herein shall be in accordance with 46 CFR Part 164.

640.2.3 Means to secure portable furniture, such as chairs and wastebaskets, in a seaway shall be provided. Such means shall not involve the use of tools.

640.2.4 Furniture shall be installed so that horizontal surfaces are parallel to the baseline of the boat. Furniture, such as berths, desks, and lockers shall be removable and shall be secured by pinned or threaded connection to hard welded studs or fasteners.

640.2.4.1 As an alternative to welding the chair tie-down fasteners to the deck, the tie-downs may be installed using Phillybond TA30 Epoxy, or equal, over fully cured deck coating systems or primer.

640.2.5 Textiles shall be chosen from standard catalogue colors and patterns. The textiles shall be colorfast, fade resistant, resistant to salt spray, free from cracking, and have a shrinkage tolerance no greater than 1.5%.

640.2.6 Mirrors shall be centered 64 inches above the deck covering, unless otherwise specified.

640.2.7 Clocks and Barometer

640.2.7.1 One 24 hour, 6 inch, mechanical clock shall be installed in the Pilothouse.

640.2.7.2 One 12 hour, 6 inch, electric clock shall be installed in each of the following locations:

640.2.7.3 Workshop

640.2.7.4 Galley

640.2.8 A barometer shall be provided and installed inside the pilothouse. The barometer shall be of the same diameter, style, and quality as the pilothouse clock.

640.3 GALLEY

640.3.1 The galley shall be provided with the following:

640.3.1.1 A sink with faucet for preparing and cleaning up meals with hot and cold potable water;

640.3.1.2 At least eight cubic feet of freezer food storage;

640.3.1.3 At least seven cubic feet of refrigerated food storage;

640.3.1.4 At least 25 cubic feet of general storage for food and supplies;

640.3.1.5 At least four square feet of counter top area;

640.3.1.6 A trash container with captive lid;

640.3.1.7 A microwave oven without a rotating table;

640.3.1.8 A paper towel holder and handwipe dispenser; and

640.3.1.9 A soap dispenser.

SECTION 644 SANITARY SPACES AND FIXTURES

644.1 GENERAL

644.1.1 Space arrangement plans and piping one-line diagrams are provided in the design package.

644.1.2 A complete plumbing system including fittings, fixtures, and trim shall be installed.

644.1.3 Hot and cold potable water and drain connections shall be installed for equipment. Where a hose connection is specified in the potable water system, see Section 532 for installation requirements.

644.1.4 Attachment of plumbing fixtures to watertight and oiltight structure shall be avoided. Where this is not possible, the attachment shall not impair the tightness of structure.

644.1.5 Piping shall be concealed behind sheathing except for terminations and connections to hoses serving the fixture. The plumbing and fixtures within sanitary spaces shall be arranged so that surfaces are accessible for cleaning.

644.2 WR/WC

644.2.1 A grab rod shall be provided adjacent to each toilet.

644.2.2 A label plate shall be provided near each toilet that states:

DO NOT DISCARD PRODUCTS OTHER THAN TOILET PAPER, TO INCLUDE FEMININE HYGIENE PRODUCTS, IN WATER CLOSET

644.3 BATH

644.3.1 A shower shall be of the individual stall type with privacy curtain and curtain rod. Minimum size shall be 30 inches wide by 30 inches deep.

644.3.2 The shower fittings shall consist of a mixed flow, hand held, low flow shower head permanently attached to a flexible hose. One shower head with valve and hose shall be installed in the shower stall. A holster shall be provided in the shower stall for stowage of the shower head. A shut off valve of the globe type shall be installed in the supply line to the shower valve.

644.3.3 Towel bar shall be provided on or within reach from the shower stall. A soap dish shall be provided within the shower stall.

644.3.4 Boat structure next to the shower enclosure shall be prepared and painted as a closed-in space.

644.4 MISCELLANEOUS

644.4.1 Emergency Eye Wash Unit

644.4.1.1 Emergency eye wash unit shall be provided in the workshop.

644.4.1.2 The root valve the eye/face wash station shall be locked open.

644.4.2 Paper towel dispensers and waste receptacles shall be surface-mounted CRES material.

SECTION 660 SEATS

660.1 GENERAL

660.1.1 Dimensions of seating shall follow the recommendations of ASTM F1166 (Series).

660.1.2 Seats shall be fabricated of corrosion resistant materials.

660.1.3 Seat covering materials shall:

660.1.3.1 Be waterproof;

660.1.3.2 Be designed for a marine environment;

660.1.3.3 Be resilient to sun and saltwater exposure; and

660.1.3.4 Resist puncture, ripping, and tearing in a high-use environment.

660.1.4 Seats shall be padded.

660.1.5 Seats shall be removable using the onboard tool kit.

SECTION 670 STOWAGE

670.1 GENERAL

670.1.1 Stowage for the outfit items as identified in the Specification and any other items the Contractor deems necessary for the safe operation of the boat shall be provided.

670.1.2 Stowage locations shall suit the items being stored, preventing damage and wear to the item. Additional guidance on stowage locations is provided in the tables of outfit as well as elsewhere in this Specification.

670.1.3 All exterior stowage lockers shall have drains leading directly to the deck and be provided with captive removable plug(s).

670.2 CREW PERSONAL GEAR STOWAGE

670.2.1 Dedicated stowage shall be provided in the habitability spaces for six aviation style duffle bags that are used by the crew for their personal protective equipment and other gear.

670.2.2 Each bag has overall dimensions that are 20 inches wide, 20 inches tall, and 24 inches deep when full.

670.3 LIFE RING BUOY, FLOATING MARKER LIGHT, AND THROW BAG

670.3.1 The 24-inch life ring buoy, the floating marker light, and the throw bag shall be stowed in locations per the arrangement drawing. A lanyard with a snap hook shall attach the floating marker light to the life ring buoy.

670.3.2 The life ring buoy shall be labeled with the assigned boat's identification number on the top half of the ring and "U.S. COAST GUARD" on the bottom half of the ring. The lettering shall be two inches in height.

670.3.3 Type II retroreflective materials as required by 46 CFR 185.604 and IMO Resolution A.658(16) shall be applied to the life ring buoy.

670.3.4 The life ring buoy stowage shall allow for the floating marker light to remain clipped to the life ring buoy. The throw bag remains unclipped from the ring buoy but can be rapidly clipped to the buoy when needed.

670.4 MOORING LINE STOWAGE

670.4.1 Mooring lines shall be stowed in a location with drainage that is readily accessible, protected from UV, and vented.

670.5 TOOL KIT AND ONBOARD REPAIR PARTS, AND STOWAGE

670.5.1 The boat shall be provided with a tool kit. The tool kit shall be contained in a watertight, airtight, dustproof, chemical resistant, corrosion proof case, approx. 14 inches x 12 inches x 6 inches, with inserts, and shall at a minimum have the following tools:

670.5.1.1 Metric Combination Wrench Set: 10mm, 11mm, 12mm, 13mm, 14mm, 15mm, 16mm, and 17mm in a durable case;

670.5.1.2 SAE Combination Wrench Set: 1/4, 5/16, 3/8, 7/16, 1/2, 9/16, 5/8, and 11/16 in. in a durable case;

670.5.1.3 3/8 in. drive SAE 6 or 12-point Socket Set: 1/4, 5/16, 3/8, 7/16, 1/2, 9/16, 5/8, 11/16, and 3/4 in.ch in a durable hard plastic case;

670.5.1.4 3/8 in. drive Metric 6 or 12-point Socket Set 10mm, 12mm, 13mm, 14mm, 15mm, 16mm, 17mm, 18mm, and 19mm in a durable hard plastic case;

670.5.1.5 3/8 in. drive ratchet with 3 inches and 6 inches extension bars in a durable hard plastic case;

670.5.1.6 10 in. Adjustable Wrench;

670.5.1.7 Strap Wrench Set (two Piece);

670.5.1.8 10 in. Curved Jaw Locking Pliers;

670.5.1.9 10 in. Straight Jaw Pushlock Pliers;

670.5.1.10 10 in. Straight Jaw Pliers;

670.5.1.11 10-in-1 Ratcheting Multi-bit Screwdriver Set;

670.5.1.12 LED Rechargeable Waterproof Spot-Beam Flashlight with at least 290 lumen;

670.5.1.13 1/8 inch x 1/8 inch x 7.5 inch black poly cable ties, 50 pack; and

670.5.1.14 One roll (20 yards) of 2-inches wide duct tape.

670.5.2 The HSC-L shall be provided with onboard repair parts to enable the crew to correct minor underway casualties (e.g., belts, filters, light bulbs, fuses, etc.).

670.5.3 The stowage for these parts shall be with the tool kit.

670.5.4 The onboard repair parts shall be in a waterproof, chemical resistant, corrosion proof case similar to the case provided for the tool kit.

670.5.5 The onboard repair parts shall include the items in Table 670-1 at a minimum.

Table 670-1: Additional Minimum Onboard Repair Parts

Part Nomenclature	QTY (Minimum)
Fuel Filters, Primary with Bowls	2
Fuel Filters/Strainer, Secondary	2
Raw Water Pump Impeller	1
Impeller Removal Tool	1
Raw Water Pump Cover Gasket	1
Alternators/Generators Belt Set	2
Electrical Tape (20 yard, roll)	1
Fuses (In-Line [See Specification Section 320-5.3], Boat Set)	1